

16 BIT 4 Channel, CCD/CMOS Sensor Analog Front End With Timing Generator and LED Driver

FEATURE

- 4-Channel CCD/CMOS Signal
2ch, 3ch and 4ch Selectable
- Power Supply
Typ 3.3 V Only (Build in LDO, 3.3 V → 1.8 V)
- Max Conversion Rate: 50Msps
- Resolution: 16 bit
- CDS/SH Selectable
- Max Input Signal Range: 2.0 V
- Gain: Analog and Digital Hybrid
Analog Gain: 0.5– 3.5 [V/V], 3/64 [V/V] Step
Digital Gain: 1– 2 [V/V], 1/256 [V/V] Step
- Offset Correction DAC: ± 250 mV, 8 Bit
- Output: Standard LVDS / CMOS Selectable
LVDS: Data ch : 2ch, 3ch, Clock ch: 1ch
8-Bit/7-Bit Serializer Selectable
CMOS: 4 Bit $\times$ 4, 8 Bit $\times$ 2
- TG (Timing Generator)
Fast Transfer Clock: 2 Signals
Slow Transfer Clock: 1 Signals
- Timing Adjustment Resolution: tMCLK/48
- Input Clamp / Input Reference Level
Internal / External Selectable
- Reference DAC = 0.5 V, 1.1 V, 1.5 V, 2 V
- SPI: 3 Wire Serial
- GPIO: 2 Port

APPLICATION

- Copier
- Facsimile
- Scanner

DESCRIPTION

VSP5640 is a high speed and high performance 16-bit analog to digital conversion device which has independent 4-channels sampling circuit for multi output CCD and CMOS line sensors. The pixel data from sensor is sampled by the SH (Sampled Hold) or CDS (Correlated Double Sampler) circuit, and is then converted to digital data by an ADC (Analog to Digital Converter). Data output is selectable in LVDS or CMOS modes.

VSP5640 has programmable gain to support the pixel level inflection caused by luminance. The integrated DAC can be used to adjust the offset level for analog input signal. Also, TG is integrated in this device for control of sensor operation.

VSP5640 uses 1.65–1.95 V for core voltage, and 3.0–3.6 V for IOs. The core voltage is supplied by built-in LDO.



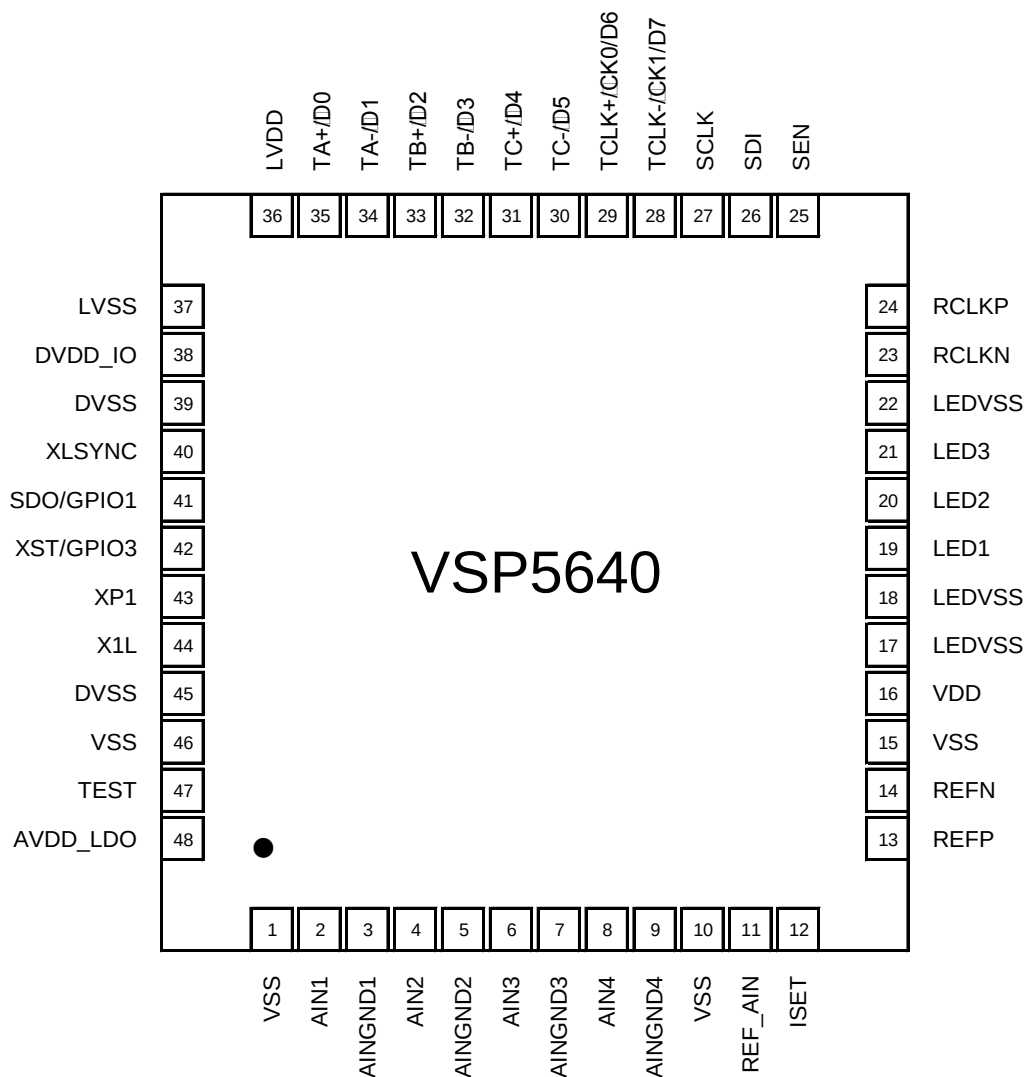
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1. PIN ASSIGNMENT

VSP5640RSL (TOP VIEW)



View from the top

Figure 1. Pin Layout

PACKAGE/ORDERING INFORMATION

PRODUCT	PACKAGE	PACKAGE CODE	OPERATION TEMPERATURE RANGE	PACKAGE MARKING	ORDERING NUMBER	TRANSPORT MEDIA
VSP5640RSL	QFN48	RSL	0°C to +85°C	VSP5640	VSP5640RSLR	Tape and Reel

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2. BLOCK DIAGRAM

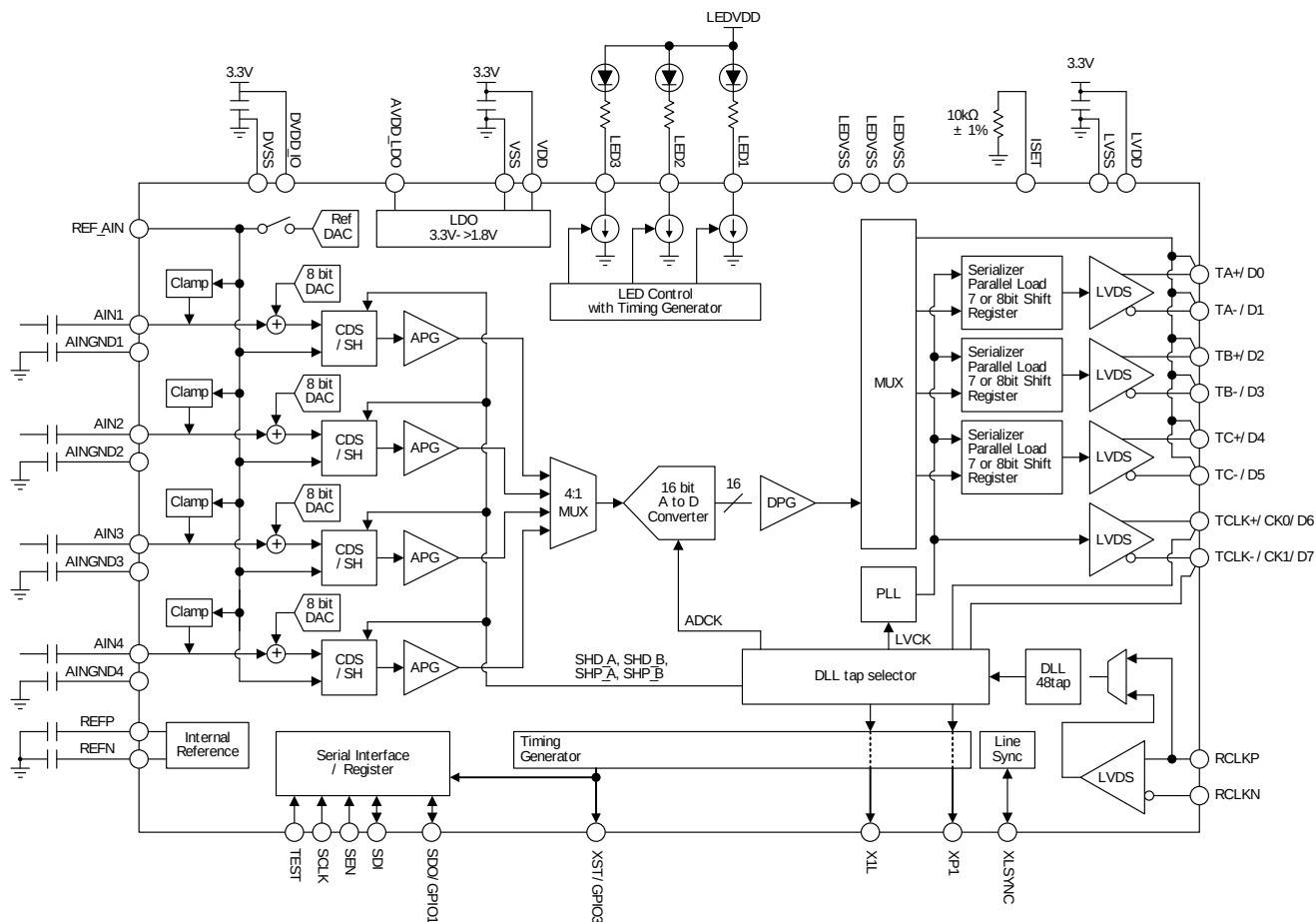


Figure 2. VSP5640 Block Diagram

3. TERMINAL FUNCTIONS

PIN NUMBER	TERMINAL NAME	TYPE ⁽¹⁾	DESCRIPTIONS
1	VSS	AGND	LDO and Analog IO ground
2	AIN1	AI3.3	1st channel analog signal input ⁽²⁾
3	AINGND1	AI3.3	1st channel analog signal ground ⁽²⁾
4	AIN2	AI3.3	2nd channel analog signal input ⁽²⁾
5	AINGND2	AI3.3	2nd channel analog signal ground ⁽²⁾
6	AIN3	AI3.3	3rd channel analog signal input ⁽²⁾
7	AINGND3	AI3.3	3rd channel analog signal ground ⁽²⁾
8	AIN4	AI3.3	4th channel analog signal input ⁽²⁾
9	AINGND4	AI3.3	4th channel analog signal ground ⁽²⁾
10	VSS	AGND	LDO and Analog IO ground
11	REF_AIN	AI3.3 /AO3.3	REF_DAC_IN="0" : Analog signal reference Monitor Output (Default) Output is intended for using in test, please don't connect any load. REF_DAC_IN="1" : Analog signal reference Input
12	ISET	LVO1.8	Internal reference voltage output (bypass to ground by 10kΩ±1% resistor)
13	REFP	AO1.8	Positive reference (Bypass to AGND by 0.1μF capacitor)
14	REFN	AO1.8	Negative reference (Bypass to AGND by 0.1μF capacitor)
15	VSS	AGND	LDO and Analog IO ground
16	VDD	AP3.3	LDO and Analog IO power supply
17	LEDVSS	LDGND	LED Driver ground
18	LEDVSS	LDGND	LED Driver ground
19	LED1	LDO5.0	LED Driver 1
20	LED2	LDO5.0	LED Driver 2
21	LED3	LDO5.0	LED Driver 3
22	LEDVSS	LDGND	LED Driver ground
23	RCLKN	LVI3.3	LVDS clock input -
24	RCLKP	LVI3.3	CMOS Master clock input / LVDS clock input +
25	SEN	DI3.3	Serial I/F enable (active low) (Internal pull up)
26	SDI	DIO3.3	SDI_BUFF_CTRL="0": Serial I/F data input SDI_BUFF_CTRL="1": Serial I/F data input/output (Internal pull down)
27	SCLK	DI3.3	Serial I/F clock (internal pull down)
28	TCLK- /CK1/D7	LVO3.3	LVDS clock output - /Clock output 1, Data output bit 7
29	TCLK+ /CK0/D6	LVO3.3	LVDS clock output + /Clock output 0, Data output bit 6

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3. TERMINAL FUNCTIONS (CONTINUED)

PIN NUMBER	TERMINAL NAME	TYPE ⁽¹⁾	DESCRIPTIONS
30	TC- /D5	LVO3.3	TC ch LVDS data output - /Data output bit 5
31	TC+ /D4	LVO3.3	TC ch LVDS data output + /Data output bit 4
32	TB- /D3	LVO3.3	TB ch LVDS data output - /Data output bit 3
33	TB+ /D2	LVO3.3	TB ch LVDS data output + /Data output bit 2
34	TA- /D1	LVO3.3	TA ch LVDS data output - /Data output bit 1
35	TA+ /D0	LVO3.3	TA ch LVDS data output + /Data output bit 0
36	LVDD	LVP3.3	LVDS / CMOS output power supply
37	LVSS	LVGND	LVDS / CMOS output ground
38	DVDD_IO	DP3.3	Digital IO power supply
39	DVSS	DGND	Digital ground
40	XLSYNC	DIO3.3	XLSYNC_SEL="0" : Internal Line synchronous signal output (Default) (In case of input, internal pull down) XLSYNC_SEL="1" : External Line synchronous signal input Polarity will be set by register "XLSYNC_POL" (Default : Active High)
41	SDO /GPIO1	DIO3.3	GPIO1_SDO_SEL="0" : GPI1(General purpose Input port 1) (Default) (In case of input, internal pull down) GPIO1_SDO_SEL="1" : GPO1(General purpose Output port 1) GPIO1_SDO_SEL="2" : Reserved (Internal Test Input) GPIO1_SDO_SEL="3" : SDO (Serial I/F data Output)
42	XST /GPIO3	DIO3.3	GPIO3_XST_SEL="0" : GPI3 (General purpose Input port 3) (Default) (In case of input, internal pull down) GPIO3_XST_SEL="1" : GPO3 (General purpose Output port 3) GPIO3_XST_SEL="2" : Reserved (Internal Test Input) GPIO3_XST_SEL="3" : ST (Storage pulse output)
43	XP1	DO3.3	Fast transfer clock output ϕ 1
44	X1L	DO3.3	Fast transfer clock output 1L
45	DVSS	DGND	Digital ground
46	VSS	AGND	LDO and Analog IO ground
47	TEST	DI3.3	Internal test pin (Connect to DGND)
48	AVDD_LDO	AP1.8	Analog core power voltage output (Non Connection, Open)

- (1) AP3.3 = 3.3 V analog power supply, AP1.8 = 1.8 V analog power supply, AGND = Analog ground, GND = Ground
 AO3.3 = 3.3 V analog output, AO1.8 = 1.8 V Analog output, AI3.3 = 3.3 V Analog input
 DP3.3 = 3.3 V digital power supply, DP1.8 = 1.8 V digital power supply, DGND = Digital ground
 DO3.3 = 3.3 V Digital output, DI3.3 = 3.3 V Digital input, DIO3.3 = 3.3 V Digital I/O
 LVP3.3 = 3.3 V LVDS power supply, LVGND = LVDS ground, LVO3.3 = 3.3 V LVDS output
 LVI3.3 = 3.3 V LVDS input, LVO = 3.3 V LVDS output
 LDO5.0 = 5.0 V LED driver output, LDGND = LED driver GND
- (2) If these pin will be unused, they can be opened or decoupled to GND by decoupling capacitor.

4. ABSOLUTE MAXIMUM RATEINGS

Over free-air Temperature range (Unless otherwise netted ⁽¹⁾)

		UNITS
Supply voltage	VDD, DVDD_IO, LVDD	4.0V
Supply voltage difference	VDD, DVDD_IO, LVDD	±0.6V
Ground voltage difference	VSS, DVSS, LVSS, LEDVSS	±0.1V
Digital voltage input		–0.3V to DVDD_IO + 0.3V
Analog voltage input		–0.3V to VDD + 0.3V
Digital input current		±10mA
LED driver voltage	LED1, LED2, LED3	7.0V
Analog input current		±10mA
Ambient temperature under bias		–40°C to +125°C
Storage temperature		–55°C to +150°C
Junction temperature		150°C
Lead temperature (soldering)		+260°C, 5s
Package temperature (IR reflow, peak)		+260°C

(1) Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these or any other conditions beyond those indicated under "recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

5. RECOMMENDATION OPERATING CONDITIONS

		MIN	TYP	MAX	UNIT
LDO and analog IO power supply voltage	VDD	3.0	3.3	3.6	V
Digital power supply voltage	DVDD_IO	3.0	3.3	3.6	V
LVDS / CMOS power supply voltage	LVDD	3.0	3.3	3.6	V
Supply Voltage difference	VDD, DVDD_IO, LVDD	–0.3		0.3	V
Digital input logic family		Low Voltage CMOS			
Master clock frequency (MCLK)		1		16.66	MHz
Serial I/O clock frequency (SCLK)				10	MHz
Operating free-air Temperature		0		85	°C

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6. ELECTRICAL CHARACTERISTICS

(All specifications at $T_A = +25^\circ\text{C}$, Supply voltage = +3.3 V and conversion rate = 12.5 MHz, 4ch mode)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
ANALOG INPUT						
Allowable input voltage			0		VDD	V
Full scale range		Gain = 1 V/V		1		Vpp
Input capacitor				10		pF
DIGITAL INPUT						
V _T ⁺	Positive-going threshold				DVDD_IO × 0.7	V
V _T ⁻	Negative-going threshold		DVDD_IO × 0.3			V
ΔV _T	Hysteresis (V _T ⁺ – V _T ⁻)			DVDD_IO × 0.13		V
I _{IN}	Input current				±1	μA
Input capacitor				5		pF
DIGITAL OUTPUT						
V _{OH}	High-level output voltage	I _{OH} = –2 mA	DVDD_IO – 0.45			V
		I _{OH} = –4 mA	DVDD_IO – 0.50			V
		I _{OH} = –8 mA	DVDD_IO – 0.50			
V _{OL}	Low-level output voltage	I _{OL} = 2 mA		0.35		V
		I _{OL} = 4 mA		0.50		V
		I _{OL} = 8 mA		0.65		V
TG output timing skew			–2		2	ns
CMOS data output bit rate					80	MHz

6. ELECTRICAL CHARACTERISTICS (CONTINUED)

(All specifications at $T_A = +25^\circ\text{C}$, Supply voltage = +3.3 V and conversion rate = 12.5 MHz, 4ch mode)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
LVDS DRIVER (TA, TB, TC, TCLK)						
$ V_{OD} $	Differential steady-state output voltage adjustment range	$R_L = 100\ \Omega$	300	350	400	mV
	Differential steady-state output adjustment step			3		step
	Differential steady-state output voltage tolerance		-30		30	%
$\Delta V_{OD} $	Change in the steady-state differential output voltage magnitude between opposite binary states				35	mV
$V_{OC(SS)}$	Steady-state common-mode output voltage	$R_L = 100\ \Omega$	1.125		1.375	V
$V_{OC(PP)}$	Peak-to-peak common mode output voltage			80	150	mV
I_{OS}	Short circuit output current	$V_O = 0$ (V_O : TA, TB, TC, TCLK)		-6	± 24	mA
I_{OZ}	Hi-Z output current	$V_O = 0$ to LVDD (V_O : TA, TB, TC, TCLK)			± 10	μA
t_{LR}/t_{LF}	Transition time, differential output voltage			0.75	1.5	ns
	TCLK clock rate		8		35	MHz
LVDS RECEIVER (RCLK)						
V_{IT+}	Positive-going differential input threshold voltage				100	mV
V_{IT-}	Negative-going differential input threshold voltage		-100			mV
	RCLK clock rate		1		11.66	MHz
LED DRIVER						
I_{LD}	Output current				60	mA/ch
	Off-state leak current			50		μA
	Output current tolerance		-10		10	%
	Total output current ⁽¹⁾	All channel			132	mA
	Channel mismatch		-5		5	%
	Current step			16		step/ch
V_{LD}	Drive voltage		0.7		4.5	V
	LED turn off, $I_{LD} = 0$ [mA]				6	V
	Temp drift	$T_A = 0-85^\circ\text{C}$, Design guarantee	-2		2	%

(1) The limitation circuitry for total of current is not incorporated

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6. ELECTRICAL CHARACTERISTICS (CONTINUED)(All specifications at $T_A = +25^\circ\text{C}$, Supply voltage = +3.3 V and conversion rate = 12.5 MHz, 4ch mode)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
POWER SUPPLY						
VDD	LDO and analog IO supply voltage		3.0	3.3	3.6	V
DVDD_IO	Digital IO supply voltage		3.0	3.3	3.6	V
LVDD	LVDS / CMOS supply voltage		3.0	3.3	3.6	V
VDD	LDO and analog IO current			100		mA
DVDD_IO	Digital IO current	CMOS 8bit (@8.75MHz), No load		12		mA
LVDD	LVDS current	7bit, 3 pair mode		13.5		mA
Power consumption		7bit, 3 pair mode		375		mW
		CMOS 8bit (@8.75MHz), No load		370		mW
		Standby mode (@0 MHz)		15		mW
TEMPERATURE RANGE						
T_A	Operation temperature		0		85	$^\circ\text{C}$
θ_{JA}	Thermal resister (junction to air)	PCB (50 × 50 mm, 4 layer), 0 airflow		32		$^\circ\text{C/W}$
θ_{JC}	Thermal resister (junction to case)			22		$^\circ\text{C/W}$
DLL, PLL						
f_{MCLK}	MCLK input frequency		1		11.66	MHz
	MCLK modulated frequency	MCLK > 5 MHz			35	kHz
	MCLK modulated amplitude		-3.5		0	%
	DLL tap number			48		tap
	Max of DLL and PLL lock-up time	MCLK = 1 MHz		10		ms
TRANSFER CHARACTERISTICS						
	Channel		2		4	channels
	Resolution			16		Bits
Conversion rate	LVDS, 2, 3-ch mode		1		16.66	MHz/ch
	LVDS, 4-ch mode		1		12.5	MHz/ch
	CMOS 8 bit × 2, 2ch mode		1		16.66	MHz/ch
	CMOS 4 bit × 4, 2ch mode		1		10	MHz/ch
	CMOS 8 bit × 2, 3ch mode		1		13.3	MHz/ch
	CMOS 4 bit × 4, 3ch mode		1		6.7	MHz/ch
	CMOS 8 bit × 2, 4ch mode		1		8.75	MHz/ch
	CMOS 4 bit × 4, 4ch mode		1		5	MHz/ch
	Maximum differential non-linearity	Gain = 1 V/V, 12 bit		±0.5		LSB
	Maximum integral non-linearity	Gain = 1 V/V, 12 bit		±2		LSB
	No missing code			Specified		
SNR	Signal to noise ratio	Gain = 1 V/V	$72^{(1)}$	76		dB
	Analog channel crosstalk	Gain = 1 V/V, 12 bit, full scale step		±6.5		LSB
	Total absolute gain error		-10		10	%

(1) Design Specified

(2) This data is based on using the JEDEC High-K board and the exposed die pad is connected to a Cu pad on the board. This is connected to the ground plane by a via matrix.

6. ELECTRICAL CHARACTERISTICS (CONTINUED)

(All specifications at $T_A = +25^\circ\text{C}$, Supply voltage = +3.3 V and conversion rate = 12.5 MHz, 4ch mode)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
ANALOG PROGRAMMABLE GAIN (APG)						
APG_x	Gain range		0.5		3.5	V/V
	Gain step			63		step
	Gain relative error	Basis gain = 1 V/V	-10		10	%
	Gain monotonicity	Only APG_x		Specified		
DIGITAL PROGRAMMABLE GAIN (DPG)						
DPG_x	Gain range		1.0		2.0	V/V
	Gain step			255		step
	Gain monotonicity	Only DPG_x		Specified		
AIN REFERENCE LEVEL (REF_AIN)						
V_{rint}	Internal DAC output	Setting code = "2"		0.5		V
		Setting code = "3"		1.1		V
		Setting code = "0" (default)		1.5		V
		Setting code = "1"		2.0		V
	Internal DAC output tolerance		-10		10	%
	Internal DAC output temp drift	$T_A = 0-85^\circ\text{C}$, design guarantee	-2		2	%
V_{rext}	External reference range		0.5		$V_{\text{DD}}-0.9$	V
INPUT CLAMP						
V_{CLP}	Clamp level	Internal reference level clamp		V_{rint}		V
		External reference level clamp		V_{rext}		V
		Fix level clamp		2.2		V
R_{CLP}	Clamp-on resistance			500		Ω
OFFSET DAC						
	Resolution			8		bits
	Output range			± 250		mV
	Setting tolerance		-10		10	%
	Temperature drift	$T_A = 0-85^\circ\text{C}$, Design guarantee	-2		2	%

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6.1. ANALOG Input Specification (AIN1, AIN2, AIN3, AIN4)

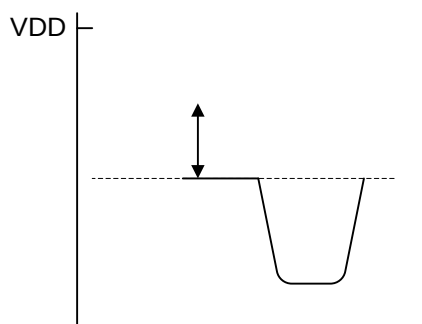


Figure 3. Analog Input Definition

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
f_{pix}	Input pixel rate		1		11.66	MHz/ch
V_{sig}	Signal range	Negative (AINx_POL ⁽¹⁾ = 0)			V_{offset}	V
		Positive (AINx_POL ⁽¹⁾ = 1)			$VDD - V_{offset}$	V
	Maximum full scale range	Gain = 0.5 V/V	1.8	2	2.2	V
V_{rst}	Reset field through noise range		$-V_{offset}$		$VDD - V_{offset}$	V
V_{offset}	Offset level	Fix level clamp mode (REF_SEL = 0)		2.2		V
		Internal reference level clamp mode (REF_SEL = 1)		V_{rnt}		V
		External reference level clamp mode (REF_SEL = 2)		V_{rnt}		V

(1) AINx_POL: Analog input polarity setting register. (x = 1, 2, 3 and 4)

6.2. LVDS Output Voltage

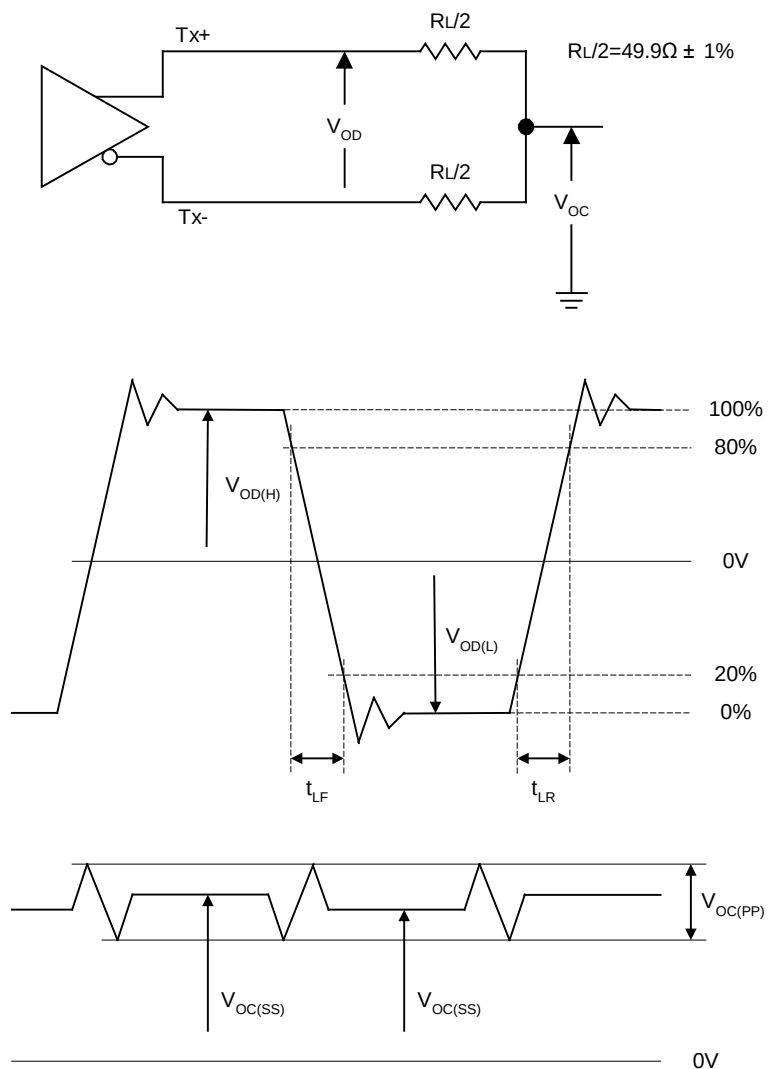


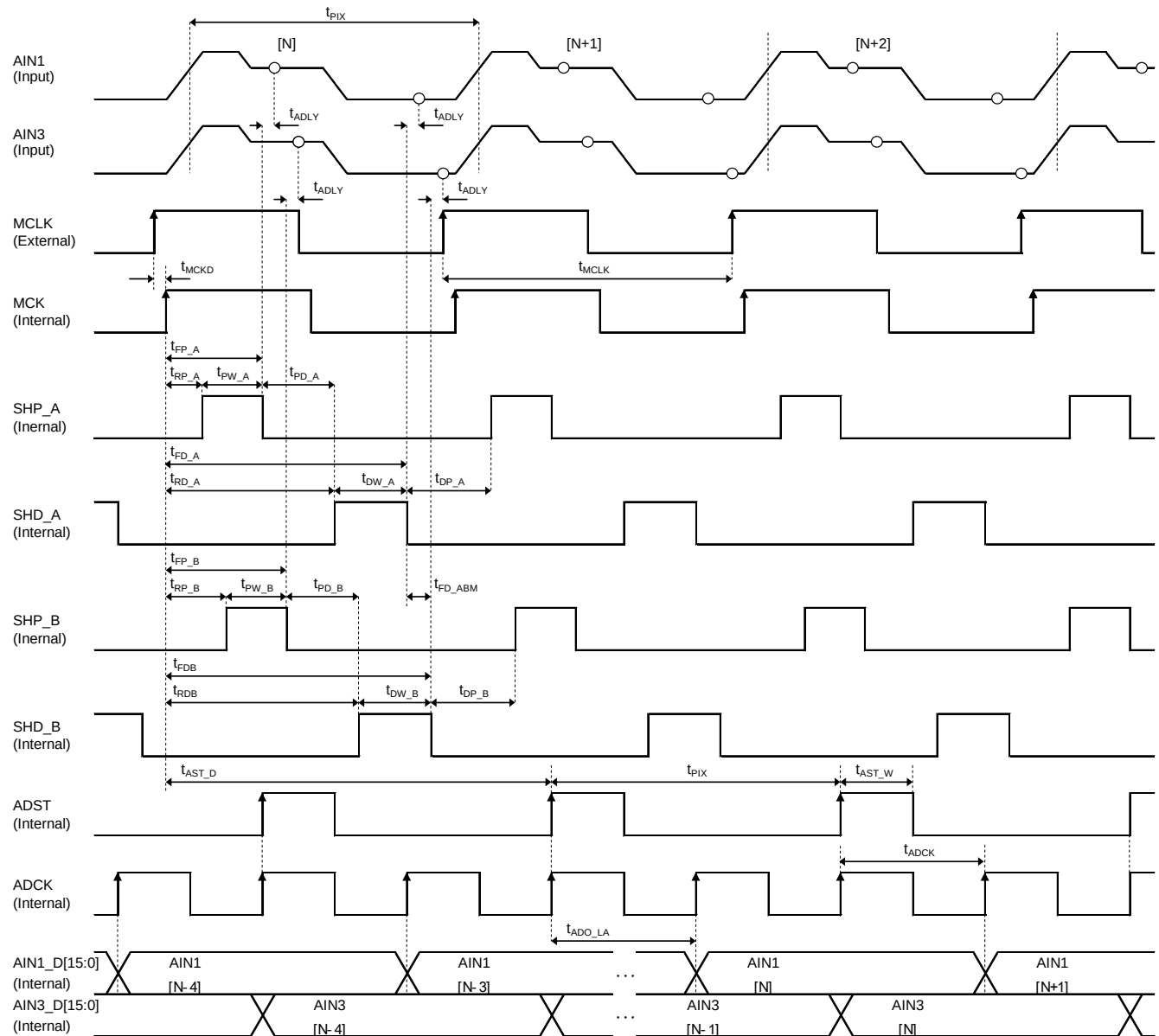
Figure 4. Test Load and Voltage Definition for LVDS Outputs

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7. TIMING SPECIFICATION

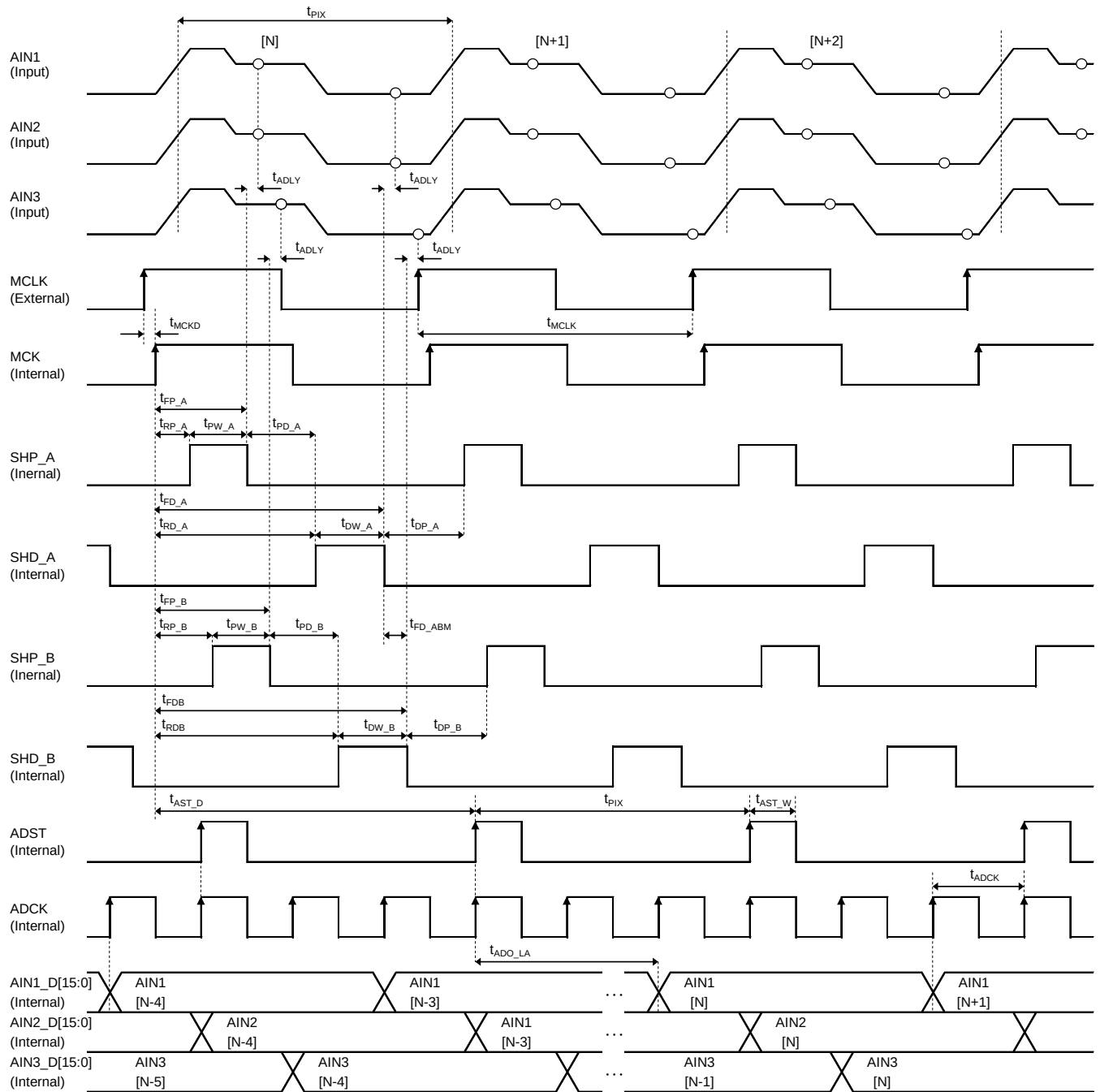
7.1. CDS Timing Specification – x1 Sampling

1) 2-Channel Analog Input Mode



7.1. CDS Timing Specification – x1 Sampling (Continued)

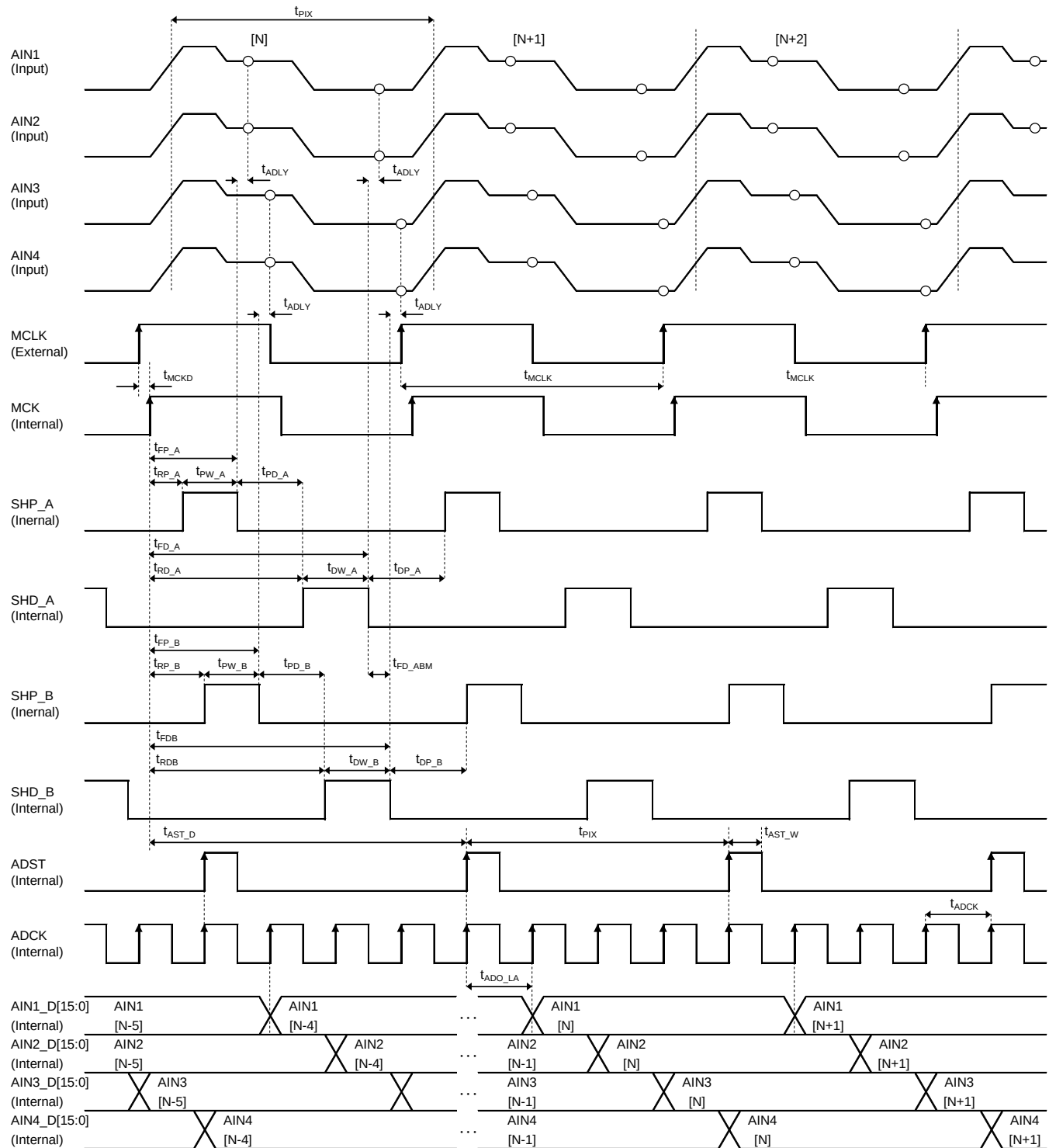
2) 3-Channel Analog Input Mode



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7.1. CDS Timing Specification – x1 Sampling (Continued)

3) 4-Channel Analog Input Mode



7.1. CDS Timing Specification – x1 Sampling (Continued)

SYMBOL	PARAMETER		MIN	TYP	MAX	UNIT
f _{MCLK}	MCLK frequency	2, 3ch mode	1		16.66	MHz
		4ch mode	1		12.5	MHz
t _{MCLK}	MCLK period			1/f _{MCLK}		ns
t _{PIX}	Pixel period			t _{MCLK}		ns
t _{MCKD}	MCLK to MCK delay			2		ns
t _{ADLY}	Aperture delay		3	5	7	ns
t _{RP_A}	SHP_A Rising Edge delay from MCK		0		t _{MCLK} · 47/48	ns
t _{FP_A}	SHP_A Falling Edge delay from MCK		0		t _{MCLK} · 47/48	ns
t _{PW_A}	SHP_A pulse width		17		t _{MCLK} · 24/48	ns
t _{RD_A}	SHD_A Rising Edge delay from MCK		0		t _{MCLK} · 47/48	ns
t _{FD_A}	SHD_A Falling Edge delay from MCK		0		t _{MCLK} · 47/48	ns
t _{DW_A}	SHD_A pulse width		17		t _{MCLK} · 24/48	ns
t _{PD_A}	SHP_A to SHD_A margin		2		t _{MCLK} · 47/48	ns
t _{DP_A}	SHD_A to SHP_A margin		2		t _{MCLK} · 47/48	ns
t _{RP_B}	SHP_B Rising Edge delay from MCK		0		t _{MCLK} · 47/48	ns
t _{FP_B}	SHP_B Falling Edge delay from MCK		0		t _{MCLK} · 47/48	ns
t _{PW_B}	SHP_B pulse width		17		t _{MCLK} · 24/48	ns
t _{RD_B}	SHD_B Rising Edge delay from MCK		0		t _{MCLK} · 47/48	ns
t _{FD_B}	SHD_B Falling Edge delay from MCK		0		t _{MCLK} · 47/48	ns
t _{DW_B}	SHD_B pulse width		17		t _{MCLK} · 24/48	ns
t _{PD_B}	SHP_B to SHD_B margin		2		t _{MCLK} · 47/48	ns
t _{DP_B}	SHD_B to SHP_B margin		2		t _{MCLK} · 47/48	ns
t _{FD_ABM}	SHD_A to SHD_B Falling Edge margin		0		t _{MCLK} · 4/48	ns
t _{ADCK}	ADCK clock Period	2ch mode		t _{MCLK} · 1/2		ns
		3ch mode		t _{MCLK} · 1/3		ns
		4ch mode		t _{MCLK} · 1/4		ns
t _{AST_D}	ADST delay from MCK	2ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.0 + t _{FD} ⁽⁴⁾		ns
			SHP Ref ⁽³⁾	t _{ADCK} · 2.0 + t _{FP} ⁽⁵⁾		ns
		3ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.0 + t _{FD} ⁽⁴⁾		ns
			SHP Ref ⁽³⁾	t _{ADCK} · 2.5 + t _{FP} ⁽⁵⁾		ns
		4ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.5 + t _{FD} ⁽⁴⁾		ns
			SHP Ref ⁽³⁾	t _{ADCK} · 3.5 + t _{FP} ⁽⁵⁾		ns
t _{AST_W}	ADST positive pulse width			t _{ADCK} · 1/2		ns
t _{ADO_LA}	ADC data output latency	2ch mode		11		clock ⁽¹⁾
		3ch mode		11		clock ⁽¹⁾
		4ch mode		13		clock ⁽¹⁾

(1) Unit Basis is ADCK

(2) Condition for synchronizing SHD falling edge to ADCK : $t_{FD}-t_{FP} > 24 \times t_{MCLK}/48$ or $t_{FP}-t_{FD} \leq 24 \times t_{MCLK}/48$

(3) Condition for synchronizing SHP falling edge to ADCK : $t_{FD}-t_{FP} \leq 24 \times t_{MCLK}/48$ or $t_{FP}-t_{FD} > 24 \times t_{MCLK}/48$

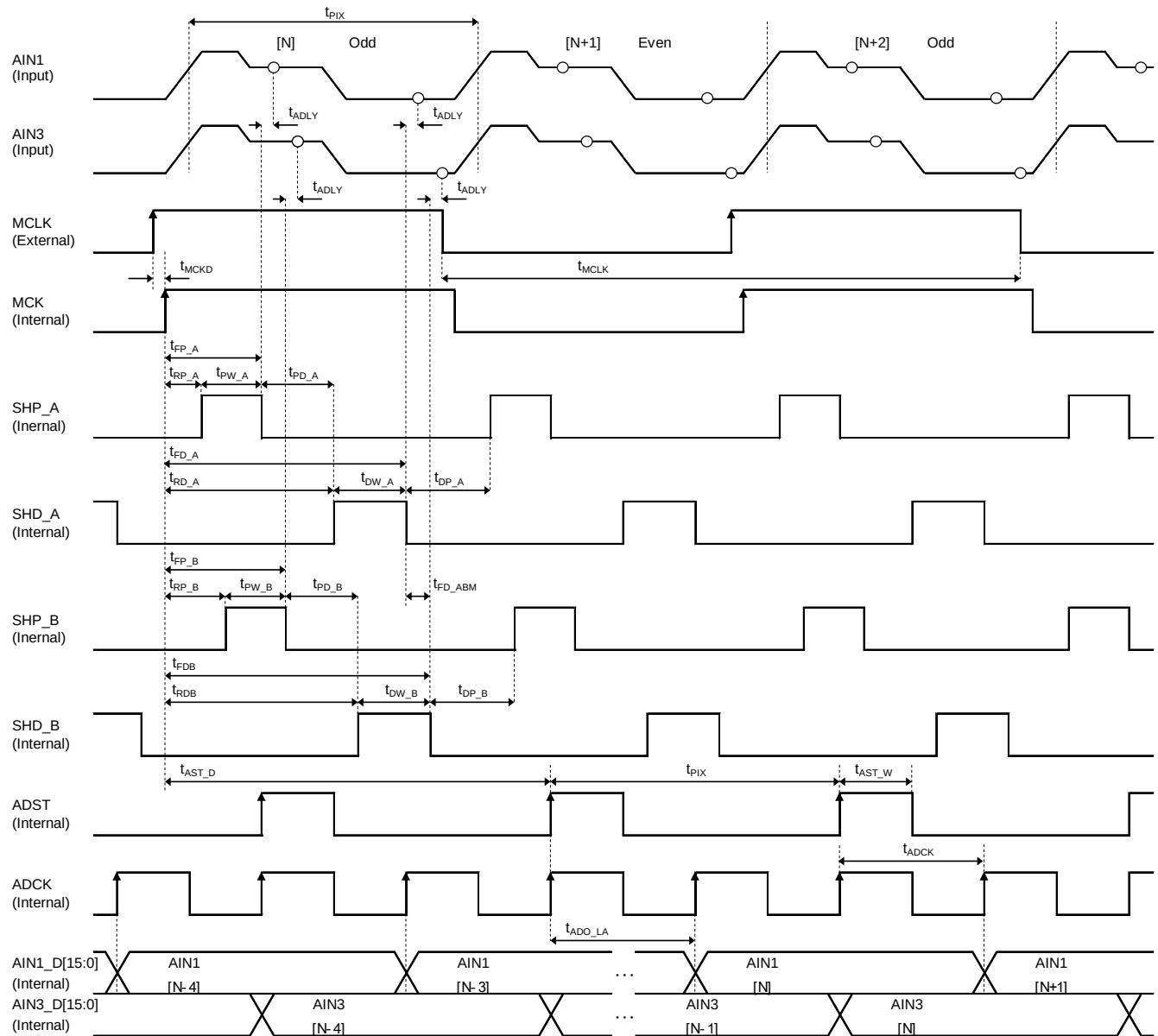
(4) t_{FD} : Reference delay time of SHD falling edge, If $t_{FD_A} \leq t_{FD_B}$, then $t_{FD} = t_{FD_A}$, If $t_{FD_A} > t_{FD_B}$, then $t_{FD} = t_{FD_B}$

(5) t_{FP} : Reference delay time of SHP falling edge, If $t_{FP_A} \leq t_{FP_B}$, then $t_{FP} = t_{FP_A}$, If $t_{FP_A} > t_{FP_B}$, then $t_{FP} = t_{FP_B}$

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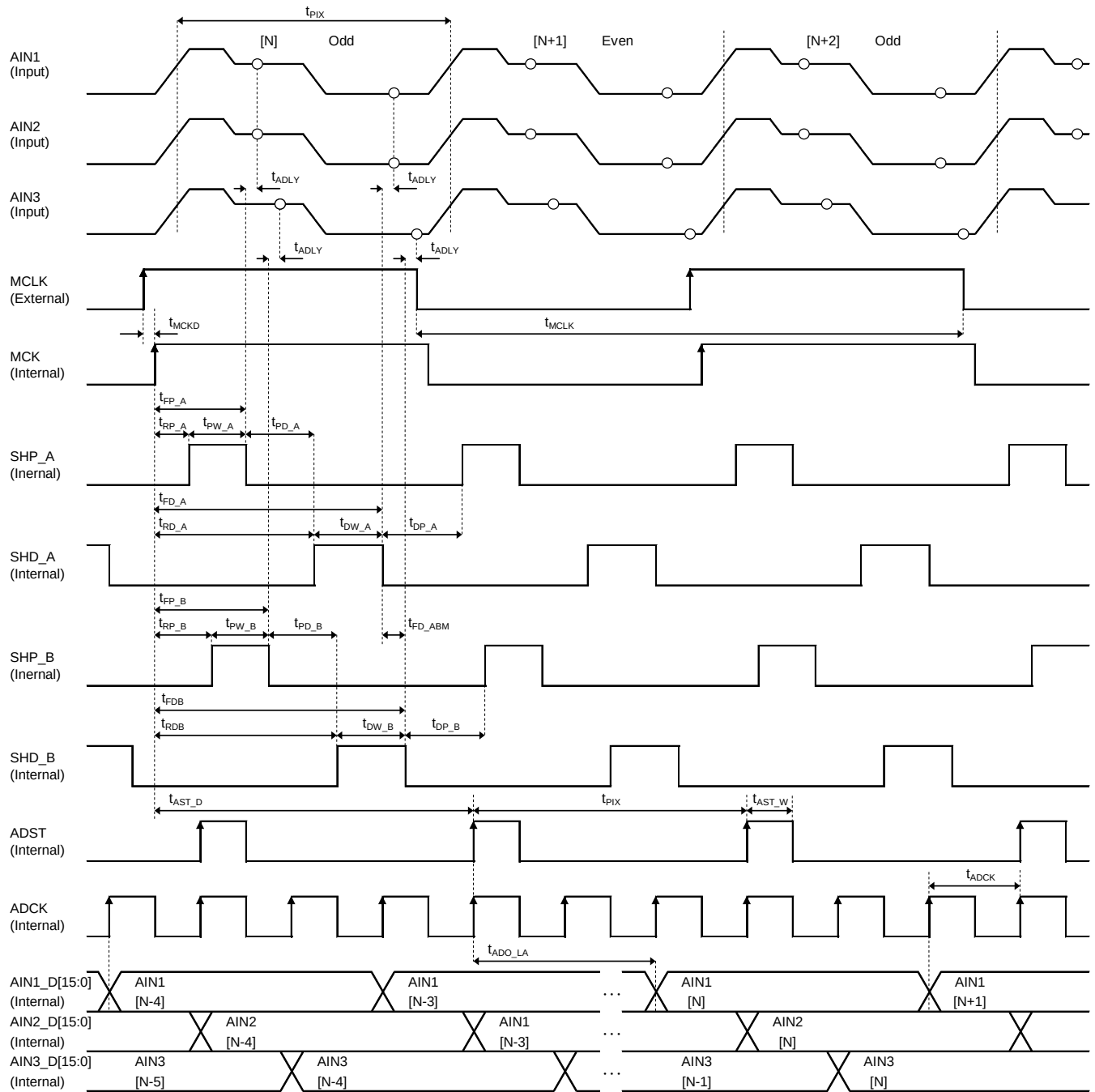
7.2. CDS Timing Specification – x2 Sampling

1) 2-Channel Analog Input Mode



7.2. CDS Timing Specification – x2 Sampling (Continued)

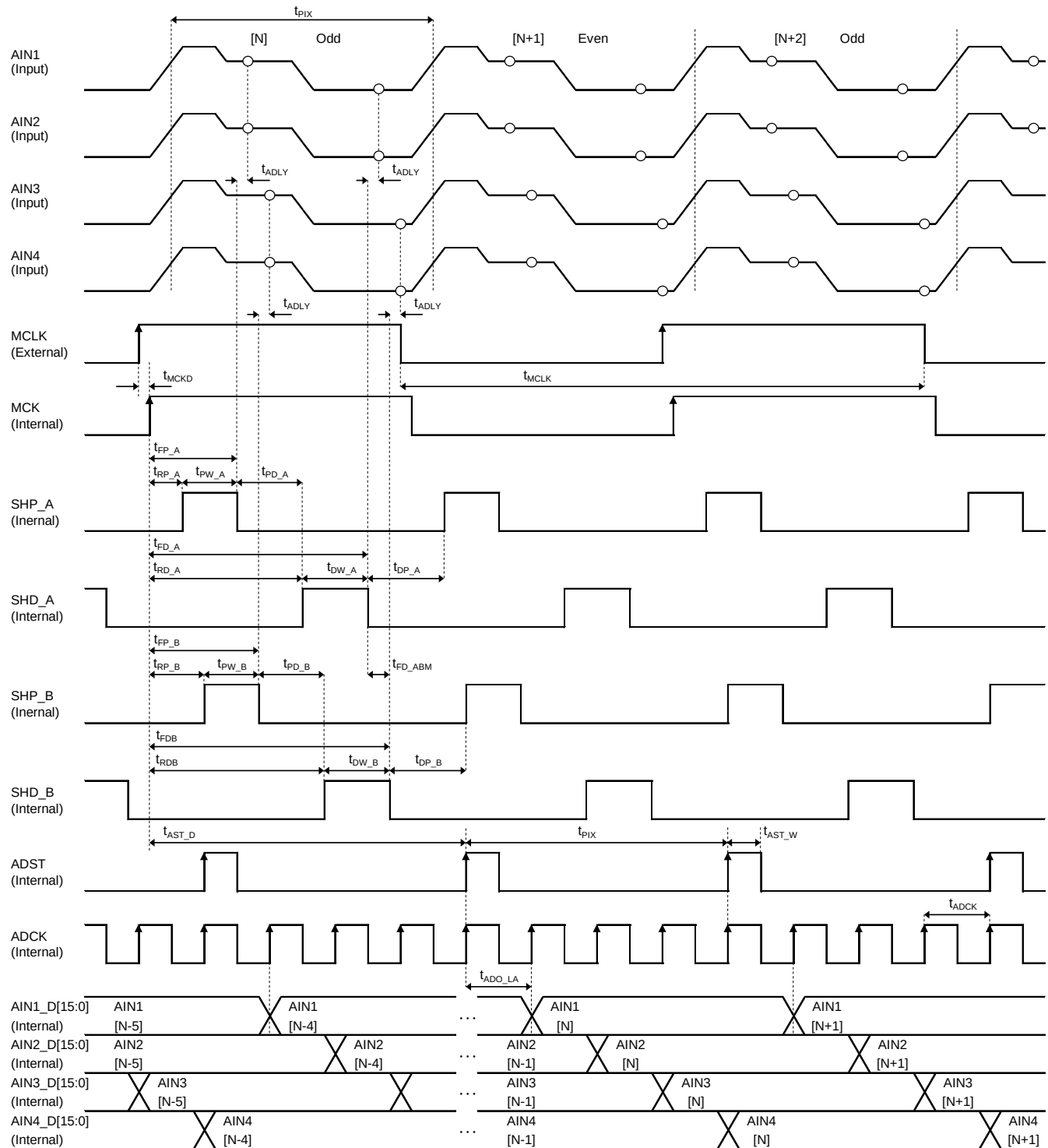
2) 3-Channel Analog Input Mode



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7.2. CDS Timing Specification – x2 Sampling (Continued)

3) 4-Channel Analog Input Mode



7.2. CDS Timing Specification – x2 Sampling (Continued)

SYMBOL	PARAMETER		MIN	TYP	MAX	UNIT
f _{MCLK}	MCLK frequency	2, 3ch mode	1		8.33	MHz
		4ch mode	1		6.25	MHz
t _{MCLK}	MCLK period			1/f _{MCLK}		ns
t _{PIX}	Pixel period			t _{MCLK} · 1/2		ns
t _{MCKD}	MCLK to MCK delay			2		ns
t _{ADLY}	Aperture delay		3	5	7	ns
t _{RP_A}	SHP_A Rising Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{FP_A}	SHP_A Falling Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{PW_A}	SHP_A pulse width		17		t _{MCLK} · 12/48	ns
t _{RD_A}	SHD_A Rising Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{FD_A}	SHD_A Falling Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{DW_A}	SHD_A pulse width		17		t _{MCLK} · 12/48	ns
t _{PD_A}	SHP_A to SHD_A margin		2		t _{MCLK} · 23/48	ns
t _{DP_A}	SHD_A to SHP_A margin		2		t _{MCLK} · 23/48	ns
t _{RP_B}	SHP_B Rising Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{FP_B}	SHP_B Falling Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{PW_B}	SHP_B pulse width		17		t _{MCLK} · 12/48	ns
t _{RD_B}	SHD_B Rising Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{FD_B}	SHD_B Falling Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{DW_B}	SHD_B pulse width		17		t _{MCLK} · 12/48	ns
t _{PD_B}	SHP_B to SHD_B margin		2		t _{MCLK} · 23/48	ns
t _{DP_B}	SHD_B to SHP_B margin		2		t _{MCLK} · 23/48	ns
t _{FD_ABM}	SHD_A to SHD_B Falling Edge margin		0		t _{MCLK} · 2/48	ns
t _{ADCK}	ADCK clock Period	2ch mode		t _{MCLK} · 1/4		ns
		3ch mode		t _{MCLK} · 1/6		ns
		4ch mode		t _{MCLK} · 1/8		ns
t _{AST_D}	ADST delay from MCK	2ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.0 + t _{FD} ⁽⁴⁾		ns
			SHP Ref ⁽³⁾	t _{ADCK} · 2.0 + t _{FP} ⁽⁵⁾		ns
		3ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.0 + t _{FD} ⁽⁴⁾		ns
			SHP Ref ⁽³⁾	t _{ADCK} · 2.5 + t _{FP} ⁽⁵⁾		ns
		4ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.5 + t _{FD} ⁽⁴⁾		ns
			SHP Ref ⁽³⁾	t _{ADCK} · 3.5 + t _{FP} ⁽⁵⁾		ns
t _{AST_W}	ADST positive pulse width			t _{ADCK} · 1/2		ns
t _{ADO_LA}	ADC data output latency	2ch mode		11		clock ⁽¹⁾
		3ch mode		11		clock ⁽¹⁾
		4ch mode		13		clock ⁽¹⁾

(1) Unit Basis is ADCK

(2) Condition for synchronizing SHD falling edge to ADCK : $t_{FD}-t_{FP} > 24 \times t_{MCLK}/48$ or $t_{FP}-t_{FD} \leq 24 \times t_{MCLK}/48$

(3) Condition for synchronizing SHP falling edge to ADCK : $t_{FD}-t_{FP} \leq 24 \times t_{MCLK}/48$ or $t_{FP}-t_{FD} > 24 \times t_{MCLK}/48$

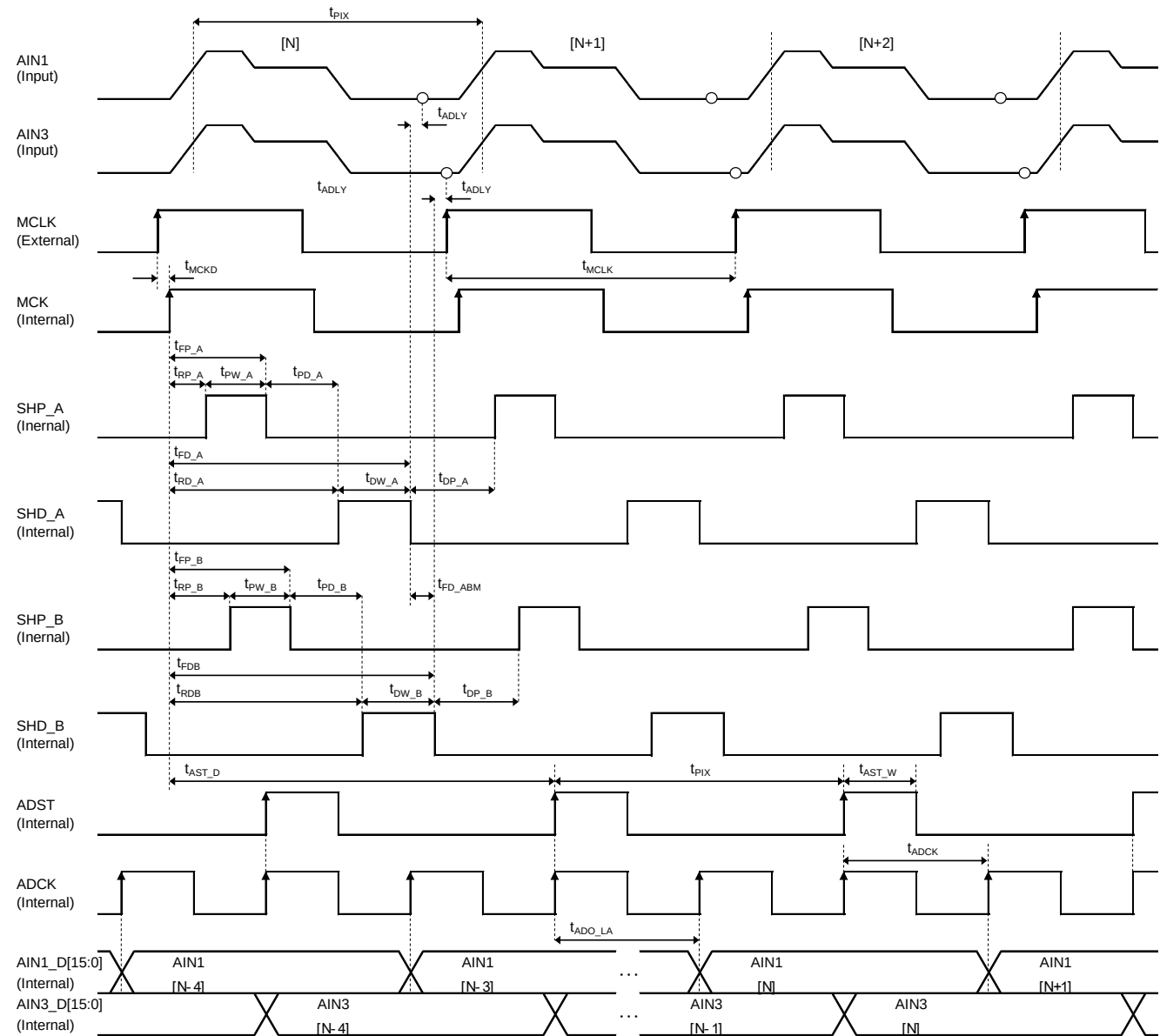
(4) t_{FD} : Reference delay time of SHD falling edge, If $t_{FD_A} \leq t_{FD_B}$, then $t_{FD} = t_{FD_A}$, If $t_{FD_A} > t_{FD_B}$, then $t_{FD} = t_{FD_B}$

(5) t_{FP} : Reference delay time of SHP falling edge, If $t_{FP_A} \leq t_{FP_B}$, then $t_{FP} = t_{FP_A}$, If $t_{FP_A} > t_{FP_B}$, then $t_{FP} = t_{FP_B}$

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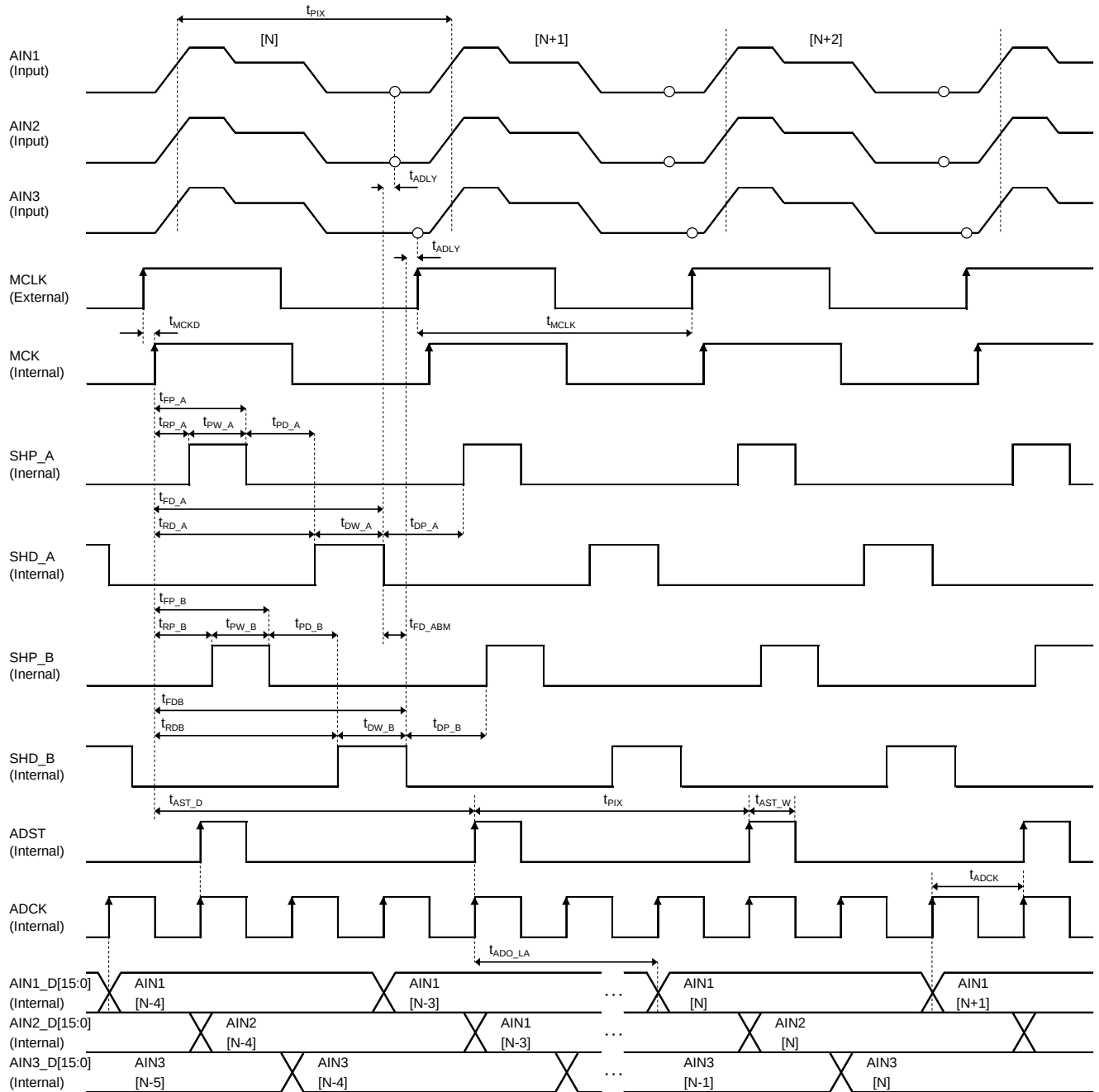
7.3. SH Timing Specification – x1 Sampling

1) 2-Channel Analog Input Mode



7.3. SH Timing Specification – x1 Sampling (Continued)

2) 3-Channel Analog Input Mode



7.3. SH Timing Specification – x1 Sampling (Continued)

SYMBOL	PARAMETER		MIN	TYP	MAX	UNIT
f _{MCLK}	MCLK frequency	2, 3ch mode	1		16.66	MHz
		4ch mode	1		12.5	MHz
t _{MCLK}	MCLK period			1/f _{MCLK}		Ns
t _{PIX}	Pixel period			t _{MCLK}		Ns
t _{MCKD}	MCLK to MCK delay			2		Ns
t _{ADLY}	Aperture delay		3	5	7	Ns
t _{RP_A}	SHP_A Rising Edge delay from MCK		0		t _{MCLK} · 47/48	Ns
t _{FP_A}	SHP_A Falling Edge delay from MCK		0		t _{MCLK} · 47/48	Ns
t _{PW_A}	SHP_A pulse width		17		t _{MCLK} · 24/48	Ns
t _{RD_A}	SHD_A Rising Edge delay from MCK		0		t _{MCLK} · 47/48	Ns
t _{FD_A}	SHD_A Falling Edge delay from MCK		0		t _{MCLK} · 47/48	Ns
t _{DW_A}	SHD_A pulse width		17		t _{MCLK} · 24/48	Ns
t _{PD_A}	SHP_A to SHD_A margin		2		t _{MCLK} · 47/48	Ns
t _{DP_A}	SHD_A to SHP_A margin		2		t _{MCLK} · 47/48	Ns
t _{RP_B}	SHP_B Rising Edge delay from MCK		0		t _{MCLK} · 47/48	Ns
t _{FP_B}	SHP_B Falling Edge delay from MCK		0		t _{MCLK} · 47/48	Ns
t _{PW_B}	SHP_B pulse width		17		t _{MCLK} · 24/48	Ns
t _{RD_B}	SHD_B Rising Edge delay from MCK		0		t _{MCLK} · 47/48	Ns
t _{FD_B}	SHD_B Falling Edge delay from MCK		0		t _{MCLK} · 47/48	Ns
t _{DW_B}	SHD_B pulse width		17		t _{MCLK} · 24/48	Ns
t _{PD_B}	SHP_B to SHD_B margin		2		t _{MCLK} · 47/48	Ns
t _{DP_B}	SHD_B to SHP_B margin		2		t _{MCLK} · 47/48	Ns
t _{FD_ABM}	SHD_A to SHD_B Falling Edge margin		0		t _{MCLK} · 4/48	Ns
t _{ADCK}	ADCK clock Period	2ch mode		t _{MCLK} · 1/2		Ns
		3ch mode		t _{MCLK} · 1/3		Ns
		4ch mode		t _{MCLK} · 1/4		Ns
t _{AST_D}	ADST delay from MCK	2ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.0 + t _{FD} ⁽⁴⁾		Ns
			SHP Ref ⁽³⁾	t _{ADCK} · 2.0 + t _{FP} ⁽⁵⁾		Ns
		3ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.0 + t _{FD} ⁽⁴⁾		Ns
			SHP Ref ⁽³⁾	t _{ADCK} · 2.5 + t _{FP} ⁽⁵⁾		Ns
		4ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.5 + t _{FD} ⁽⁴⁾		Ns
			SHP Ref ⁽³⁾	t _{ADCK} · 3.5 + t _{FP} ⁽⁵⁾		Ns
t _{AST_W}	ADST positive pulse width			t _{ADCK} · 1/2		Ns
t _{ADO_LA}	ADC data output latency	2ch mode		11		clock ⁽¹⁾
		3ch mode		11		clock ⁽¹⁾
		4ch mode		13		clock ⁽¹⁾

(1) Unit Basis is ADCK

(2) Condition for synchronizing SHD falling edge to ADCK : $t_{FD}-t_{FP} > 24 \times t_{MCLK}/48$ or $t_{FP}-t_{FD} \leq 24 \times t_{MCLK}/48$

(3) Condition for synchronizing SHP falling edge to ADCK : $t_{FD}-t_{FP} \leq 24 \times t_{MCLK}/48$ or $t_{FP}-t_{FD} > 24 \times t_{MCLK}/48$

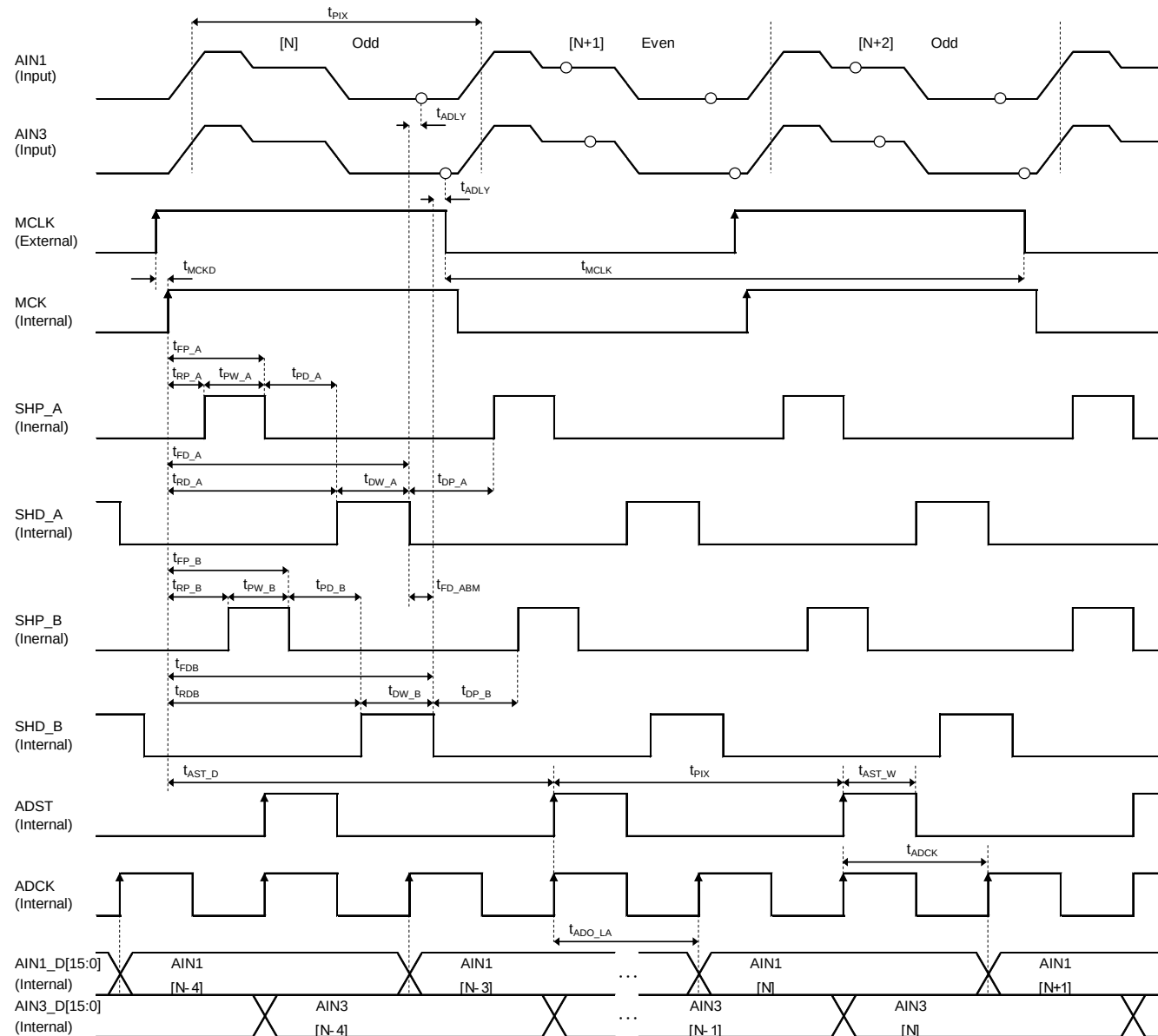
(4) t_{FD} : Reference delay time of SHD falling edge, If $t_{FD_A} \leq t_{FD_B}$, then $t_{FD} = t_{FD_A}$, If $t_{FD_A} > t_{FD_B}$, then $t_{FD} = t_{FD_B}$

(5) t_{FP} : Reference delay time of SHP falling edge, If $t_{FP_A} \leq t_{FP_B}$, then $t_{FP} = t_{FP_A}$, If $t_{FP_A} > t_{FP_B}$, then $t_{FP} = t_{FP_B}$

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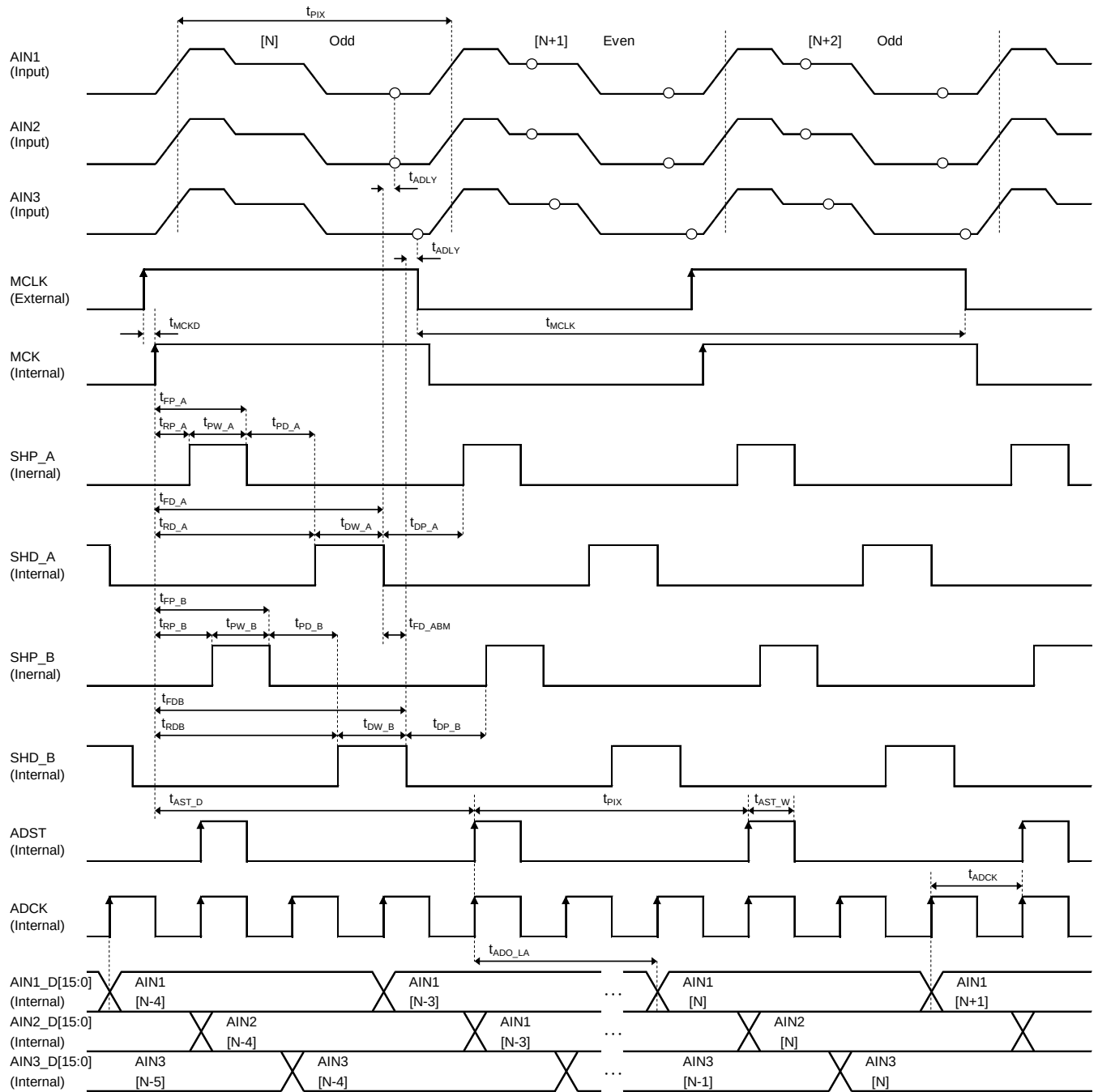
7.4. SH Timing Specification – x2 Sampling

1) 2-Channel Analog Input Mode



7.4. SH Timing Specification – x2 Sampling (Continued)

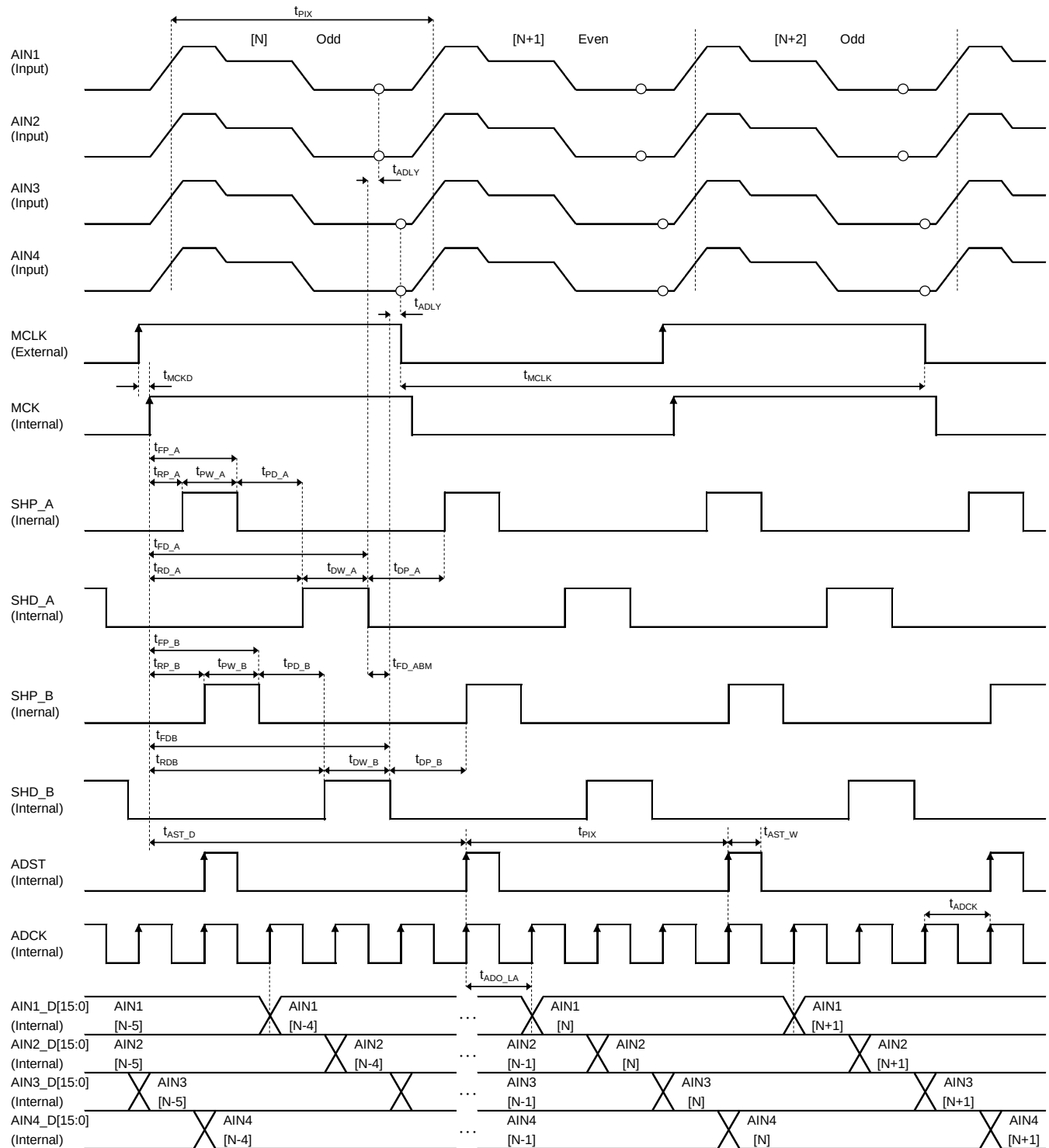
2) 3-Channel Analog Input Mode



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7.4. SH Timing Specification – x2 Sampling (Continued)

3) 4-Channel Analog Input Mode



7.4. SH Timing Specification – x2 Sampling (Continued)

SYMBOL	PARAMETER		MIN	TYP	MAX	UNIT
f _{MCLK}	MCLK frequency	2, 3ch mode	1		8.33	MHz
		4ch mode	1		6.25	MHz
t _{MCLK}	MCLK period			1/f _{MCLK}		ns
t _{PIX}	Pixel period			t _{MCLK} · 1/2		ns
t _{MCKD}	MCLK to MCK delay			2		ns
t _{ADLY}	Aperture delay		3	5	7	ns
t _{RP_A}	SHP_A Rising Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{FP_A}	SHP_A Falling Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{PW_A}	SHP_A pulse width		17		t _{MCLK} · 12/48	ns
t _{RD_A}	SHD_A Rising Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{FD_A}	SHD_A Falling Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{DW_A}	SHD_A pulse width		17		t _{MCLK} · 12/48	ns
t _{PD_A}	SHP_A to SHD_A margin		2		t _{MCLK} · 23/48	ns
t _{DP_A}	SHD_A to SHP_A margin		2		t _{MCLK} · 23/48	ns
t _{RP_B}	SHP_B Rising Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{FP_B}	SHP_B Falling Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{PW_B}	SHP_B pulse width		17		t _{MCLK} · 12/48	ns
t _{RD_B}	SHD_B Rising Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{FD_B}	SHD_B Falling Edge delay from MCK		0		t _{MCLK} · 23/48	ns
t _{DW_B}	SHD_B pulse width		17		t _{MCLK} · 12/48	ns
t _{PD_B}	SHP_B to SHD_B margin		2		t _{MCLK} · 23/48	ns
t _{DP_B}	SHD_B to SHP_B margin		2		t _{MCLK} · 23/48	ns
t _{FD_ABM}	SHD_A to SHD_B Falling Edge margin		0		t _{MCLK} · 2/48	ns
t _{ADCK}	ADCK clock Period	2ch mode		t _{MCLK} · 1/4		ns
		3ch mode		t _{MCLK} · 1/6		ns
		4ch mode		t _{MCLK} · 1/8		ns
t _{AST_D}	ADST delay from MCK	2ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.0 + t _{FD} ⁽⁴⁾		ns
			SHP Ref ⁽³⁾	t _{ADCK} · 2.0 + t _{FP} ⁽⁵⁾		ns
		3ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.0 + t _{FD} ⁽⁴⁾		ns
			SHP Ref ⁽³⁾	t _{ADCK} · 2.5 + t _{FP} ⁽⁵⁾		ns
		4ch mode	SHD Ref ⁽²⁾	t _{ADCK} · 1.5 + t _{FD} ⁽⁴⁾		ns
			SHP Ref ⁽³⁾	t _{ADCK} · 3.5 + t _{FP} ⁽⁵⁾		ns
t _{AST_W}	ADST positive pulse width			t _{ADCK} · 1/2		ns
t _{ADO_LA}	ADC data output latency	2ch mode		11		clock ⁽¹⁾
		3ch mode		11		clock ⁽¹⁾
		4ch mode		13		clock ⁽¹⁾

(1) Unit Basis is ADCK

(2) Condition for synchronizing SHD falling edge to ADCK : $t_{FD}-t_{FP} > 24 \times t_{MCLK}/48$ or $t_{FP}-t_{FD} \leq 24 \times t_{MCLK}/48$

(3) Condition for synchronizing SHP falling edge to ADCK : $t_{FD}-t_{FP} \leq 24 \times t_{MCLK}/48$ or $t_{FP}-t_{FD} > 24 \times t_{MCLK}/48$

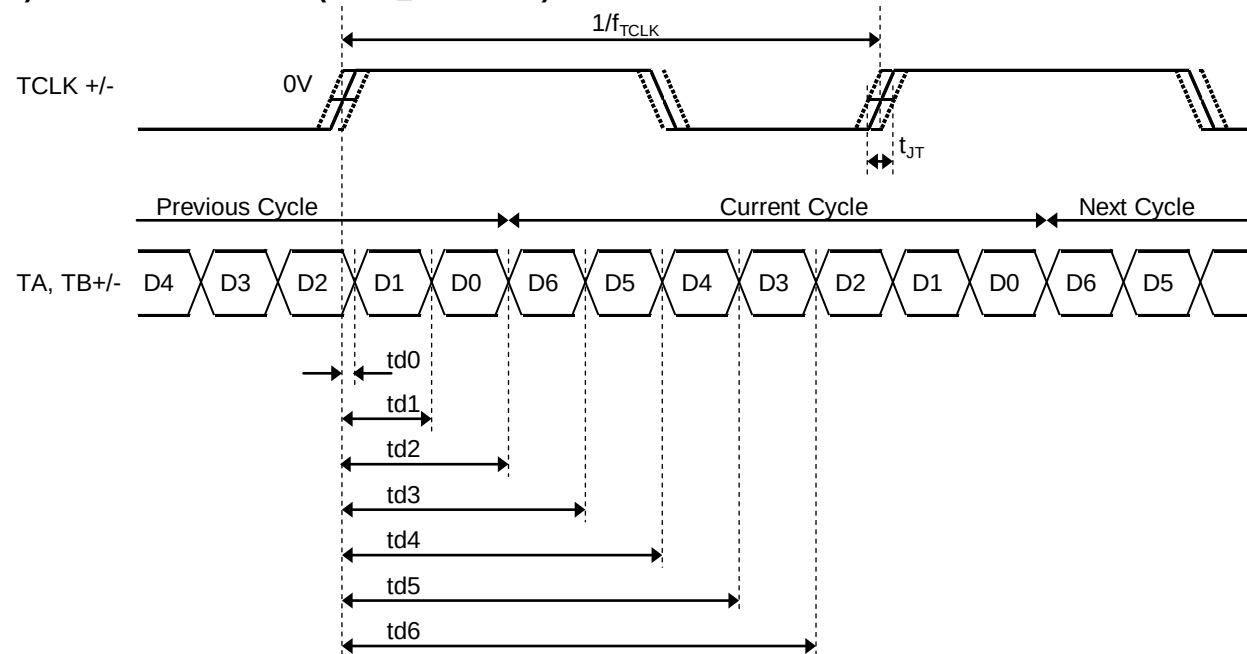
(4) t_{FD} : Reference delay time of SHD falling edge, If $t_{FD_A} \leq t_{FD_B}$, then $t_{FD} = t_{FD_A}$, If $t_{FD_A} > t_{FD_B}$, then $t_{FD} = t_{FD_B}$

(5) t_{FP} : Reference delay time of SHP falling edge, If $t_{FP_A} \leq t_{FP_B}$, then $t_{FP} = t_{FP_A}$, If $t_{FP_A} > t_{FP_B}$, then $t_{FP} = t_{FP_B}$

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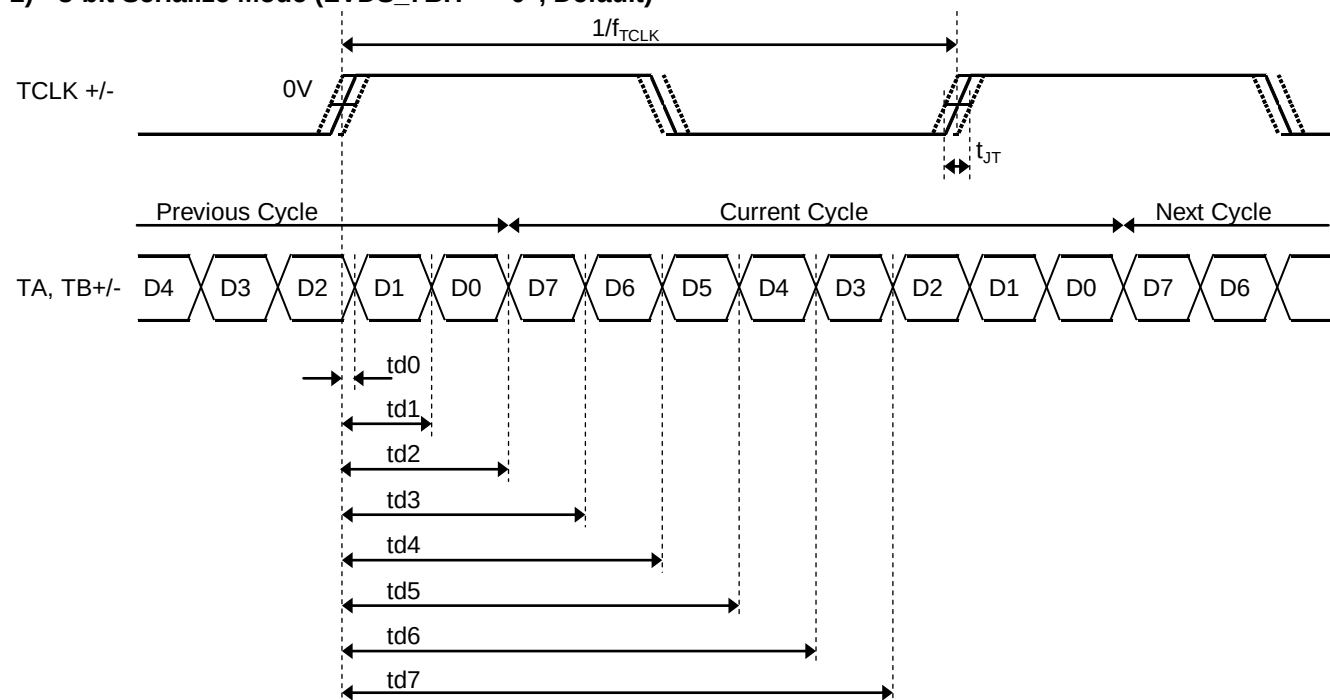
7.5. LVDS Output Timing

1) 7-Bit Serialize Mode (LVDS_7BIT = "1")



Note) TCLK_FRM_SET[7:0]="1100011x"

2) 8-bit Serialize Mode (LVDS_7BIT = "0", Default)



Note) TCLK_FRM_SET[7:0]="11000011"

7.5. LVDS Output Timing (Continued)

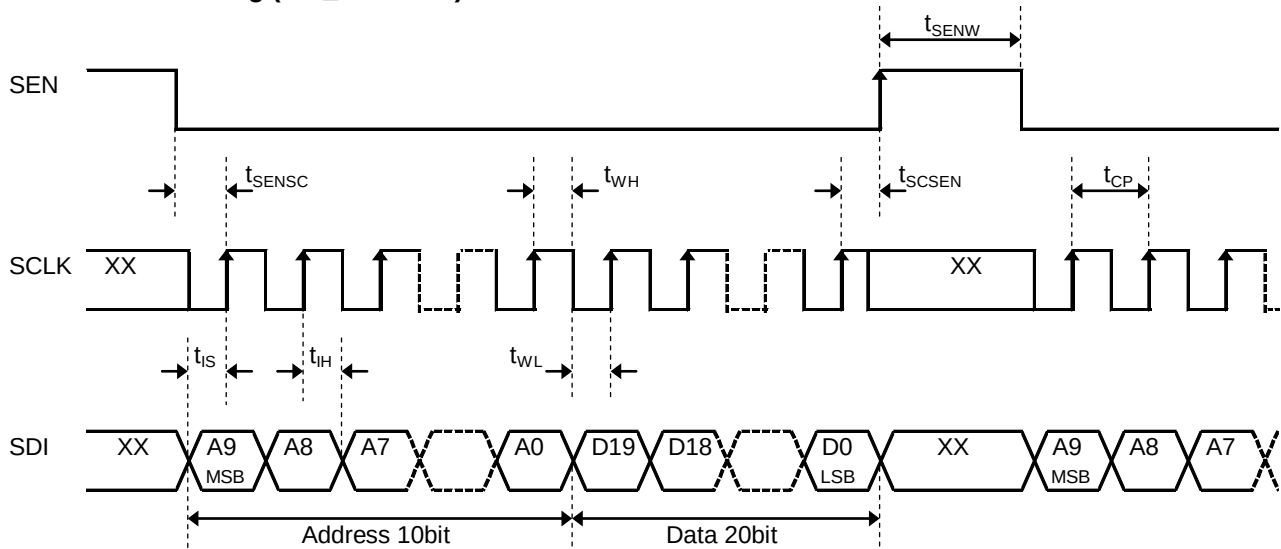
SYMBOL	PARAMETER		MIN	TYP	MAX	UNIT
	TCLK high level duty	7-bit mode	45.7	57.1	68.6	%
		8-bit mode	40.0	50.0	60.0	%
t_{d0}	Delay time TCLKx rising edge to serial bit position 0		$-t_{SK}$		t_{SK}	ns
t_{d1}	Delay time TCLKx rising edge to serial bit position 1	7-bit mode	$1/7 \cdot t_{TCLK} - t_{SK}$		$1/7 \cdot t_{TCLK} + t_{SK}$	ns
		8-bit mode	$1/8 \cdot t_{TCLK} - t_{SK}$		$1/8 \cdot t_{TCLK} + t_{SK}$	ns
t_{d2}	Delay time TCLKx rising edge to serial bit position 2	7-bit mode	$2/7 \cdot t_{TCLK} - t_{SK}$		$2/7 \cdot t_{TCLK} + t_{SK}$	ns
		8-bit mode	$2/8 \cdot t_{TCLK} - t_{SK}$		$2/8 \cdot t_{TCLK} + t_{SK}$	ns
t_{d3}	Delay time TCLKx rising edge to serial bit position 3	7-bit mode	$3/7 \cdot t_{TCLK} - t_{SK}$		$3/7 \cdot t_{TCLK} + t_{SK}$	ns
		8-bit mode	$3/8 \cdot t_{TCLK} - t_{SK}$		$3/8 \cdot t_{TCLK} + t_{SK}$	ns
t_{d4}	Delay time TCLKx rising edge to serial bit position 4	7-bit mode	$4/7 \cdot t_{TCLK} - t_{SK}$		$4/7 \cdot t_{TCLK} + t_{SK}$	ns
		8-bit mode	$4/8 \cdot t_{TCLK} - t_{SK}$		$4/8 \cdot t_{TCLK} + t_{SK}$	ns
t_{d5}	Delay time TCLKx rising edge to serial bit position 5	7-bit mode	$5/7 \cdot t_{TCLK} - t_{SK}$		$5/7 \cdot t_{TCLK} + t_{SK}$	ns
		8-bit mode	$5/8 \cdot t_{TCLK} - t_{SK}$		$5/8 \cdot t_{TCLK} + t_{SK}$	ns
t_{d6}	Delay time TCLKx rising edge to serial bit position 6	7-bit mode	$6/7 \cdot t_{TCLK} - t_{SK}$		$6/7 \cdot t_{TCLK} + t_{SK}$	ns
		8-bit mode	$6/8 \cdot t_{TCLK} - t_{SK}$		$6/8 \cdot t_{TCLK} + t_{SK}$	ns
t_{d0}	Delay time TCLKx rising edge to serial bit position 7	8-bit mode	$7/8 \cdot t_{TCLK} - t_{SK}$		$7/8 \cdot t_{TCLK} + t_{SK}$	ns
t_{SK}	Skew		-0.2		0.2	ns

Note) Design Guarantee

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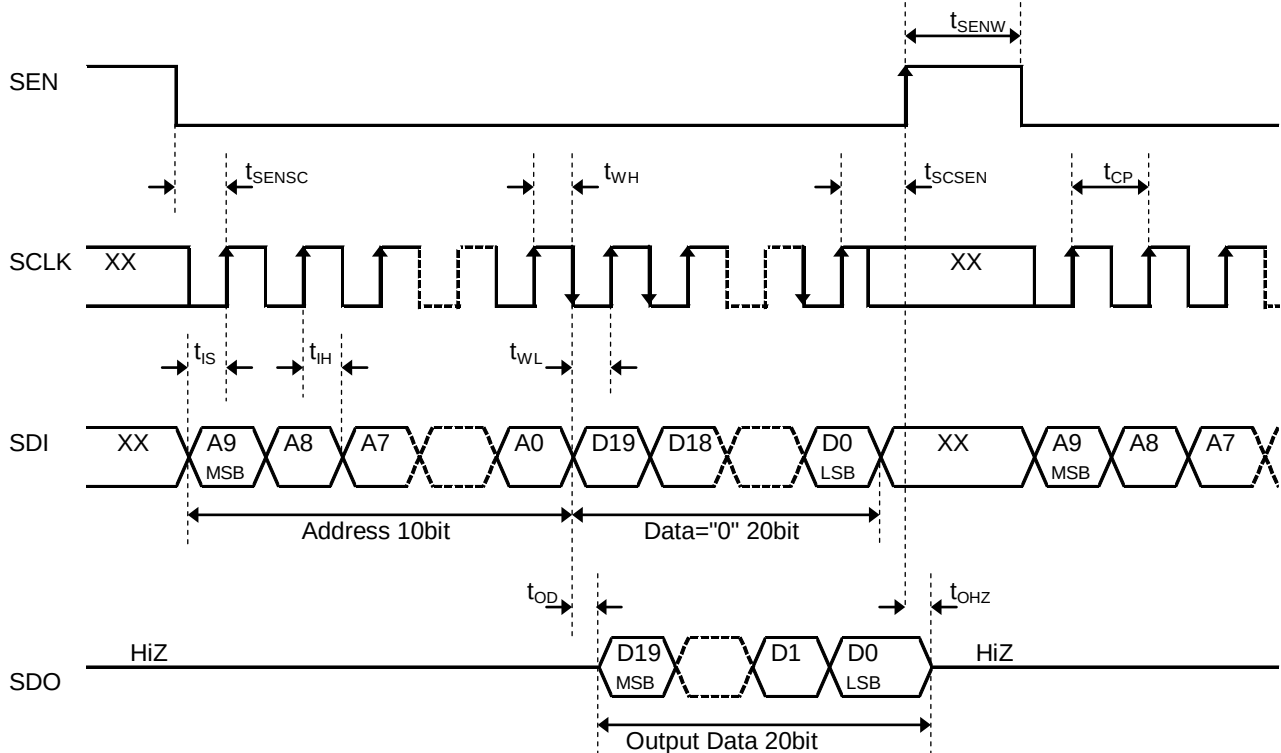
7.6. Serial Interface Timing Specification

1) Write Mode Timing (SPI_RW = "0")



2) Read Mode Timing (SPI_RW = "1")

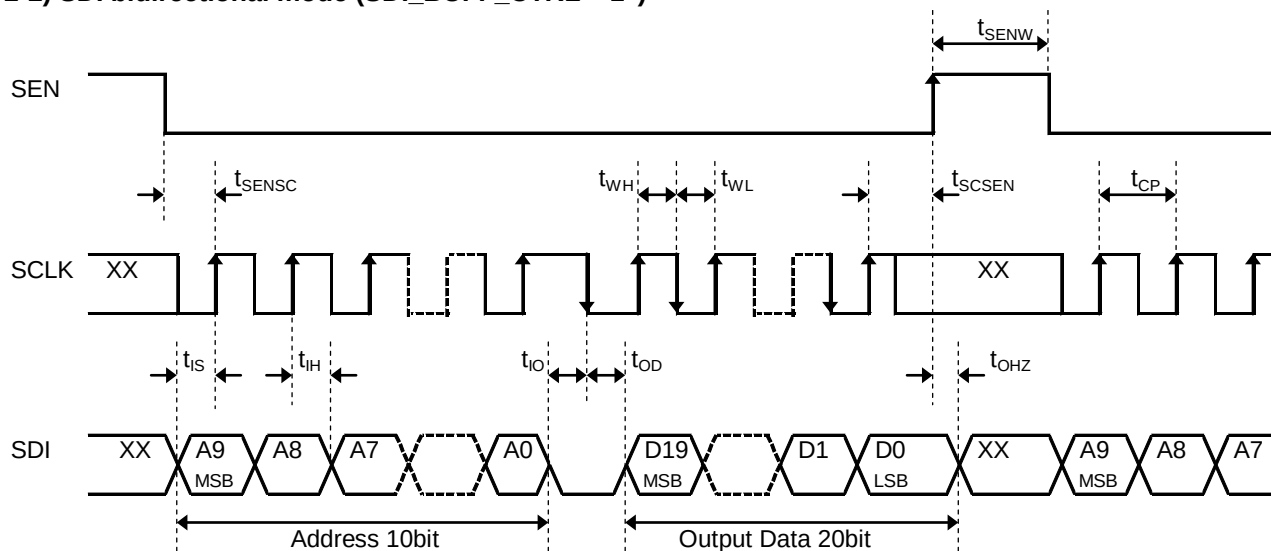
2-1) SDI input mode (SDI_BUFF_CTRL="0", default)



(1) SDO signal outputs with SDO/GPIO1 pin.

(2) This mode needs to set GPIO1_SDO_SEL[1:0] = "3"

2-2) SDI bidirectional mode (SDI_BUFF_CTRL="1")



7.6. Serial Interface Timing Specification (Continued)

SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{CP}	SCLK cycle period	100			ns
t_{WH}	SCLK high width	40			ns
t_{WL}	SCLK low width	40			ns
	SCLK transition rise or fall			4	ns
t_{IS}	Data settling time	15			ns
t_{IH}	Data hold time	15			ns
t_{SENSC}	SEN leading edge to SCLK rising edge	20			ns
t_{SCSEN}	SCLK rising edge to SEN trailing edge	20			ns
t_{SENW}	SEN disable width	20			ns
t_{OD}	Output delay of SDO			30	ns
t_{OHZ}	Delay of SDO changed output to Hi-Z			40	ns
t_{IO}	Interval between "A0" and clock falling edge	20			ns

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7.7. Power Up Sequence

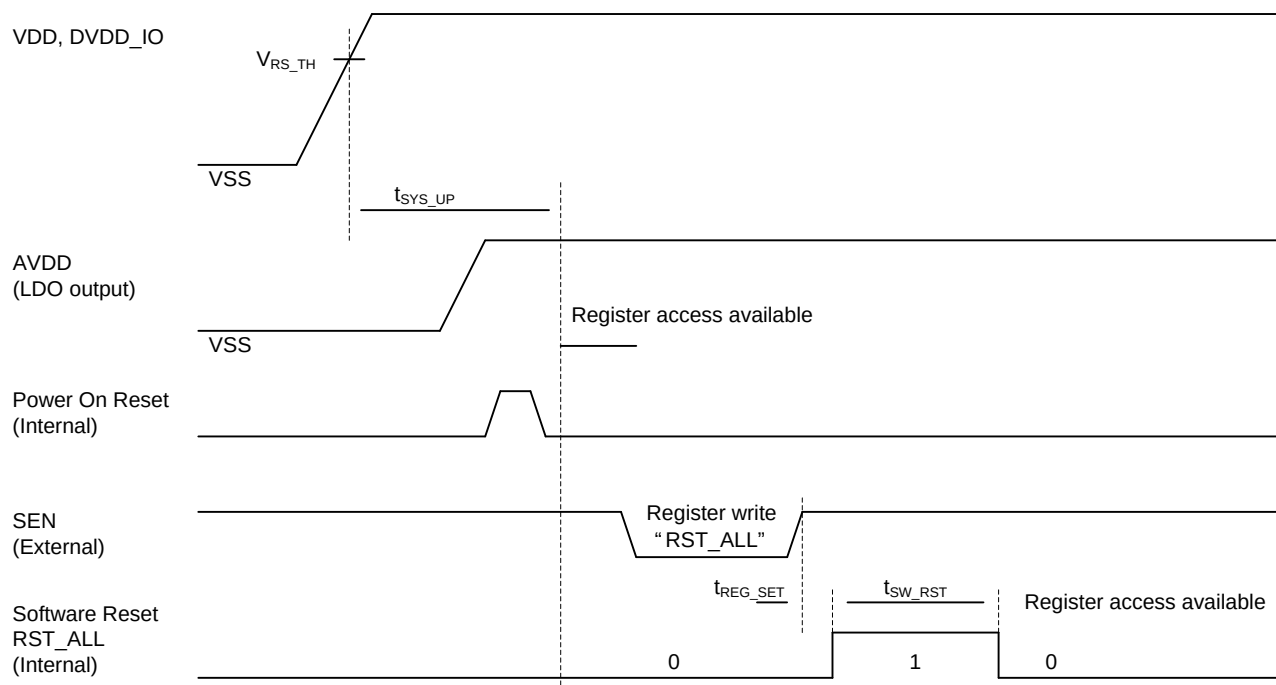


Figure 5. Power Up Sequence Timing

SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
V_{RS_T}	Reset threshold level of VDD		2.2		V
t_{SYS_UP}	System up time			100	msec
t_{REG_SET}	Actual register setting time			6	μ sec
t_{SW_RST}	Software reset period			1	msec

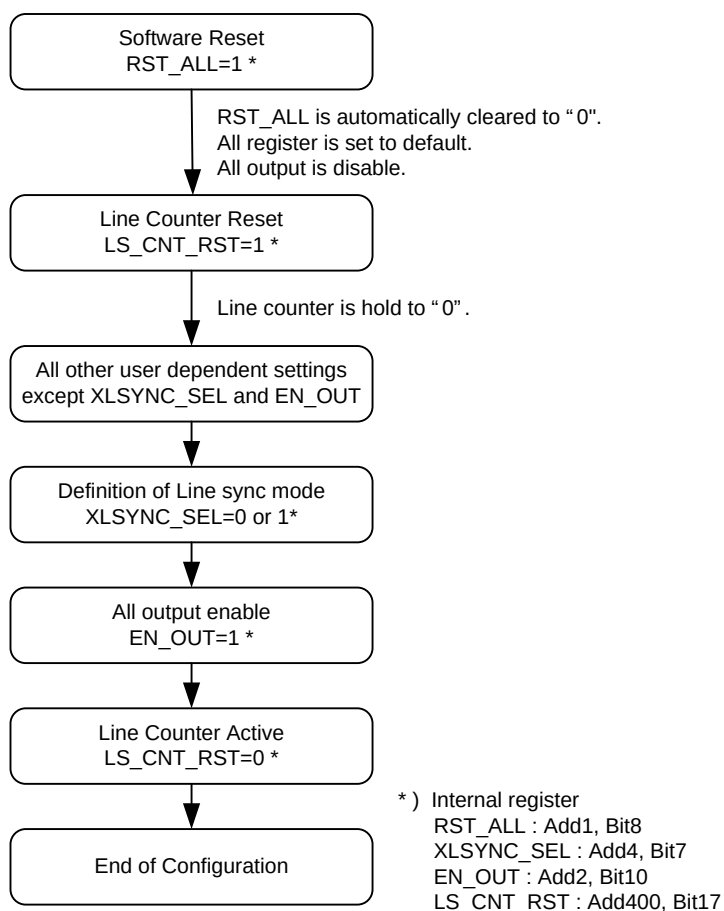


Figure 6. Recommendation sequence after software reset

8. SYSTEM OVERVIEW

8.1. Introduction

VSP5640 is developed as an analog front-end for CCD and CMOS line image sensors applications such as copier, facsimiles and others. VSP5640 provides independent 4 channels of data processing.

The data from each channel of image sensor is sampled and held by either SH or CDS circuit and is then converted into digital data by an ADC. The digital data for each channel is later converted into serial data that can be outputted in either LVDS mode or CMOS mode.

9. AFE BLOCK

9.1. Analog Signal Input

This product has 4-channels as analog input ports for image sensor. Besides 4-channel input, this AFE also supports 3-channel and 2-channel inputs. Table 1 shows the register setting required to select different channel modes.

Table 1. Analog Input Channel Mode Selection

MODE	AIN_CH_SEL	AIN1	AIN2	AIN3	AIN4
2 channel	2	Active	Standby	Active	Standby
3 channel	1	Active	Active	Active	Standby
4 channel	0	Active	Active	Active	Active

Each analog input supports the CDS (Correlated Double Sampling) and the simple SH (Sample and Hold) circuits to accommodate CCD and CMOS image sensors. The sampling-mode can be selected independently for each channel by configuring internal registers. As showing in Table 2, if AINx_SH_CDS is set to "0", then the corresponding channel operates in CDS mode.

Table 2. CDS/SH Mode Selection

AINX_SH_CDS ⁽¹⁾	SH/CDS
0	CDS
1	SH

(1) AINx_POL: Analog input polarity setting register. (x = 1, 2, 3 and 4)

In addition, this device also supports independent selection of the input signal polarity for each channel. Input signal polarity can be set using the register "AINx_POL" (x=1, 2, 3 or 4). The input signal range and polarity are defined in chapter 6.1.

1) Correlated Double Sampler (CDS) Mode (AINx_SH_CDS = "0")

CDS mode is designed to accommodate inputs from CCD sensor. The output signal of a CCD image sensor is sampled twice during one pixel period. First, the reference interval is sampled by SHP pulse and then the data interval is sampled by SHD pulse. Subtracting these two samples provides the video information of the pixel as well as removes any noise which is common to both the intervals. Thus, CDS plays an important role to reduce the reset-noise and other low frequency noises that are present on the CCD output signal. Figure 7 shows block diagram of CDS mode.

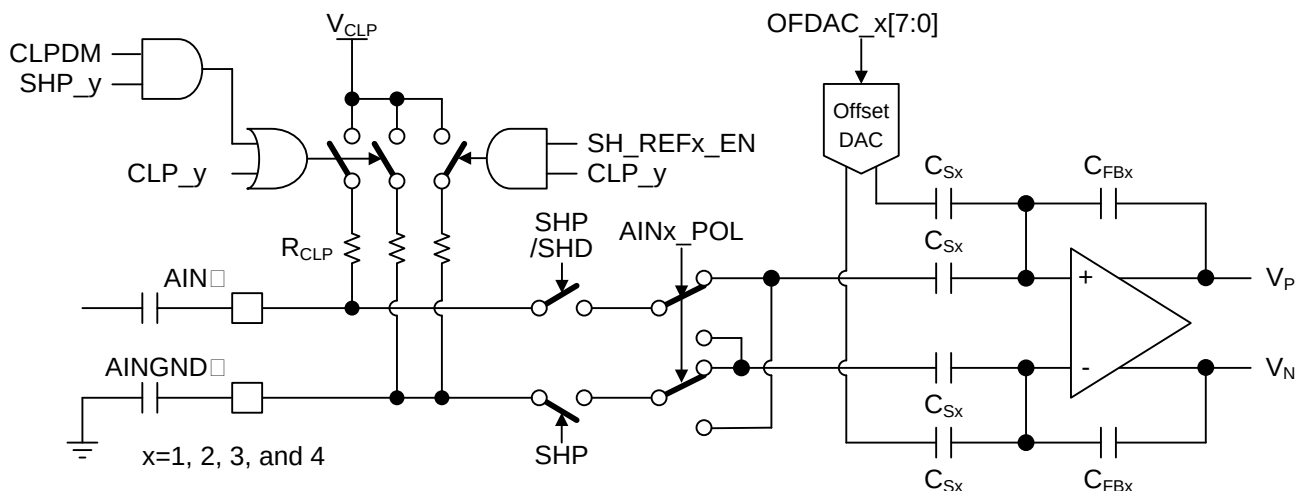


Figure 7. Input Circuit of CDS Mode for CCD Signal

2) Sample Hold (SH) Mode (AINx_SH_CDS = "1")

SH mode supports CCD and CMOS sensors. In case of CCD, the pedestal level of sensor signal is clamped to V_{CLP} level using an internal clamp circuit. SH samples only once during one pixel period. The SHD pulse is used to sample data interval of the CCD signal. After sampling, SH circuit takes the difference of data and V_{CLP} levels to extract video information.

In case of CMOS input, input clamp function should be set as per the requirements. If sensor output is within the allowable input range, AC coupling capacitor for analog input might not be needed. When sensor signal is directly input to AFE, SH circuit needs a reference voltage to set the black level. To use V_{CLP} as reference, SH_REFx_EN should be enabled, and then, AINGNDx should be opened or coupled to GND with capacitor. To use external reference, it can be input to AINGNDx with sensor signals connected to AINx. Figure 8 shows SH mode block diagram.

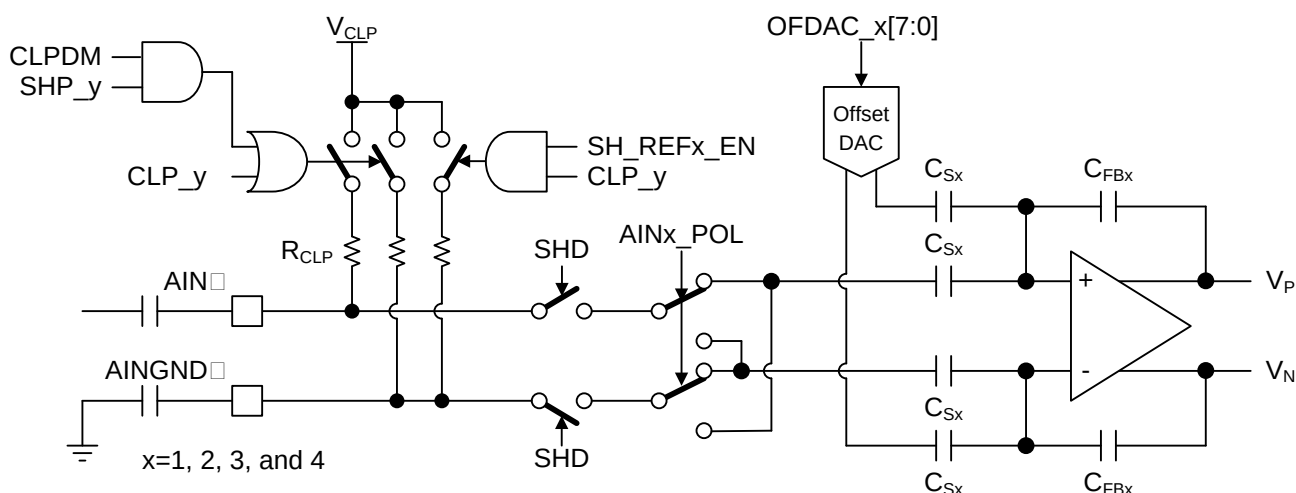


Figure 8. Input Circuit of SH Mode for CCD or CMOS Signal

Note:

- (1) Under some conditions the sensor signal can be directly input to AFE and the external capacitor is not required..
- (2) In SH mode, SHP clock still should be programmed so that it does not overlap with SHD clock.

9.2. Input Clamp and Sensor Reference

CCD output signal has large DC offset that might be beyond the input range of the AFE input circuit. This output signal is thus AC coupled to AFE through a capacitor and the internal DC level is set to Clamp voltage (V_{CLP}) by an internal clamp circuit. VSP5640 provides 3-modes for clamp operation as shown in Table 3. The clamp mode can be set independently for each channel by configuring register AINx_CLP_SEL.

In case of "Pixel Clamp" mode, CLP_A/B is used for clamping. The input signal is clamped to V_{CLP} using CLP_A/B pulse during each pixel period. Since AC coupling CAP is charged on a pixel to pixel basis, the droop of clamp level can be controlled by clamp pulse width.

Meanwhile, in "Line Clamp" mode, SHP_A/B is used for clamping during CLPDM active. The input signal is clamped only in CLPDM period within one line cycle of sensor. The charge leaks from the coupling capacitor for a longer period except during the "CLPDM" period. Accordingly, as there may be a large droop in the clamp level, this product does not support Line Clamp in the SH mode.

"Non Clamp" mode is mainly used in case of CMOS sensor input. If sensor signal is directly connected to AFE, this mode should be configured without AC coupling cap at the input port. This mode has two options to select reference for sensor's black-level. One is to use internal reference, V_{CLP} , with AINx_CLP_SEL="2" and the other is to use external input from AINGNDx with AINx_CLP_SEL="3".

Table 3. Clamp Mode Selection

CLAMP MODE	MODE SETTING	CLAMP ACTIVE CONDITION & SETING			
	AINx_CLP_SEL ⁽¹⁾	CDS/SH	CLP_y ⁽²⁾	CLPDM AND SHP_y ⁽²⁾	SH_REF_EN
PIXEL CLAMP	0 (default)	CDS/SH	Active	Active	OFF
LINE CLAMP	1	CDS/SH	-	Active	OFF
NON CLAMP	2	Only SH	-	-	ON
	3	Only SH	-	-	OFF

(1) AINx_CLP_SEL (x=1, 2, 3 and 4)

(2) y=A, B

As shown in Figure 9, the internal node V_{CLP} provides the clamp reference voltage. As shown, it is possible to select three reference voltage modes by setting register AINx_REF_SEL. The first mode provides fixed 2.2V, the second mode provides selectable outputs (0.5, 1.1, 1.5 and 2.0V) of an internal DAC, and the third mode allows external input from REF_AIN pin as clamp reference. This pin REF_AIN is bidirectional and also acts as output of internal DAC. Table 4 shows the relationship between register and clamp level, and Table 5 shows DAC configuration.

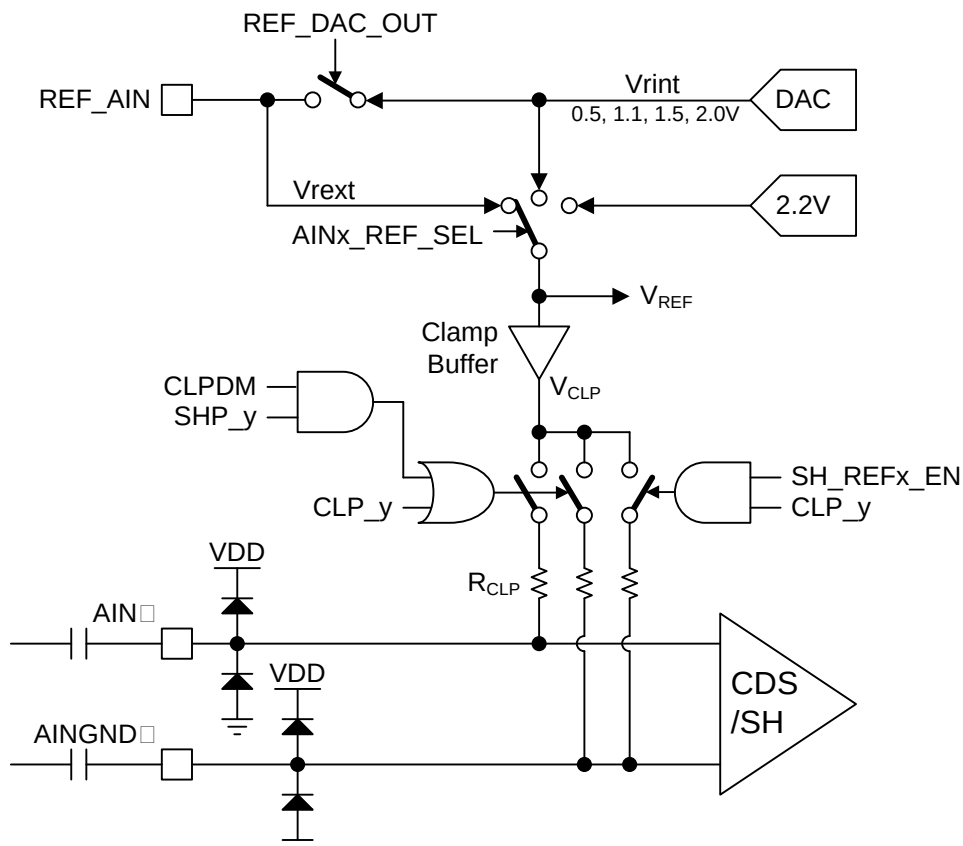
Table 4. Clamp Level Selection

MODE SETTING AINX_REF_SEL[1:0]	CLAMP LEVEL
0	2.2 V
1	V_{rint} Reference DAC (0.5, 1.1, 1.5, 2.0 V)
2	V_{ext} REF_AIN external input

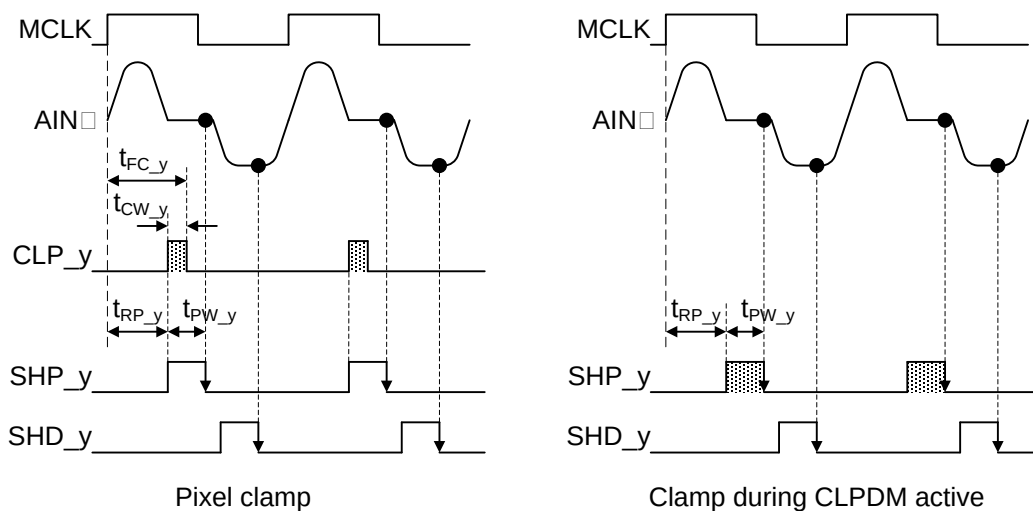
Note: AINx_CLP_SEL (x = 1, 2, 3 and 4)

Table 5. V_{rint} Voltage Selection

SETTING CODE VRINT_SEL[1:0]	REF DAC V_{RINT} [V]
2	0.5
3	1.1
0	1.5 (default)
1	2.0



Note) If sensor signal is directly input to AFE, the external capacitor shouldn't be connected.



Note)

x : AIN channel number, x=1, 2, 3 and 4

y : Group code of sample pulse signals, x=1 or 2 : y=A, x=3 or 4 : y=B

Figure 9. Input Clamp Circuit

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If “Line Clamp” mode is used, the period of CLPDM should be configured by internal registers. The period of CLPDM is determined with reference to “LS”, the signal of line cycle for sensor. Thus the start and end of CLPDM are defined as number of pixels from falling edge of “LS”. Since CLPDM is used as clamp period, it should be assigned for the interval of dummy or optical black pixels. Figure 10 shows the relationship between “LS” and “CLPDM”.

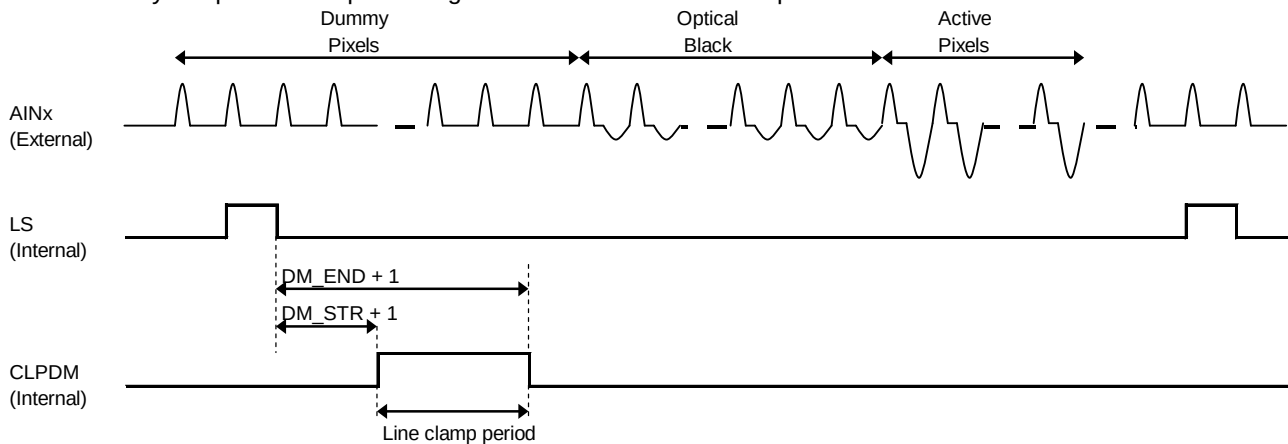
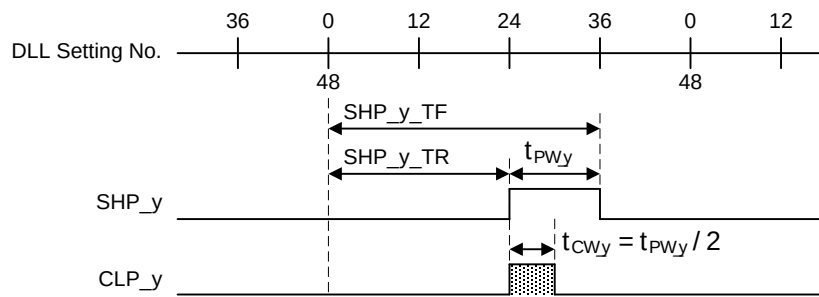


Figure 10. Line Clamp Period Setting

In case of “Pixel Clamp” mode without “CLPDM”, sensor signal is clamped with “CLP_A” and “CLP_B” pulses. CLP_A corresponds to AIN1 and AIN2, whereas, CLP_B corresponds to AIN3 and AIN4. The start of these pulses is synchronized with the rising edge of “SHP_y” (y=A or B). There are two options to configure the end position. One is to automatically set the pulse width to 50% that of SHP_y and the other is to manually configure the end position using an internal register. Figure 11 explains the details of the clamp pulse function.

1) Automatic mode (CLP_TF_AT_DIS=“0”)



2) Manual mode (CLP_TF_AT_DIS=“1”)

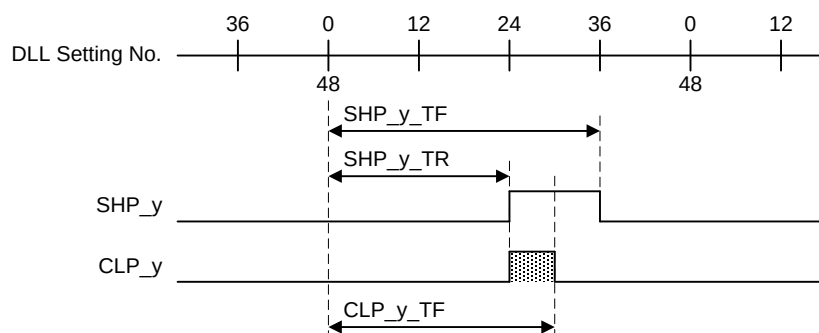


Figure 11. Pixel Clamp Period Setting

In case of "Pixel Clamp" mode during "CLPDM" active, sensor signal is clamped with "SHP_y". Therefore the pixel clamp operation is closely related with the status of "CLPDM" and the condition of "CLPDM" should be properly defined with the internal registers. Since CLPDM is always high in case of default condition after reset or power up, you should define the status of CLPDM according to this sequence. And it is necessary to do it at the 3 step "All other user dependent settings except XLSYNC_SEL and EN_OUT" of software reset sequence described in Figure 6.

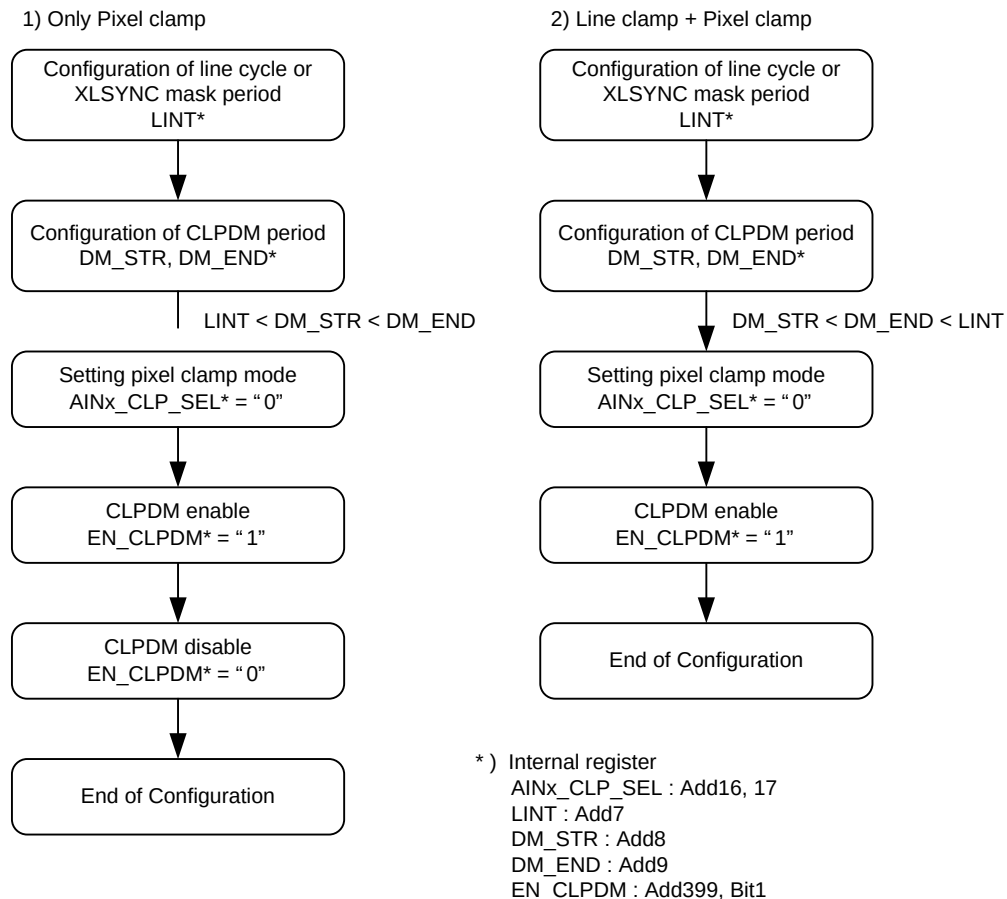


Figure 12. Configuration sequence for pixel clamp

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9.3. Analog Programmable Gain (APG)

SH output can be amplified using programmable analog gain. This gain can be set from 0.5 to 3.5 [V/V] with a step size of 3/64 [V/V].

The gain setting can be controlled by an internal register (APG_x). The following equation shows the relationship between setting-code and gain. The gain of each of the four channels can be set independently using different registers. Note that there is a possibility of change in the black pixel level due to change in the gain, therefore, the appropriate timing of gain change should be used to avoid degradation in image quality.

$$\text{APG}[\text{V/V}] = \frac{3}{63} \times \text{Code} + 0.5 \quad (\text{Code} = 0\text{--}63[\text{LSB}])$$

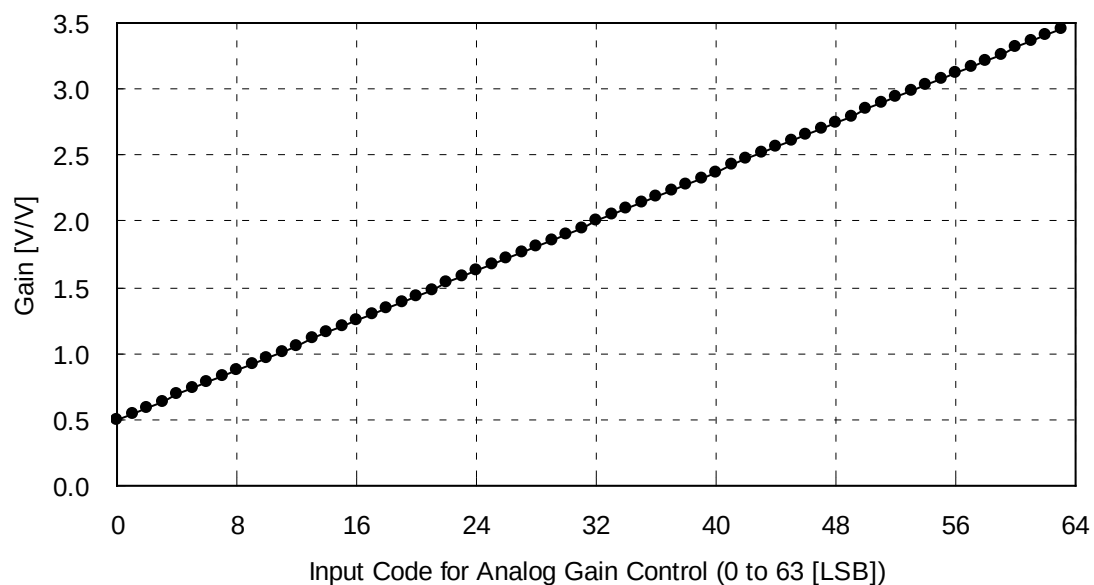


Figure 13. Analog Gain vs. Setting Code

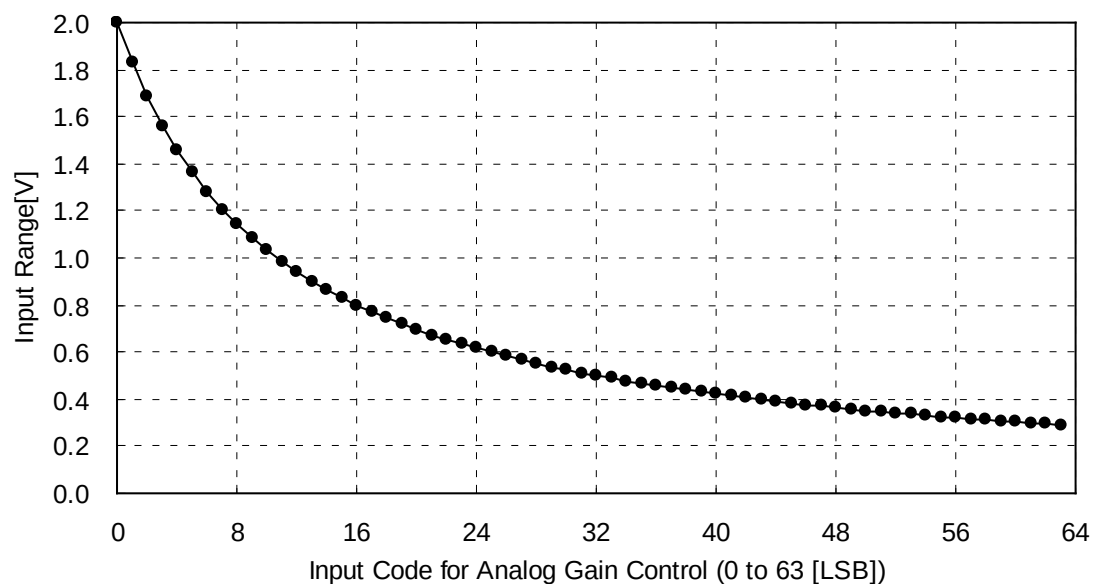


Figure 14. Input Range vs. Analog Gain Setting Code

9.4. Digital Programmable Gain (DPG)

VSP5640 provides a maximum digital gain of 2 [V/V]. The total gain is fixed by the combination of Analog Gain (APG) of CDS/SH and Digital Gain (DPG). DPG is controlled by an 8-bit internal register (DPG_x) that can set the gain from 1 to 2 [V/V] as defined by the formula shown below. Each of the four channels has this register, so gain of each channel can be set independently.

Figure 15 shows the relationship between Digital Gain and register code. Note that the default value is 1[V/V].

$$\text{DPG[V/V]} = \frac{1}{256} \times \text{Code} + 1 \quad (\text{Code} = 0\text{--}255[\text{LSB}])$$

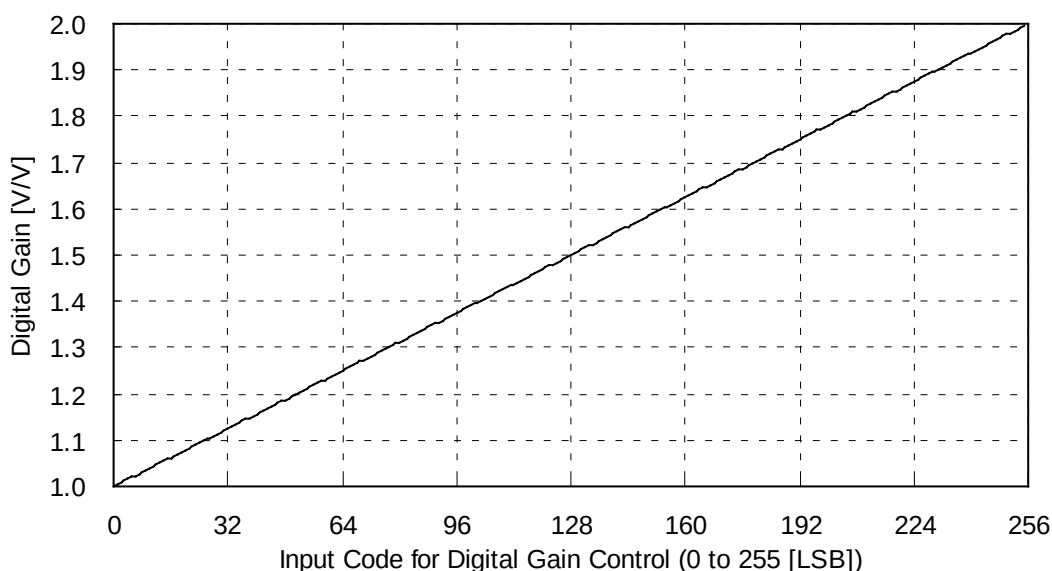


Figure 15. Digital Gain Setting Code

9.5. ADC

ADC output format is selectable from two's complement or offset binary by configuring register. The following table shows the relationship between register setting and condition.

Table 6. ADC Data format configuration

ADC_DAT_FRM	MODE
0 (Default)	Two's complement
1	Offset binary

9.6. OFFSET DAC

VSP5640 has an independent DAC in each channel for correcting offset level of input signal. Its correction range is ± 250 mV and resolution is 8 bit. DAC output voltage can be set by register settings. Table 7 and Figure 16 show the relationship between output and setting codes. Setting code is defined in 2's complement format.

Table 7. Offset DAC Setting Code

SETTING CODE OFDAC_X[7:0] ⁽¹⁾	OUTPUT [mV]
7Fh	248.05
7Eh	246.09
⋮	⋮
01h	1.95
00h	0.00
FFh	-1.95
⋮	⋮
81h	-248.05
80h	-250.00

(1) $\times = 1, 2, 3$ and 4

$$\text{DAC Output [mV]} = \frac{250}{128} \times \text{OFDAC_x[7:0]}, (\text{x}: 1, 2, 3 \text{ and } 4)$$

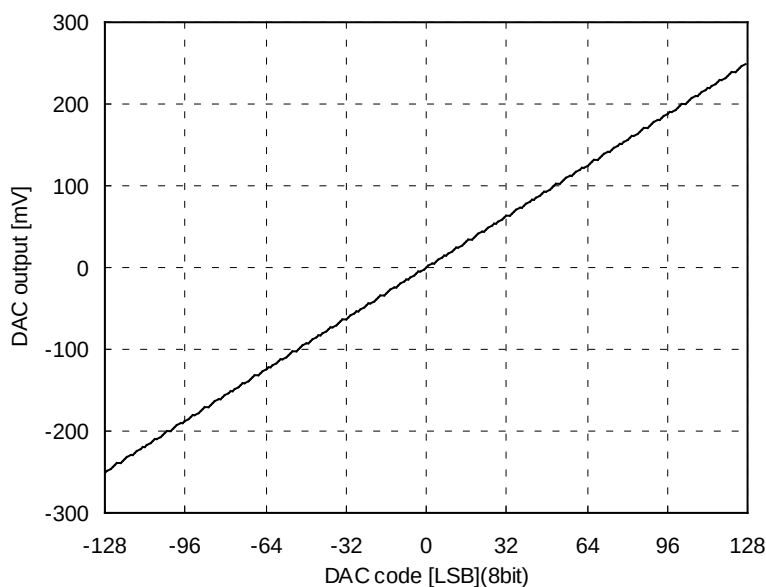


Figure 16. Offset DAC Setting Code vs. Output Voltage

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Table 9. Output Clock Rate on Each Mode for LVDS

Sample rate	Analog Input	LVDS Data Output	Data ⁽¹⁾	MCLK				LVCK, TCLK						SERCLK LVDS Serialize clock						
				7bit mode		8bit mode		7bit mode			8bit mode			7bit mode			8bit mode			
				min [MHz]	max [MHz]	min [MHz]	max [MHz]		min [MHz]	max [MHz]		min [MHz]	max [MHz]		min [MHz]	max [MHz]		min [MHz]	max [MHz]	
x1	2ch	1pair	Single			2.0	12.5					x4	8	50				x32	64	400
			Double			1.0	6.3					x8	8	50				x64	64	400
		2pair	Single			4.0	16.7					x2	8	33.33				x16	64	266.7
			Double			2.0	8.3					x4	8	33.33				x32	64	266.7
		3pair	Single	4.0	16.7			x2	8	33.33				x14	56	233.3				
			Double																	
	3ch	1pair	Single																	
			Double																	
		2pair	Single	2.0	16.7	2.7	16.7	x4	8	66.67	x3	8	50	x28	56	466.7	x24	64	400	
			Double	1.0	8.3	1.3	8.3	x8	8	66.67	x6	8	50	x56	56	466.7	x48	64	400	
		3pair	Single	2.7	16.7			x3	8	50				x21	56	350				
			Double																	
	4ch	1pair	Single																	
			Double																	
		2pair	Single			2.0	12.5				x4	8	50					x32	64	400
			Double			1.0	6.3				x8	8	50					x64	64	400
		3pair	Single	2.0	12.5			x4	8	50				x28	56	350				
			Double																	
x2	2ch	1pair	Single			1.0	6.3					x8	8	50				x64	64	400
			Double																	
		2pair	Single																	
			Double																	
		3pair	Single	2.0	8.3			x4	8	33.33				x28	56	233.3				
			Double																	
	3ch	1pair	Single																	
			Double																	
		2pair	Single	1.0	8.3	1.3	8.3	x8	8	66.67	x6	8	50	x56	56	466.7	x48	64	400	
			Double																	
		3pair	Single	1.3	8.3			x6	8	50				x42	56	350				
			Double																	
	4ch	1pair	Single																	
			Double																	
		2pair	Single			1.0	6.3				x8	8	50					x64	64	400
			Double																	
		3pair	Single	1.0	6.3			x8	8	50				x56	56	350				
			Double																	

(1) Double means that same data will come out by 2 times.

Figure 18 shows the configuration of TCLK waveform. Since TCLK has same serializer as TX for output data, it is possible to set the waveform of TCLK by unit of LVDS data bits. Therefore, before capturing it is necessary to set register "TCLK_FRM_SET[7:0]" which defines waveform of TCLK.

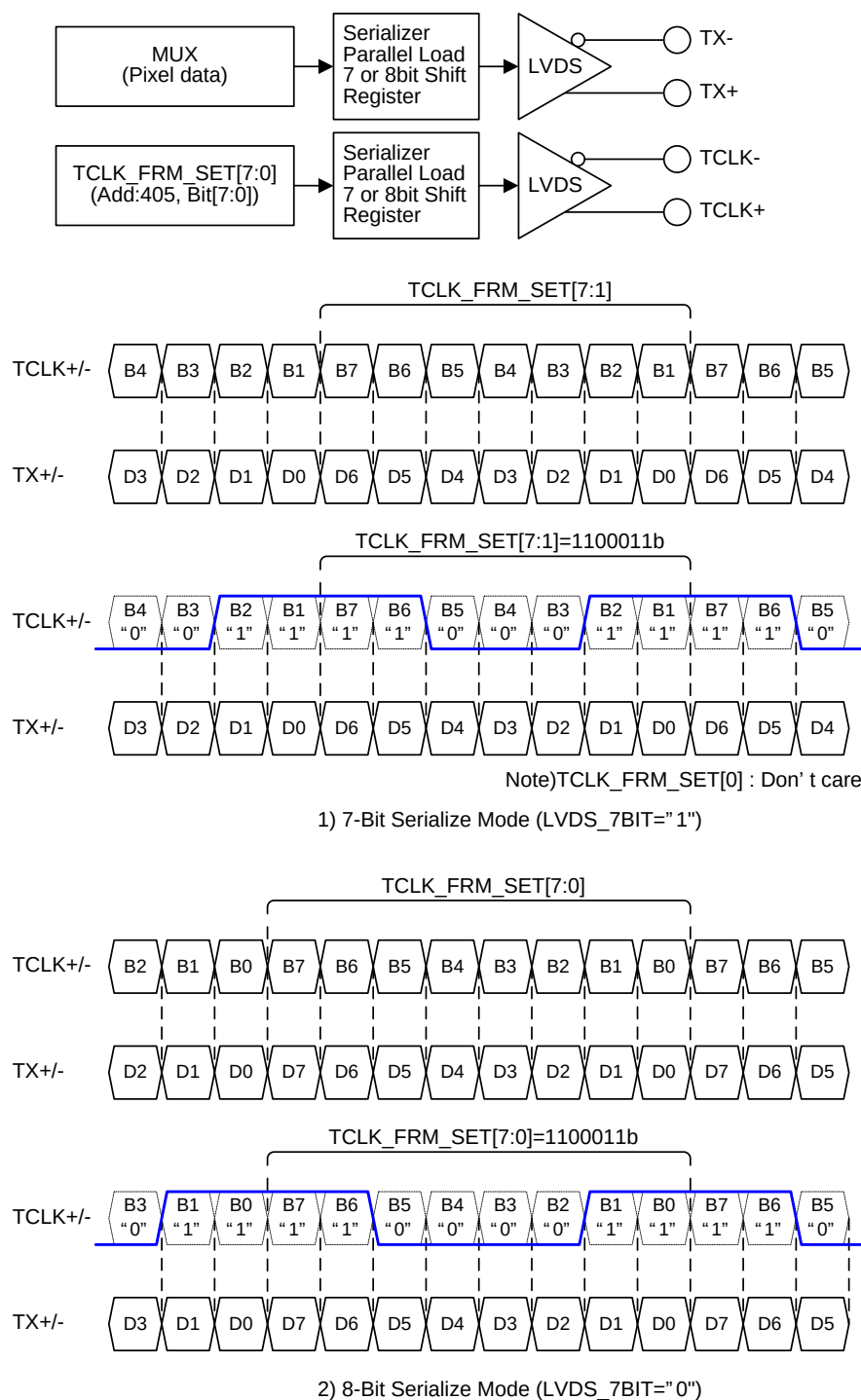


Figure 18. Configuration of TCLK

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[CMOS Output Mode (CMOS_OUT = "1")]

In CMOS output mode, as shown in Figure 19, this device has a function to output data in either 4-bit mod or 8-bit mode. Table 10 shows the pin assignments for CMOS mode. In 4-bit mode, pins CK0 and CK1 can be used to output internal clocks LS and LVCK respectively (Table 11). For each mode, as shown in Table 12, there are clock rate limitations due to CMOS driver characteristic.

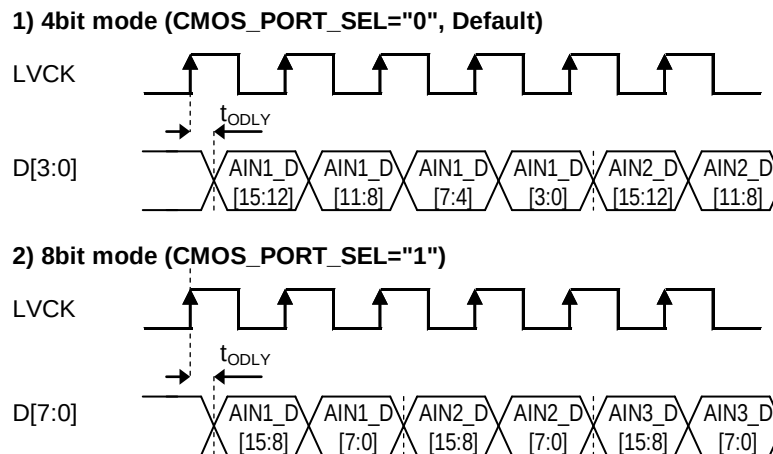


Figure 19. Data Bus for CMOS Output

Table 10. CMOS Output Port Setting

PIN NAME	4 PORT MODE	8 PORT MODE
	CMOS_PORT = "0"	CMOS_PORT = "1"
TA+/D0	D[0]	D[0]
TA-/D1	D[1]	D[1]
TB+/D2	D[2]	D[2]
TB-/D3	D[3]	D[3]
TC+/D4	"L"	D[4]
TC-/D5	"L"	D[5]
TCLK+/CK0	CK[0]	D[6]
TCLK-/CK1	CK[1]	D[7]

Table 11. CK0, CK1 Output Clock in 4-bit Mode

PIN NAME	OUTPUT
CK0	LS
CK1	LVCK

Table 12. Output Clock Rate on Each Mode for CMOS Output

Sample rate	Analog Input	Data	MCLK		LVCK	min		max	Remark
			min	max		min	max		
			[MHz]	[MHz]		[MHz]	[MHz]		
x1	2 ch	8 bit x 2	1.0	16.7	x4	4	66.67		
		4 bit x 4	1.0	10.0	x8	8	80		
	3 ch	8 bit x 2	1.0	13.3	x6	6	80		
		4 bit x 4	1.0	6.7	x12	12	80		
	4ch	8 bit x 2	1.0	10.0	x8	8	80		
		4 bit x 4	1.0	5.0	x16	16	80	Duty 33%	

10.1. 7 bit Serialize Mode (LVDS 3 pair Output)

In 7-bit serialize mode with LVDS 3-pair output, the total number of data bits is 21. 16-bits represent the pixel data and remaining 5-bits are used as status flag (ST[4:0]) as shown in Table 14. Data from four channels is output in rotation. Output data also includes a header code synchronized to "LS" signal for identifying the top. The header code is showed in Table 15.

Figure 20. LVDS Output Data Format With 7-Bit Serialize Mode by 3-Pair of Buffer

LVDS CHANNEL	BIT NO	BIT ASSIGNMENT
TC	6	ST[4]
	5	ST[3]
	4	AINx[15]
	3	AINx[14]
	2	AINx[13]
	1	AINx[12]
	0	AINx[11]
TB	6	AINx[10]
	5	AINx[9]
	4	AINx[8]
	3	AINx[7]
	2	AINx[6]
	1	AINx[5]
	0	AINx[4]
TA	6	AINx[3]
	5	AINx[2]
	4	AINx[1]
	3	AINx[0]
	2	ST[2]
	1	ST[1]
	0	ST[0]

(1) AIN x [15:0]: x = 1, 2, 3 and 4 channel analog input data
(0: LSB, 15: MSB)

(2) ST[4:0]: Status bit

Table 13. Status Flag for 7-Bit Serialize Mode Output

STATUS BIT	MODE0	MODE1	MODE2	MODE3
ST[0]	AINF[0]	AINF[0]	"L"	AINF[0]
ST[1]	GPI[1]	GPI[1]	GPI[1]	GPI[1]
ST[2]	"L"	"L"	"L"	"L"
ST[3]	GPI[3]	GPI[3]	GPI[3]	GPI[3]
ST[4]	AINF[1]	AINF[1]	AINF[1]	CLPDM

(1) GPIO[3:0]: General purpose input

(2) AINF[1:0]: Data channel flag

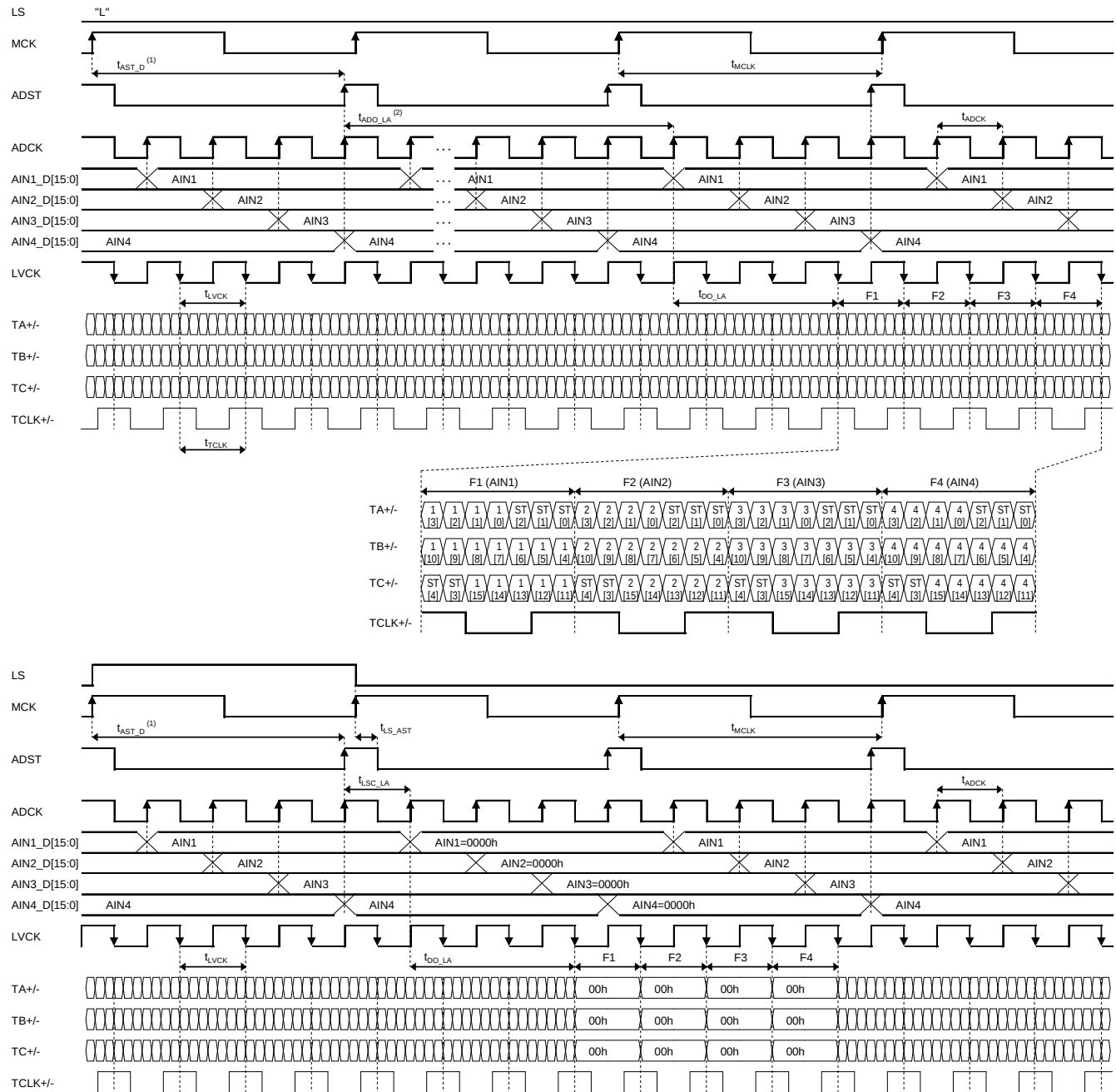
("00" = 1 channel, "01" = 2 channel, "10" = 3 channel, "11" = 4 channel)

Table 14. Line Synchronize Code for 7-Bit Serialize Mode Output

DATA	DESCRIPTION
0000h	Line synchronize code
0001h-0002h	Reserved
0003h-FFFFh	Actual picture data

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1) 7-Bit Serialize, 4 Channel, ×1 Sample, 3 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot 3/48$		$t_{MCLK} \cdot 50/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

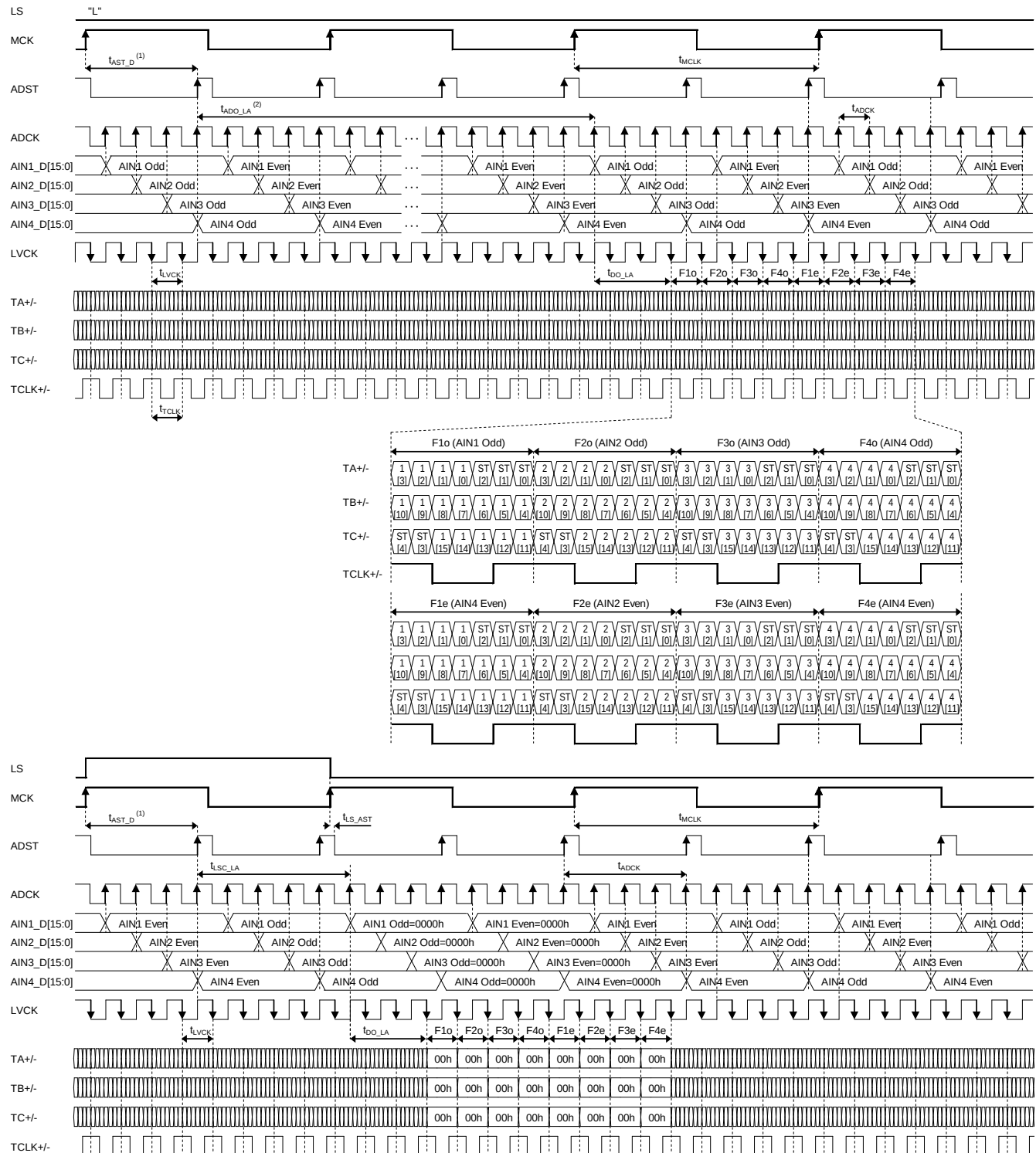
(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

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2) 7-Bit Serialize, 4 Channel, ×2 Sample, 3 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{TCLK}	TCCK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot 2/48$		$t_{MCLK} \cdot 25/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

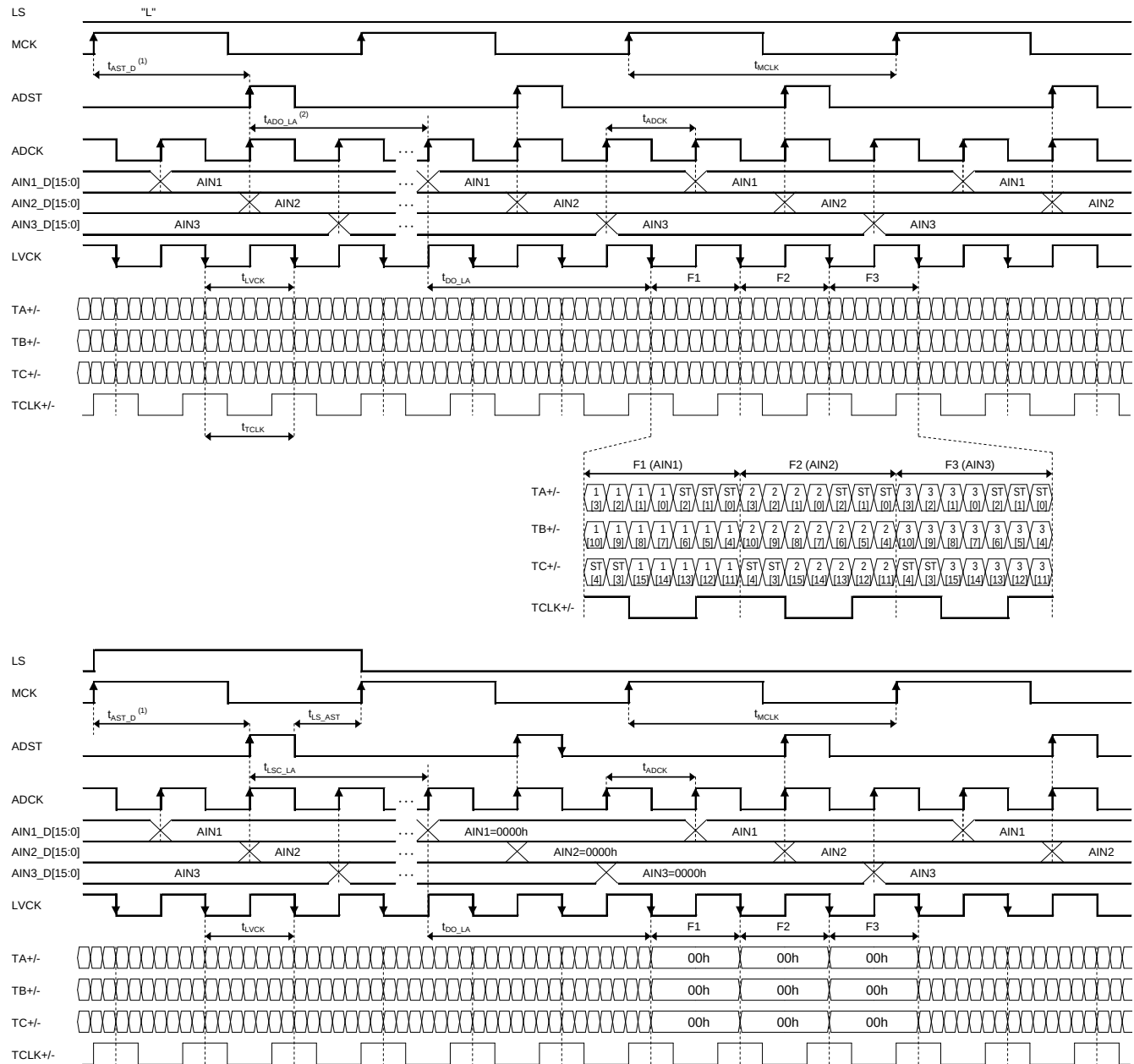
(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

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3) 7-Bit Serialize, 3 Channel, ×1 Sample, 3 Pair LVDS, Single Output



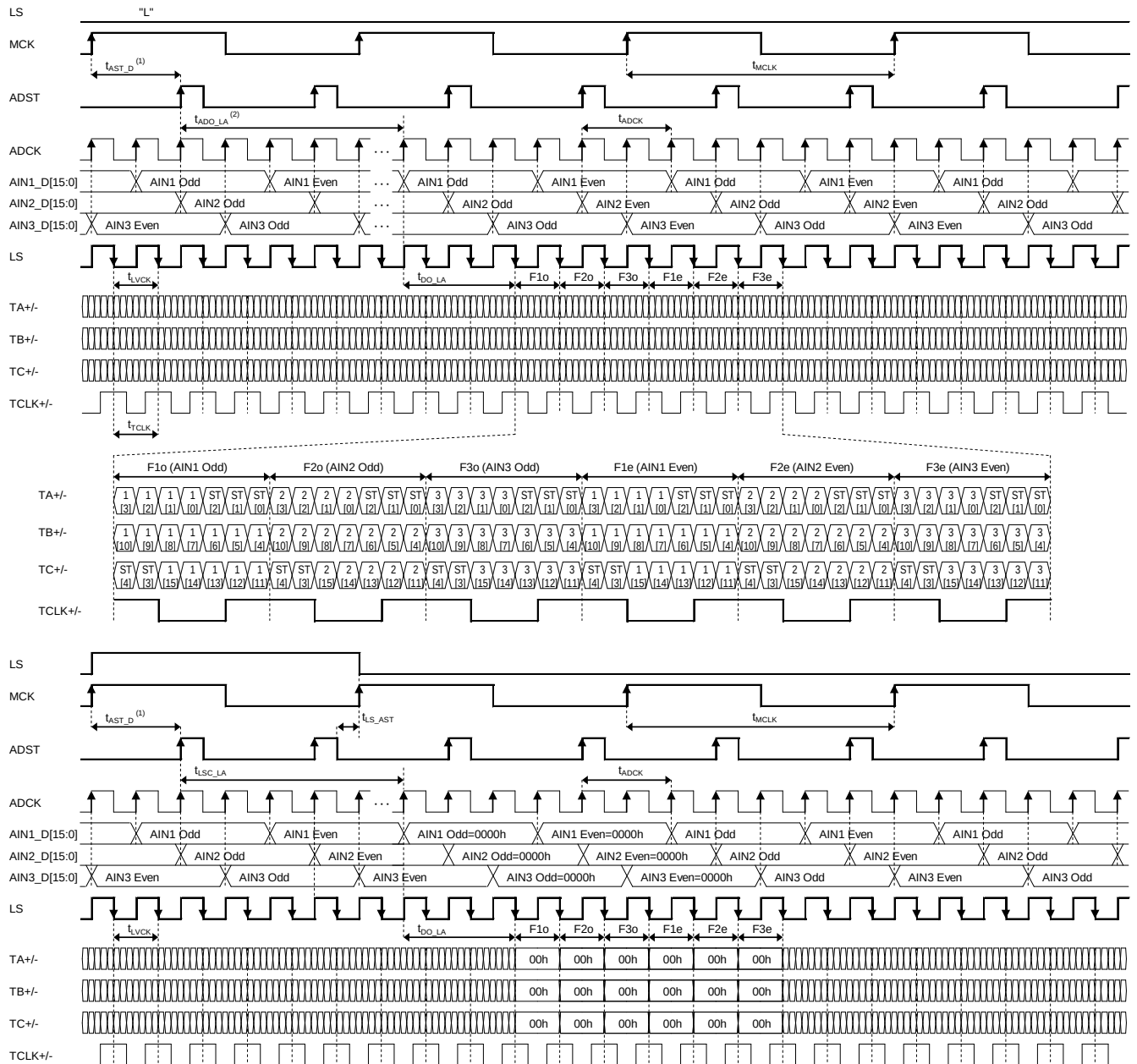
SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/3$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/3$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot -13/48$		$t_{MCLK} \cdot 35/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

4) 7-Bit Serialize, 3 Channel, x2 Sample, 3 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t _{LVCK}	LVCK cycle period		t _{MCLK} • 1/6		ns
t _{TCLK}	TCLK cycle period		t _{MCLK} • 1/6		ns
t _{LS_AST}	Delay time from LS falling edge to AST falling edge	t _{MCLK} • -6/48		t _{MCLK} • 17/48	ns
t _{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t _{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

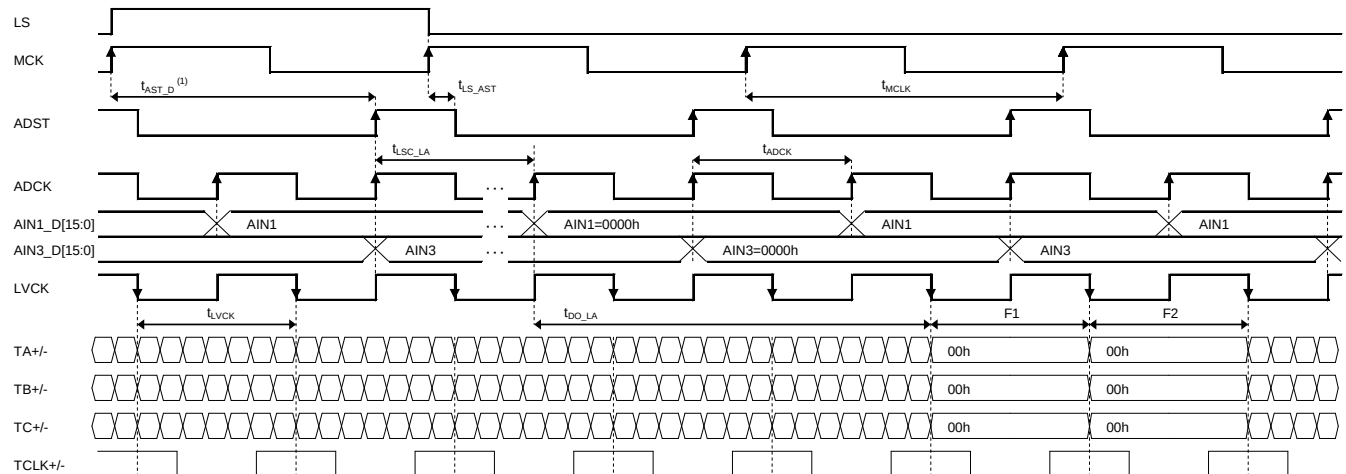
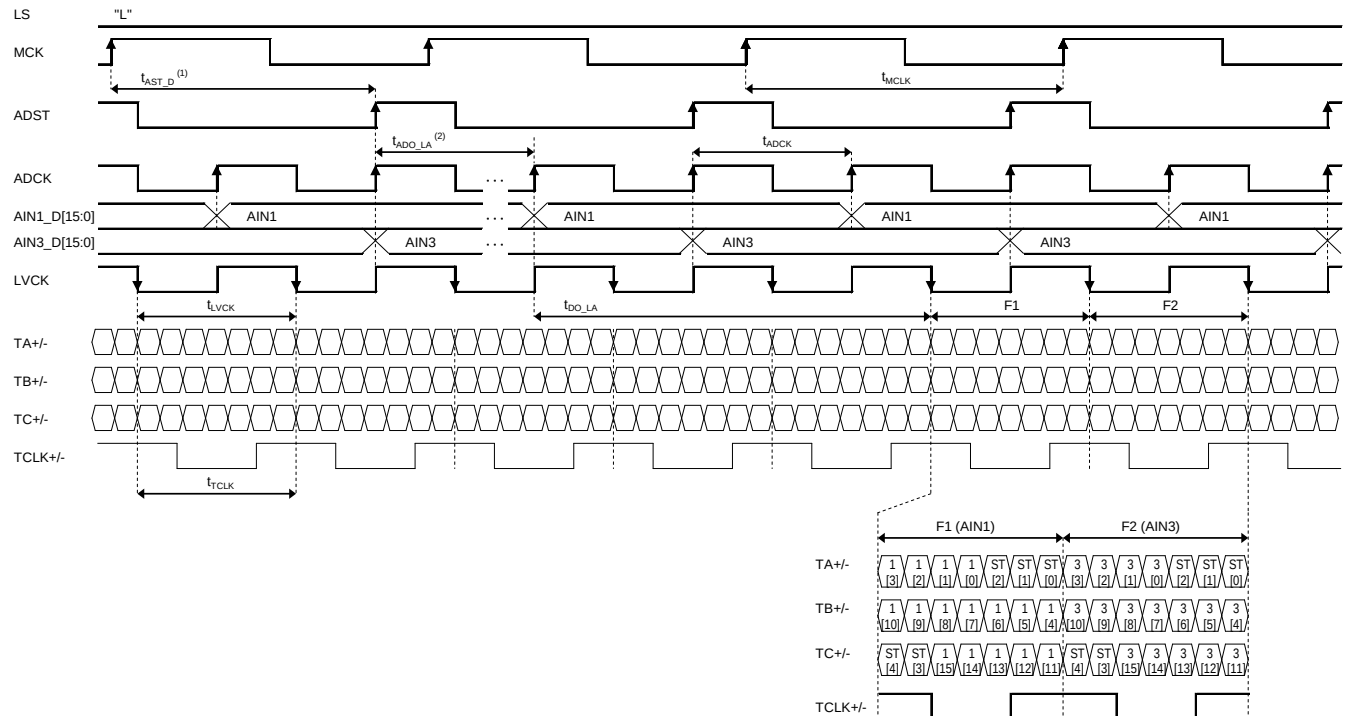
(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

5) 7-Bit Serialize, 2 Channel, x1 Sample, 3 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t _{LVCK}	LVCK cycle period		t _{MCLK} · 1/2		ns
t _{TCLK}	TCCK cycle period		t _{MCLK} · 1/2		ns
t _{LS_AST}	Delay time from LS falling edge to AST falling edge	t _{MCLK} · 3/48		t _{MCLK} · 50/48	ns
t _{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t _{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

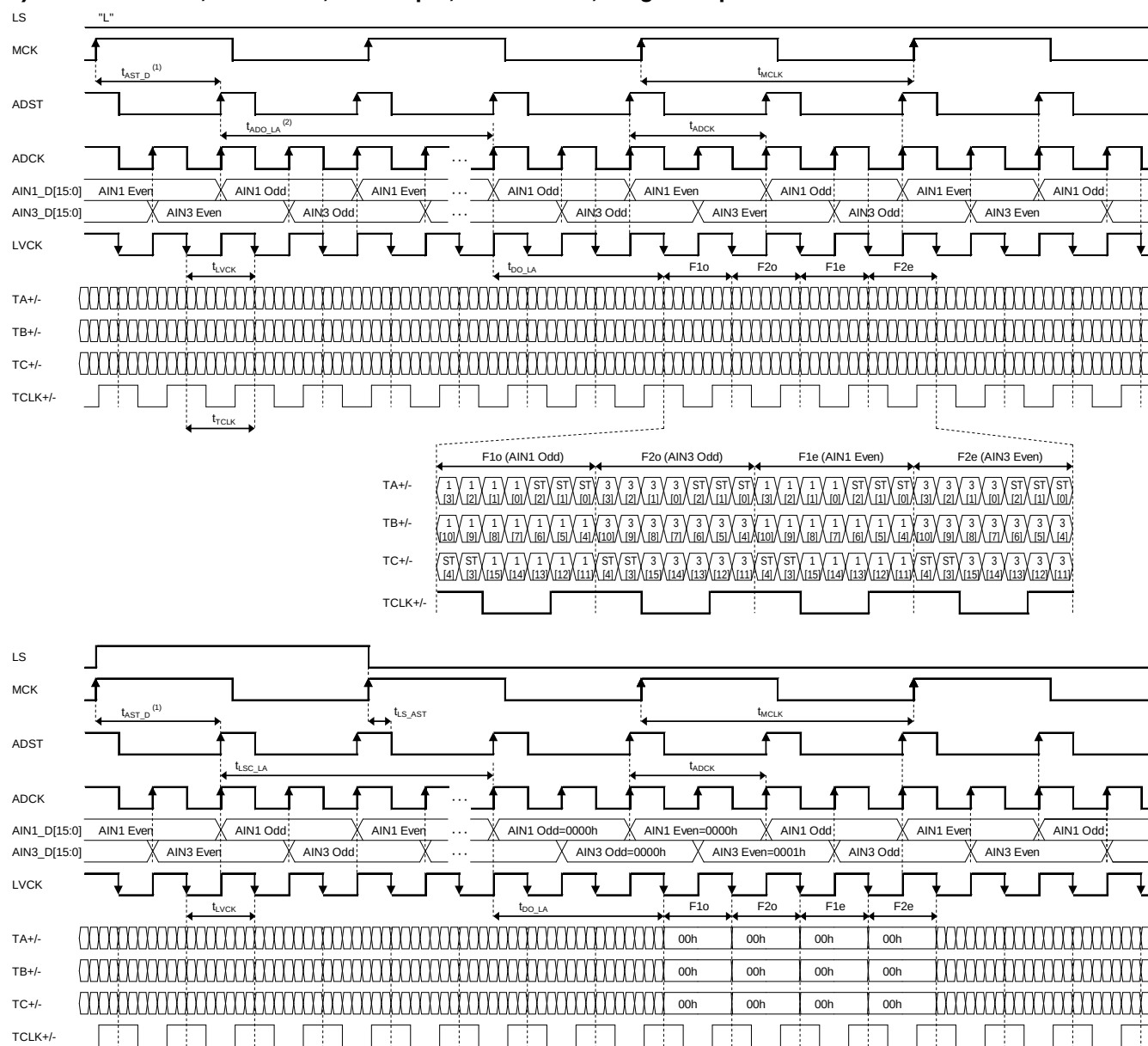
(1) $t_{\text{AST-D}}$: ADST delay from MCK, please refer the CDS/SH timing

(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

6) 7-Bit Serialize, 2 Channel, x2 Sample, 3 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t _{LVCK}	LVCK cycle period		t _{MCLK} · 1/4		ns
t _{TCLK}	TCCK cycle period		t _{MCLK} · 1/4		ns
t _{LS_AST}	Delay time from LS falling edge to AST falling edge	t _{MCLK} · 2/48		t _{MCLK} · 25/48	ns
t _{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t _{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

- (1) $t_{\text{AST_D}}$: ADST delay from MCK, please refer the CDS/SH timing
- (2) $t_{\text{ADO_LA}}$: ADC data output latency, please refer the CDS/SH timing
- (3) Unit Basis is ADCK
- (4) Unit Basis is LVCK

10.2. 7 Bit Serialize Mode (LVDS 2 Pair Output)

7-bit serialize mode with 2 pair LVDS supports only 3ch input mode. If you need 2 or 4ch input, please apply other mode, 7-bit serialize with 3 pair LVDS or 8bit serialize.

In this mode, the top bit of output data is the header flags for identifying the first data for each packet. Table 16 shows the header flag code and its description. H0 means the code of TA, and H1 is TB. And this mode supports line synchronized code. The output data bits are masked to all "0" with "LS". Therefore, main system can recognize start position of each line by checking output code. Line synchronize code is shown in Table 17.

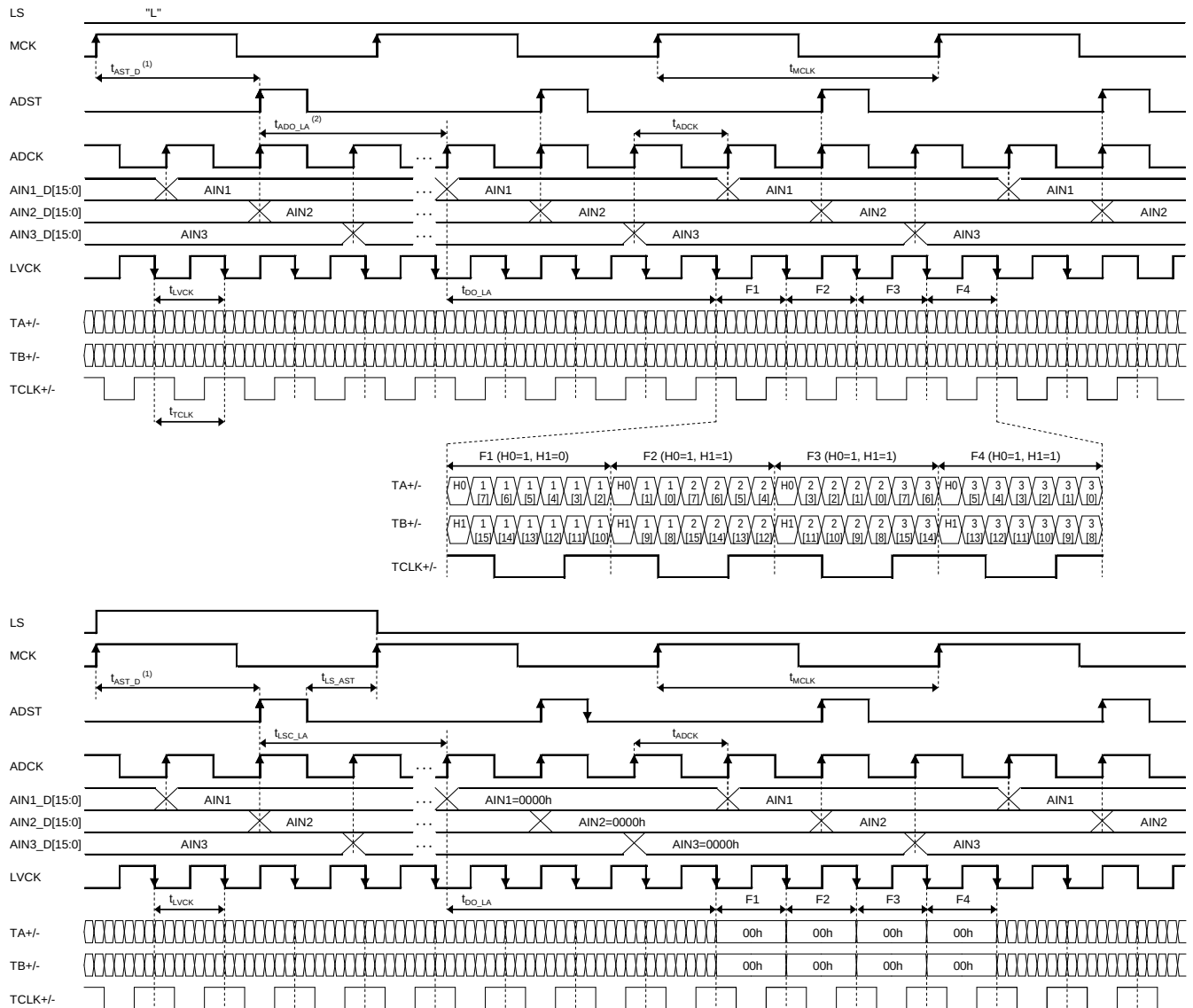
Table 15. Header Flag for 7-Bit Serialize Mode Output

HEADER		DESCRIPTION
H0	H1	
0	0	Line Synchronous
0	1	Reserved
1	0	1st frame data (F1, F1a, F1b and F1o)
1	1	Other frame data

Table 16. Line Synchronize Code for 7-Bit Serialize Mode Output

DATA	DESCRIPTION
0000h	Line synchronize code
0001h-0002h	Reserved
0003h-FFFFh	Actual picture data

1) 7 Bit Serialize, 3 Channel, ×1 Sample, 2 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{AST_LS}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot -13/48$		$t_{MCLK} \cdot 35/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		3.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

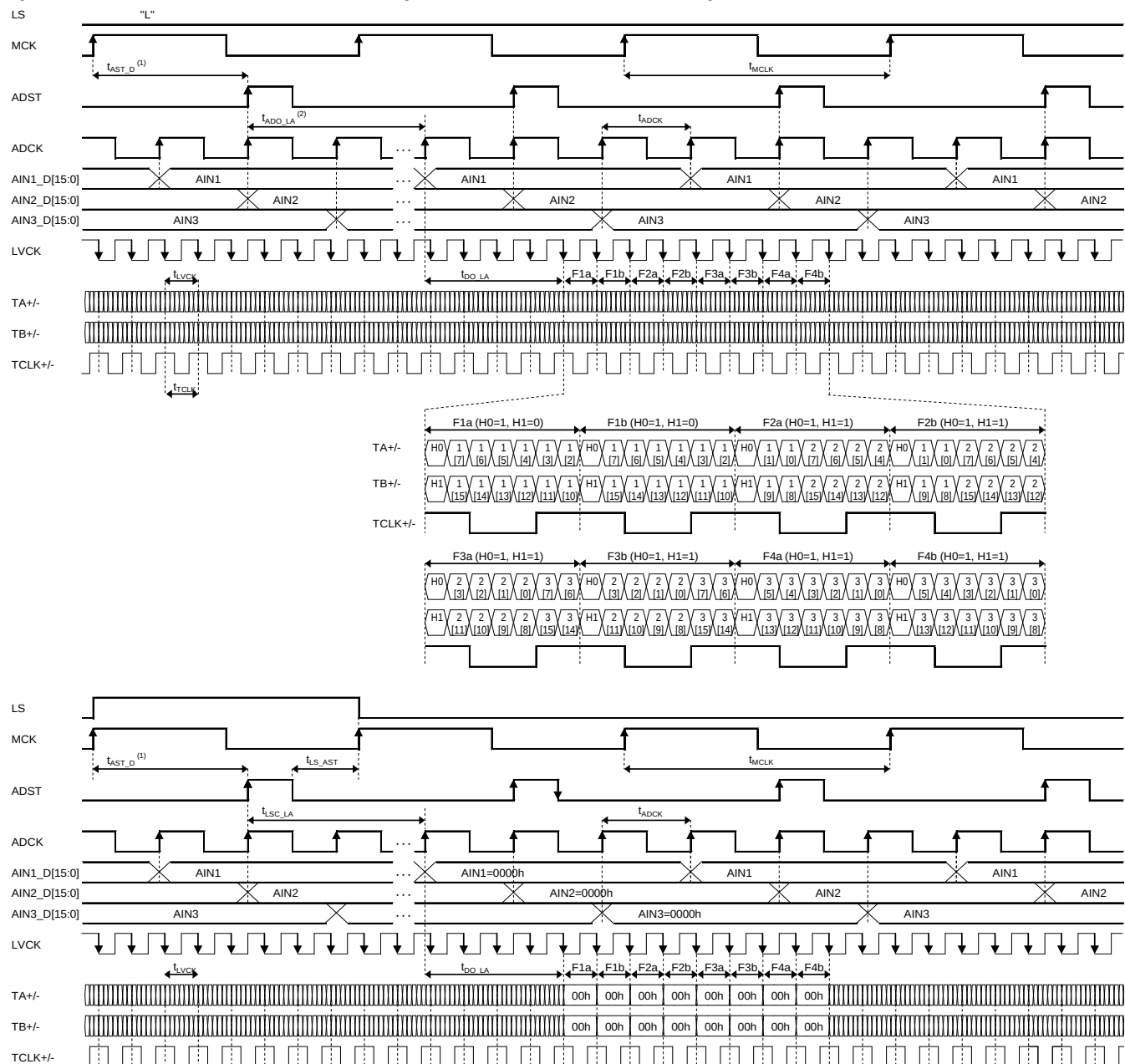
(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

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2) 7-Bit Serialize, 3 Channel, ×1 Sample, 2 Pair LVDS, Double Output



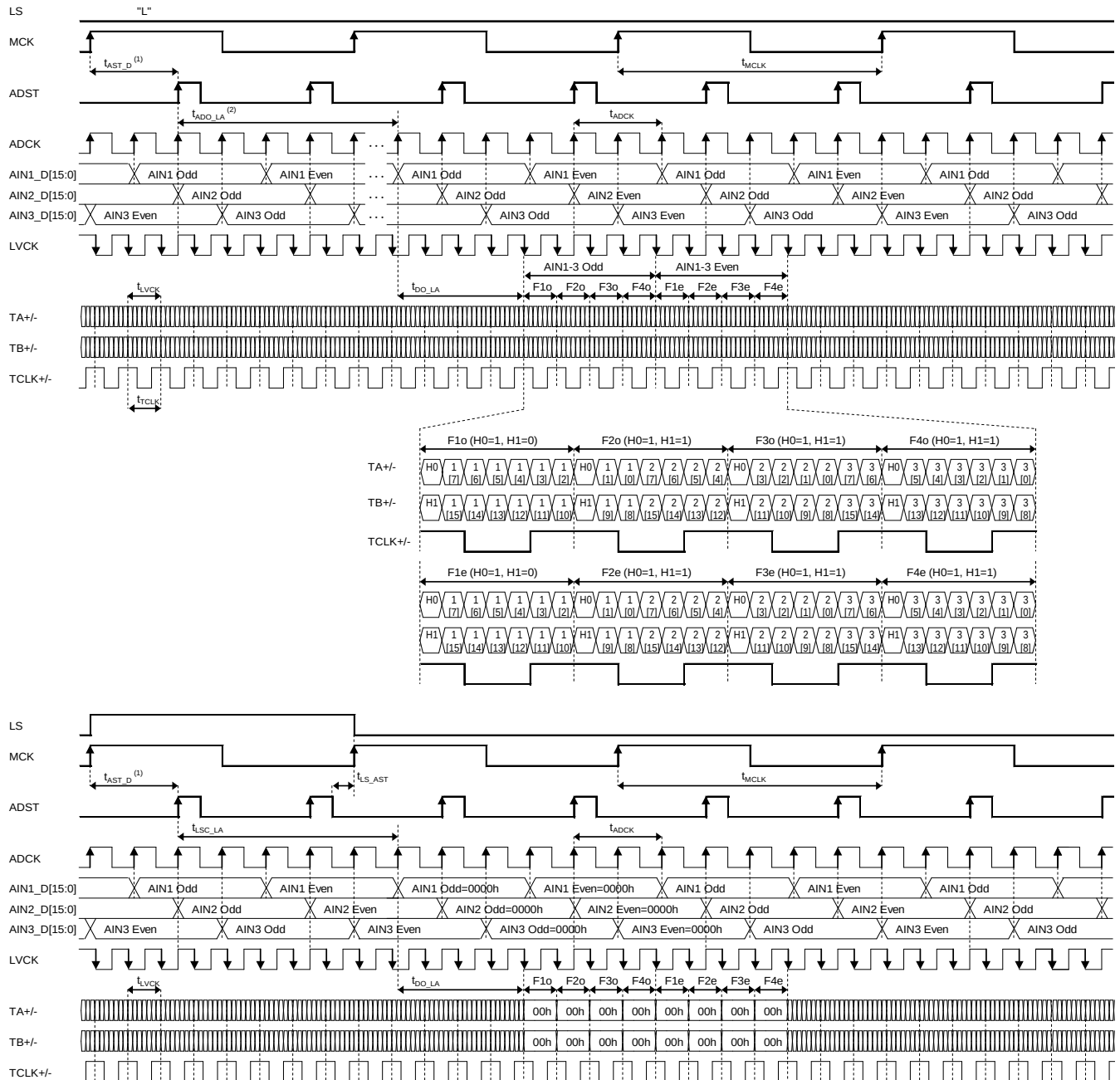
SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{AST_LS}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot -13/48$		$t_{MCLK} \cdot 35/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		3.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

3) 7-Bit Serialize, 3 Channel, x2 Sample, 2 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{AST_LS}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot -6/48$		$t_{MCLK} \cdot 17/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		3.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

10.3. 8-Bit Serialize Mode Data Output

This mode can support 2, 3 and 4ch input. Since only TA and TB LVDS channels are used in 8-bit serialize mode, TC channel is always disabled. However, 1pair LVDS output with TA is supported in case of only 2ch input. TB buffer is automatically standby in 2ch input with 1pair LVDS output.

And this mode supports line synchronized code. The output data bits are masked to all "0" with "LS". Therefore, main system can recognize start position of each line by checking output code. Line synchronize code is shown in Table 18 and Table 19.

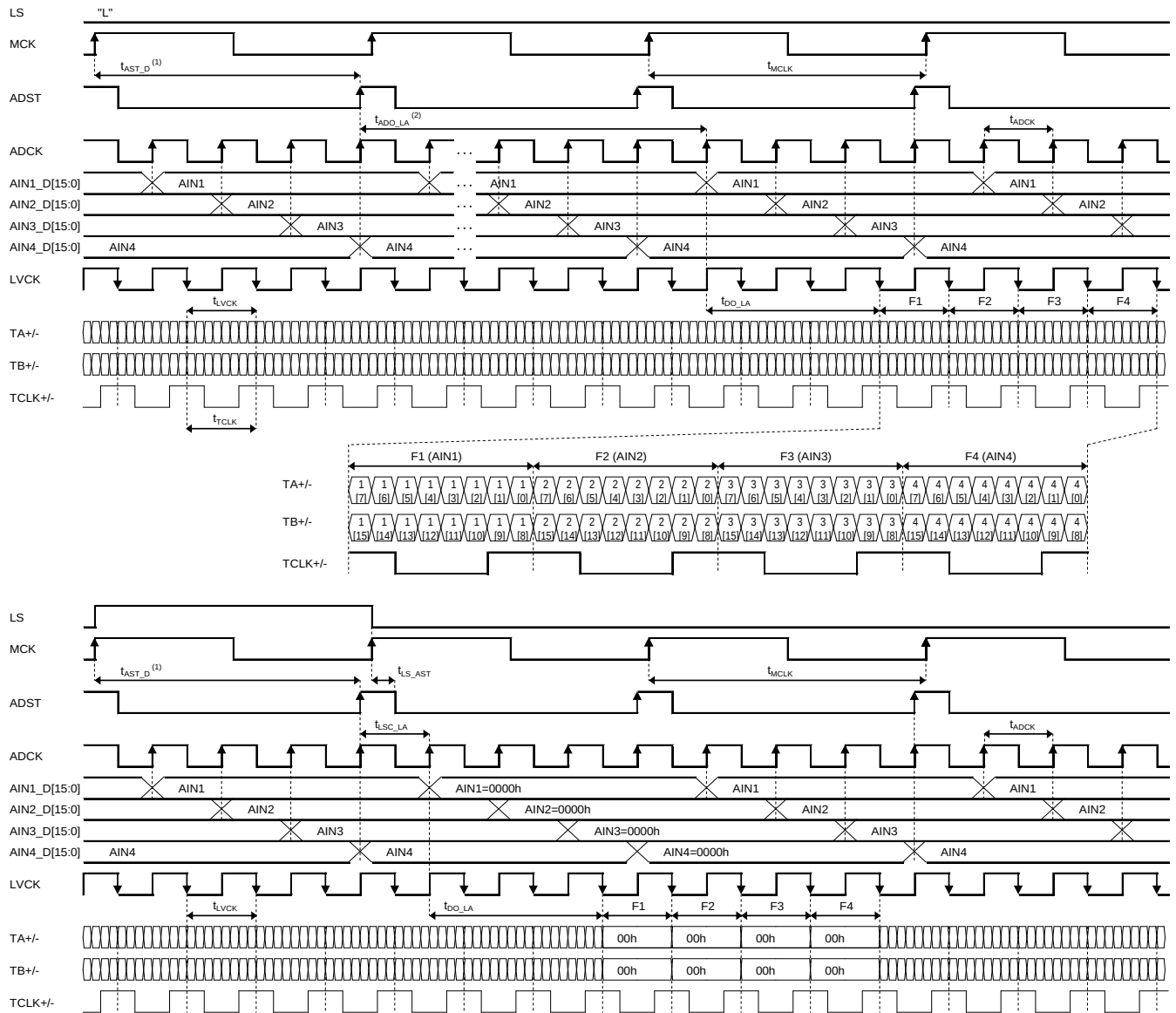
Table 17. Line Synchronize Code for 8-Bit Serialize Mode Output (4-Channel, 3-Channel Mode)

DATA	DESCRIPTION
0000h	Line synchronize code
0001h-0002h	Reserved
0003h-FFFFh	Actual picture data

Table 18. Line Synchronize Code for 8-Bit Serialize Mode Output (2-Channel Mode)

DATA	DESCRIPTION
0000h	Line synchronize code
0001h	End of line synchronize code
0002h	Reserved
0003h-FFFFh	Actual picture data

1) 8-Bit Serialize, 4 Channel, ×1 Sample, 2 Pair LVDS, Single Output

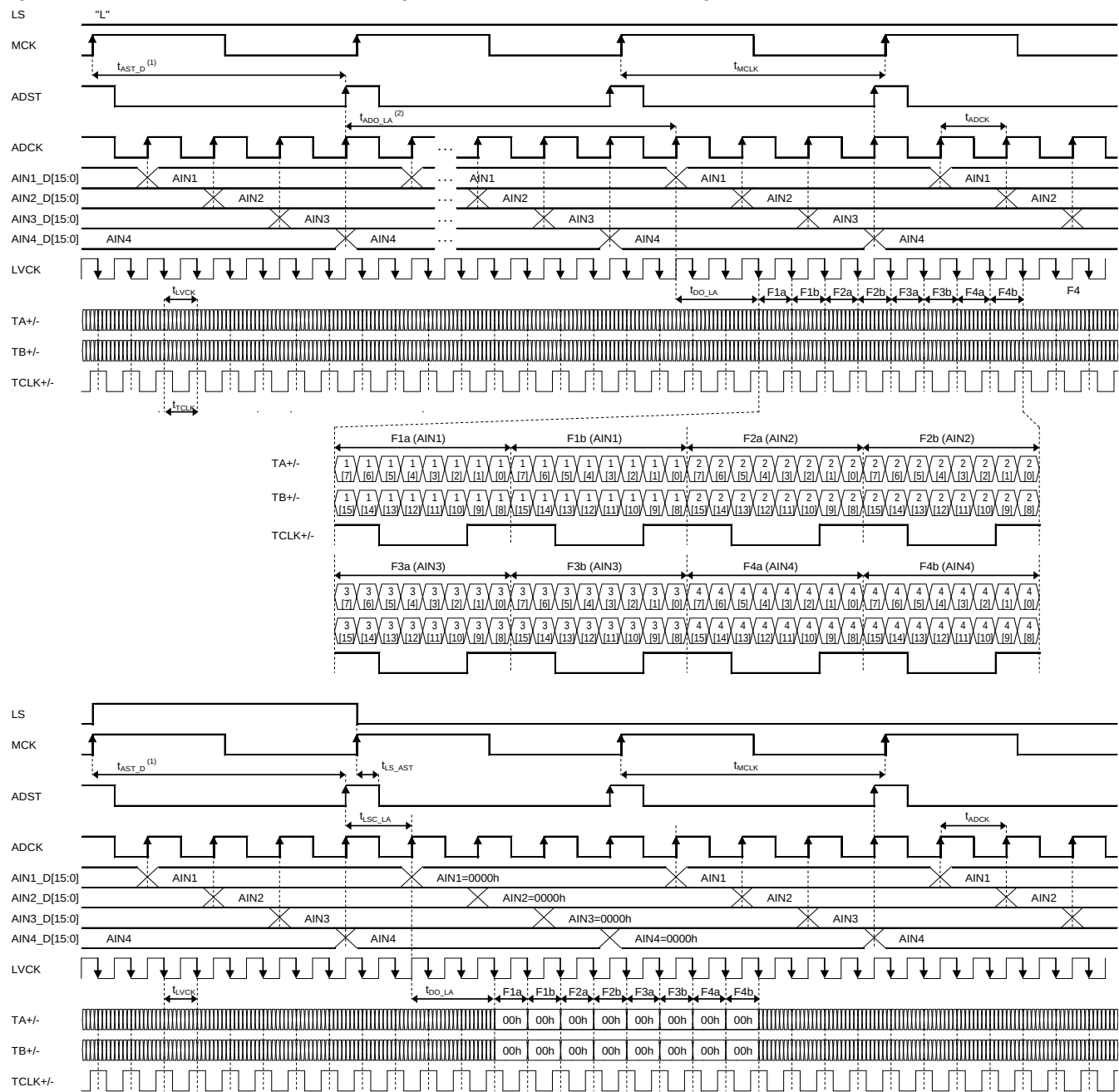


SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{LS_AST}	Delay time from LS falling edge to ADST falling edge	$t_{MCLK} \cdot 3/48$		$t_{MCLK} \cdot 50/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

- (1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing
(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing
(3) Unit Basis is ADCK
(4) Unit Basis is LVCK

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2) 8-Bit Serialize, 4 Channel, ×1 Sample, 2 Pair LVDS, Double Output



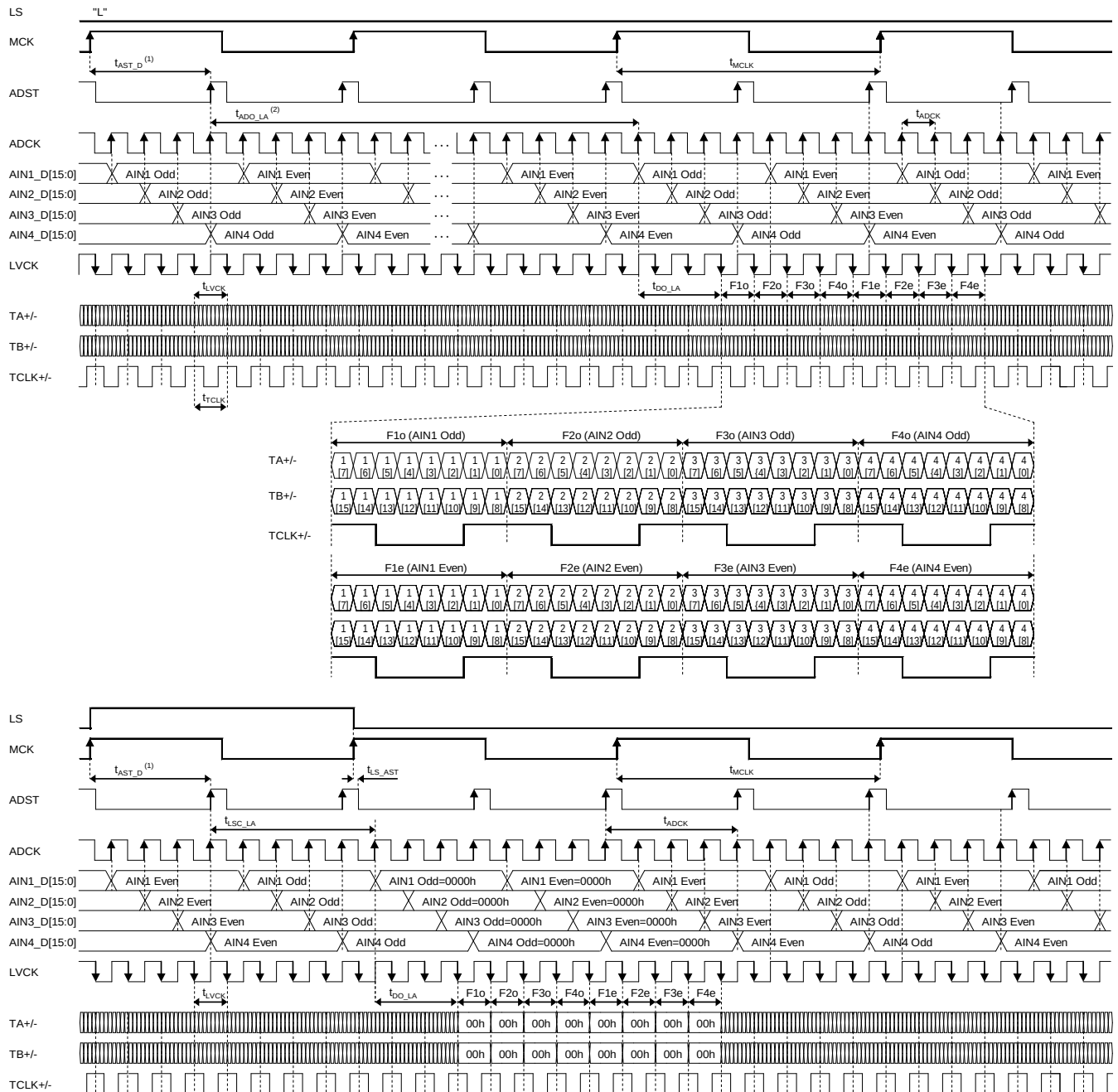
SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot 3/48$		$t_{MCLK} \cdot 50/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

3) 8-Bit Serialize, 4 Channel, x2 Sample, 2 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot 2/48$		$t_{MCLK} \cdot 25/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

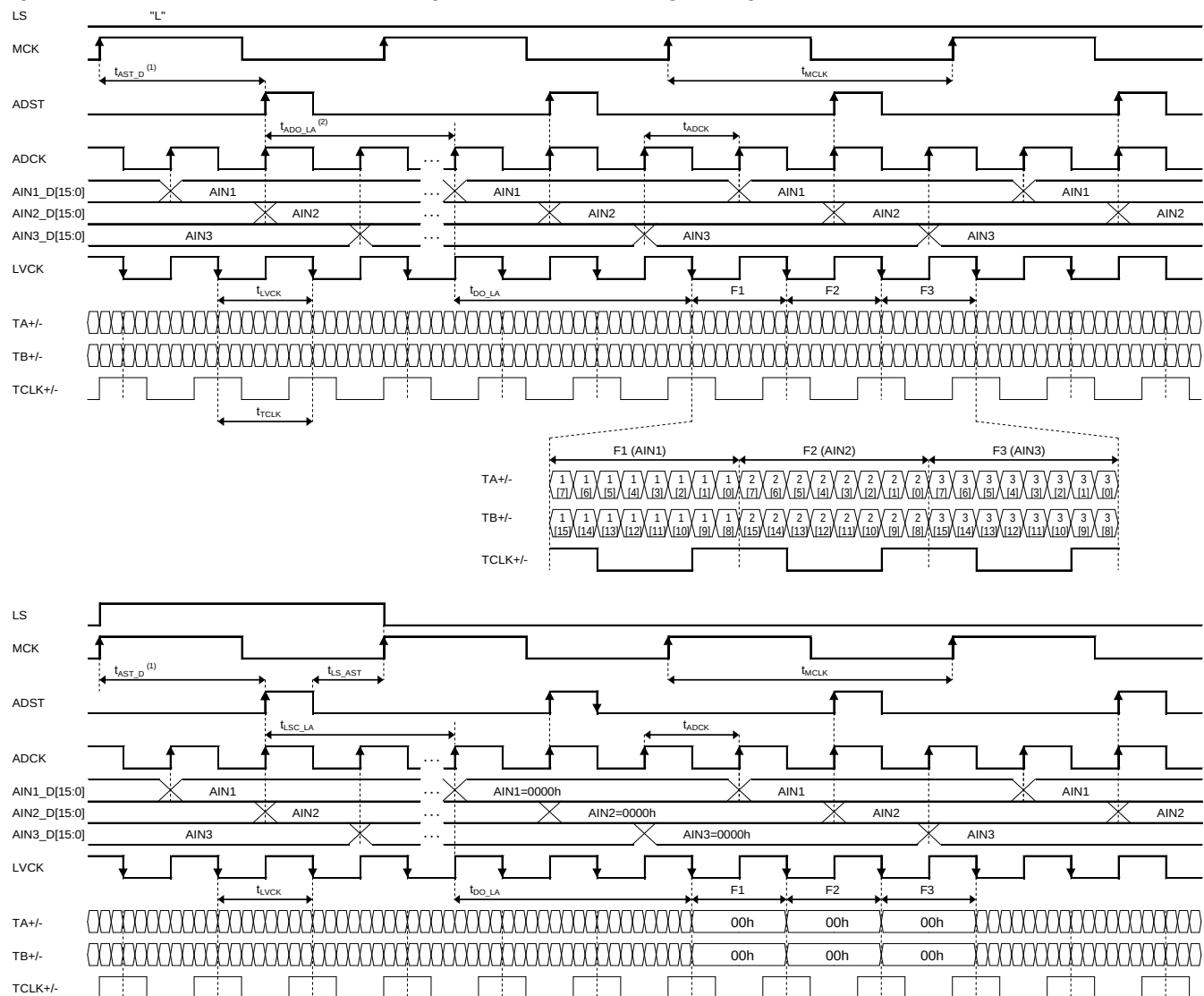
(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

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4) 8-Bit Serialize, 3 Channel, ×1 Sample, 2 Pair LVDS, Single Output



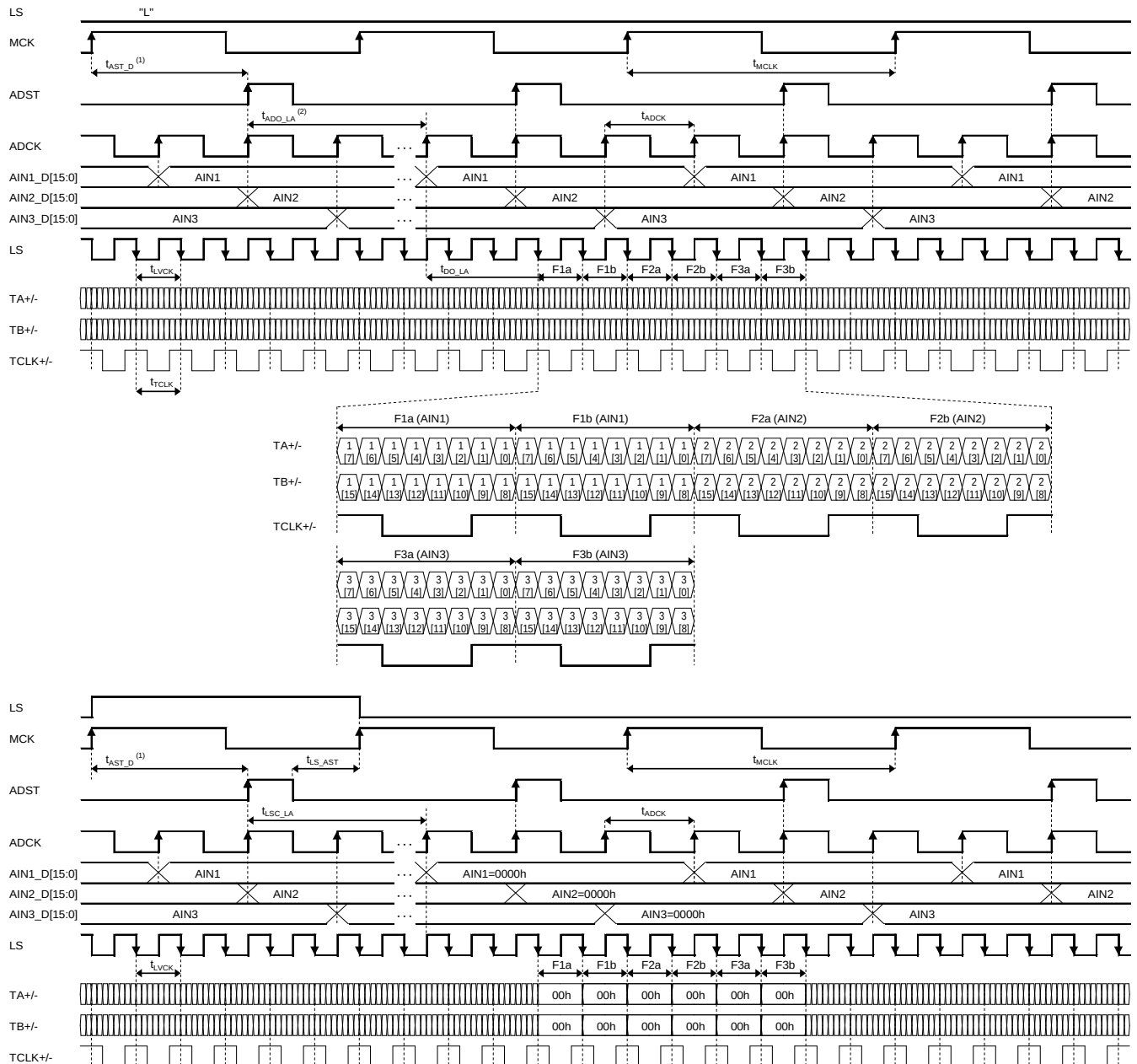
SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot \frac{1}{3}$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot \frac{1}{3}$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot -13/48$		$t_{MCLK} \cdot 35/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

5) 8-Bit Serialize, 3 Channel, ×1 Sample, 2 Pair LVDS, Double Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t _{LVCK}	LVCK cycle period		t _{MCLK} · 1/6		ns
t _{TCLK}	TCCK cycle period		t _{MCLK} · 1/6		ns
t _{LS_AST}	Delay time from LS falling edge to AST falling edge	t _{MCLK} · -13/48		t _{MCLK} · 35/48	ns
t _{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t _{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

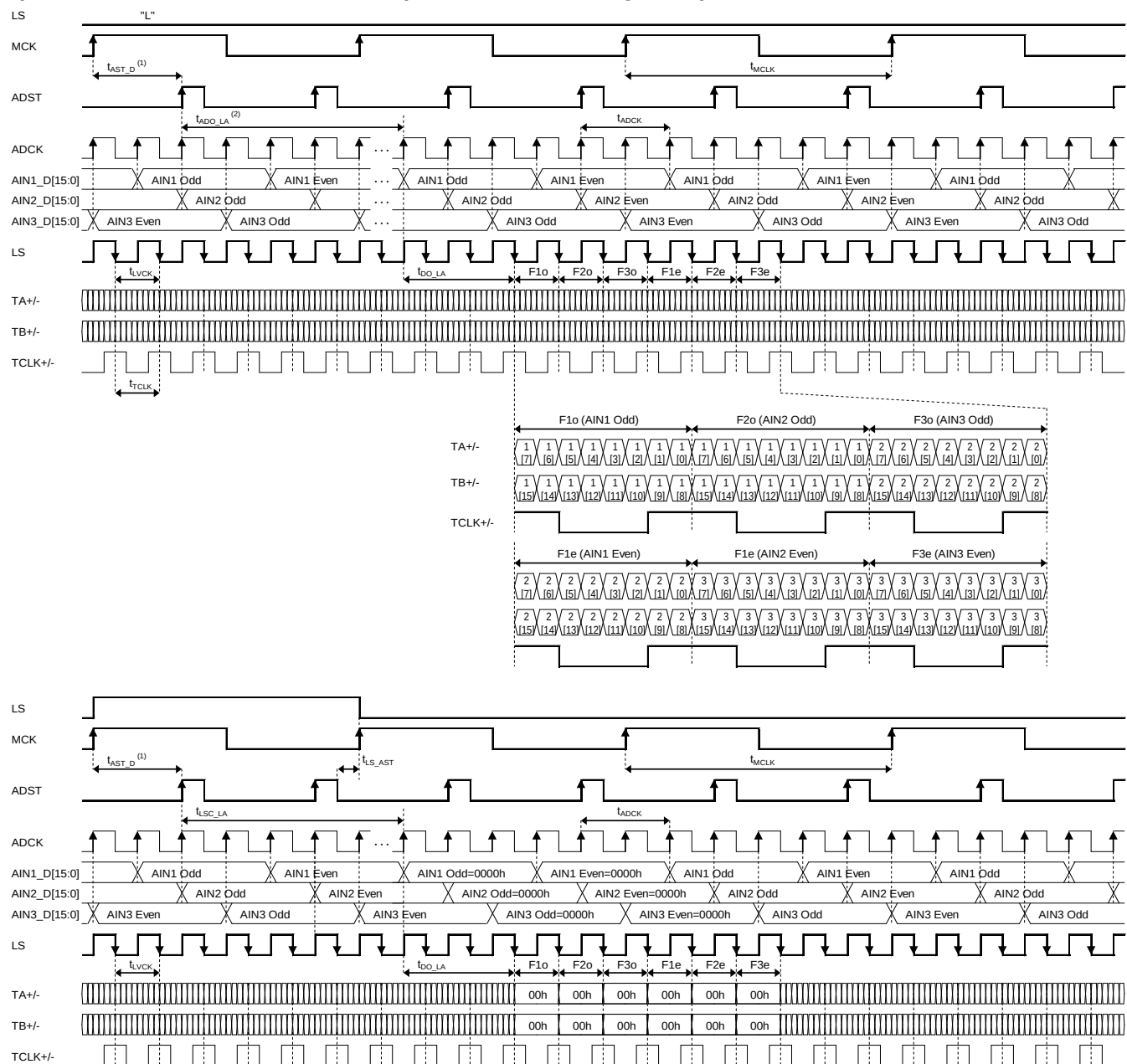
(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

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6) 8 Bit Serialize, 3 Channel × 2 Sample, 2 Pair LVDS, Single Output



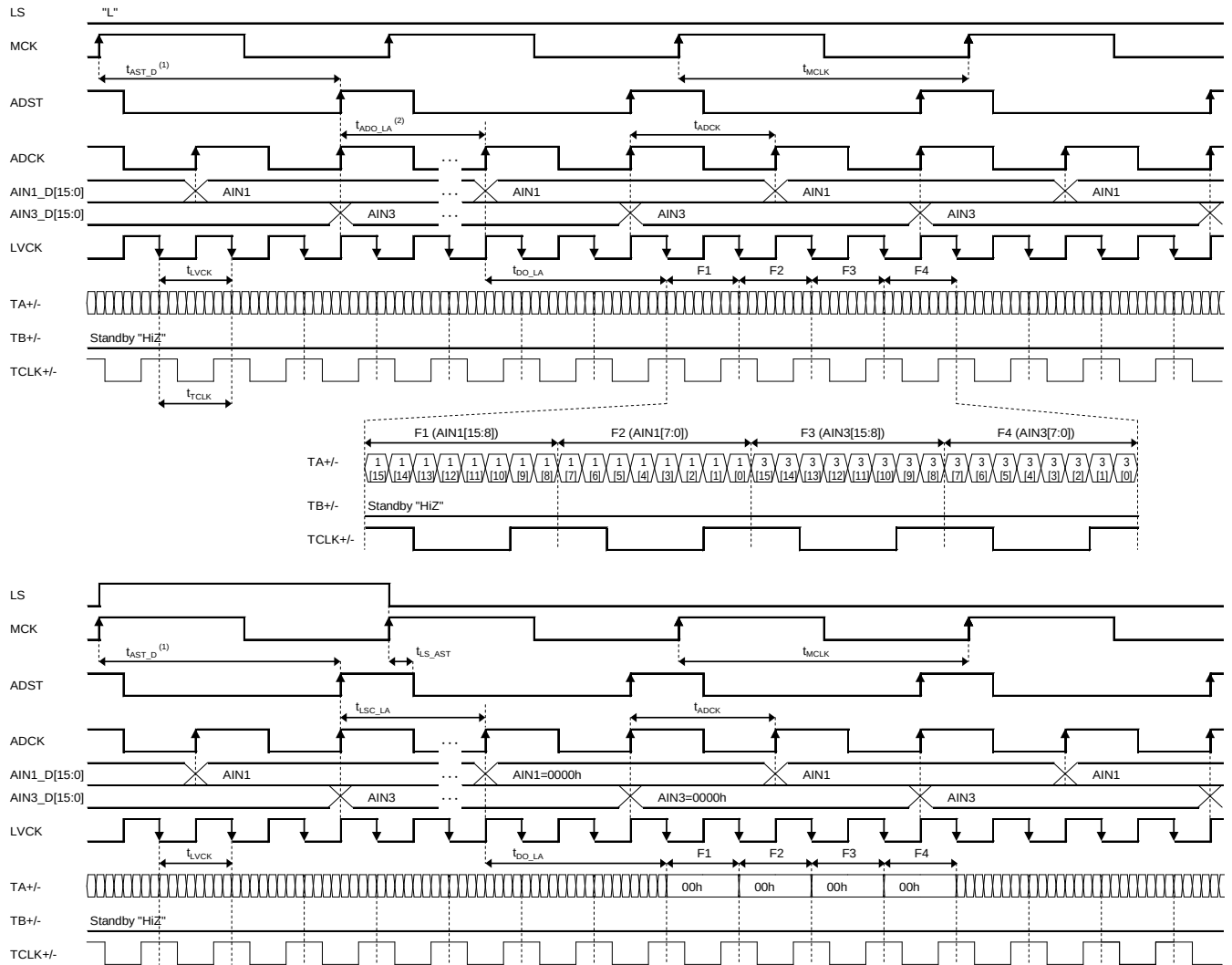
SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/6$		ns
t_{TCCK}	TCCK cycle period		$t_{MCLK} \cdot 1/6$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot -6/48$		$t_{MCLK} \cdot 17/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

7) 8-Bit Serialize, 2 Channel, ×1 Sample, 1 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{LS_AST}	Delay time from LS falling edge to ADST falling edge	$t_{MCLK} \cdot 3/48$		$t_{MCLK} \cdot 50/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

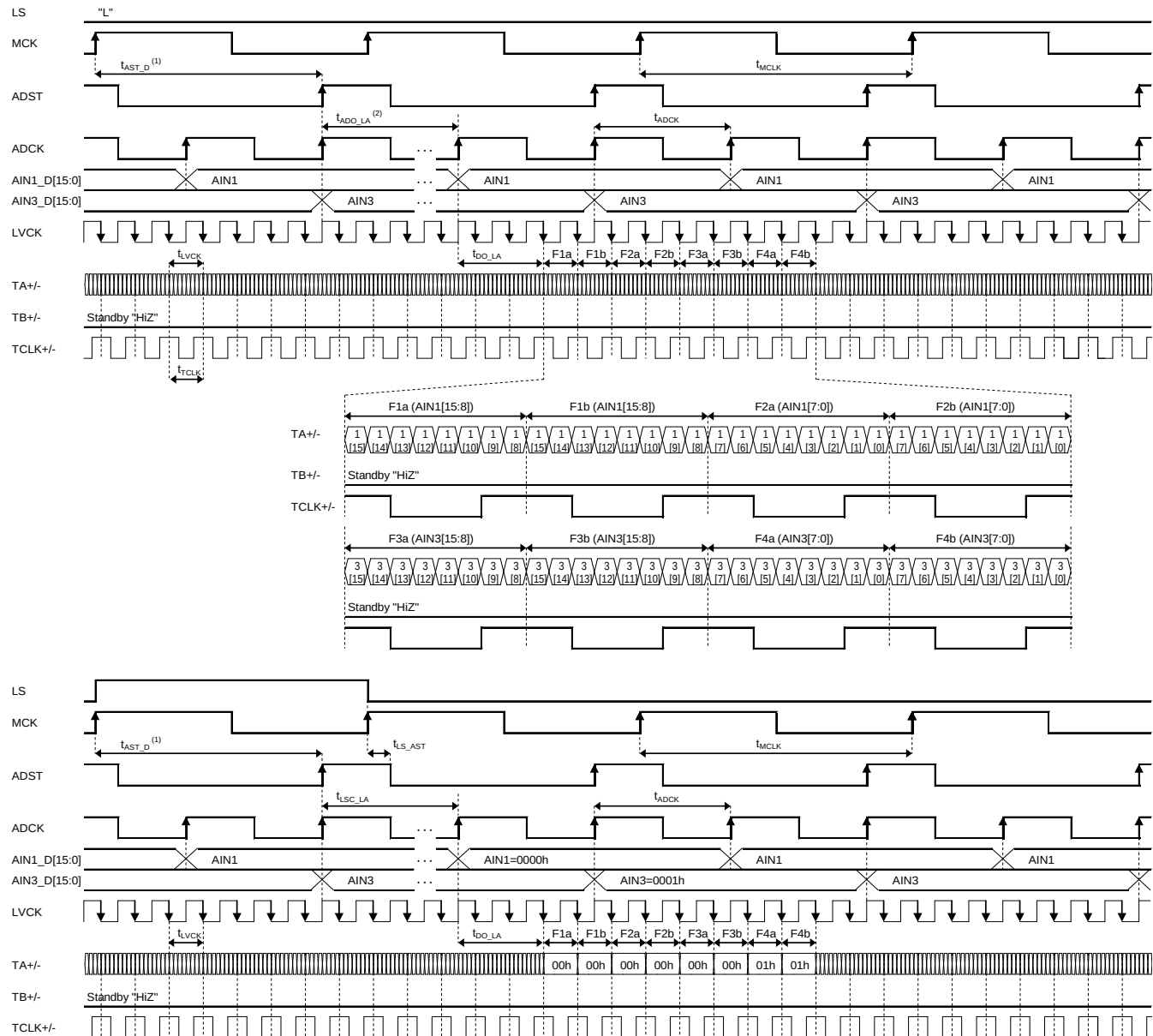
(2) t_{ADD_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

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8) 8-Bit Serialize, 2 Channel, ×1 Sample, 1 Pair LVDS, Double Output



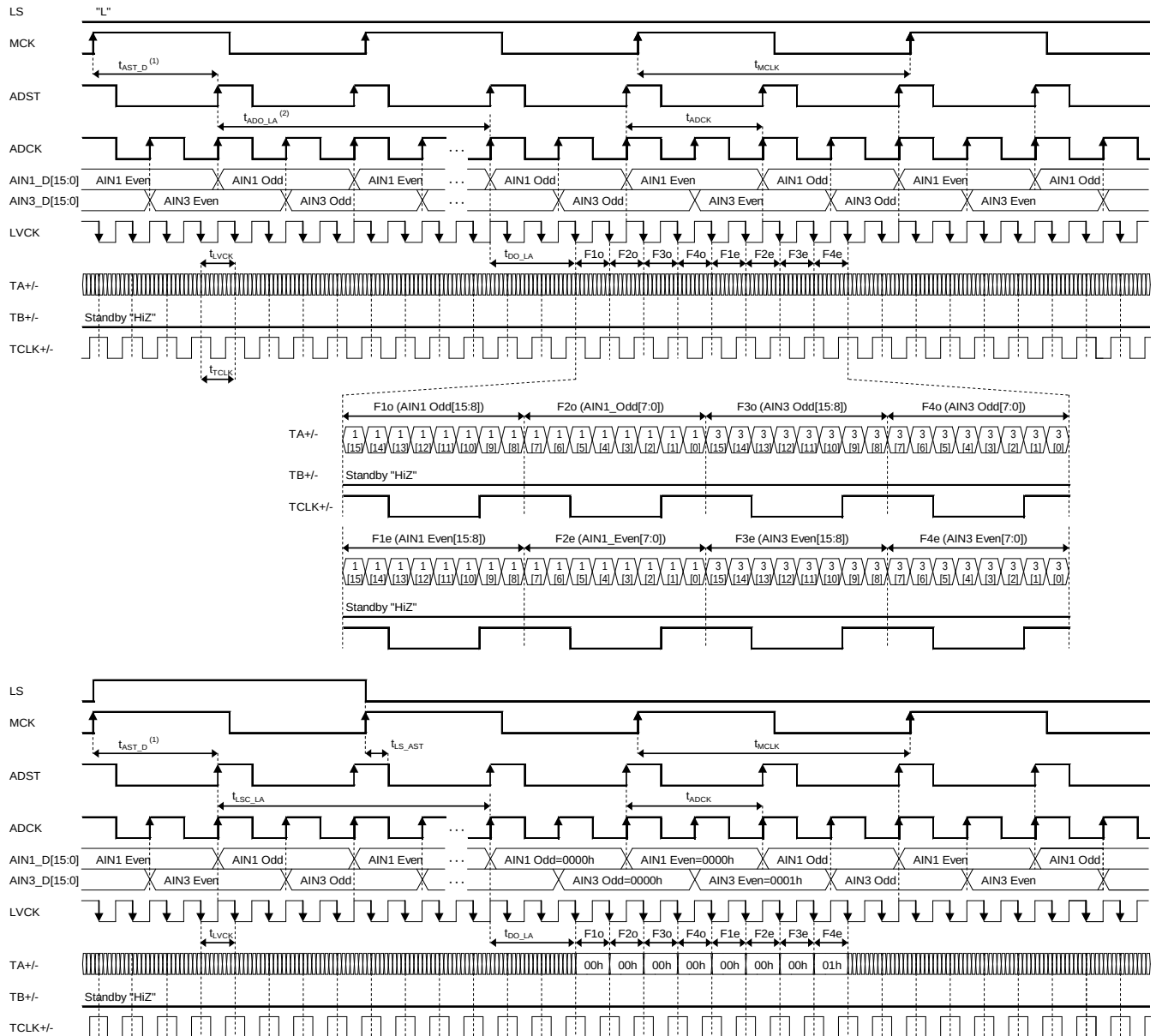
SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot 3/48$		$t_{MCLK} \cdot 50/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

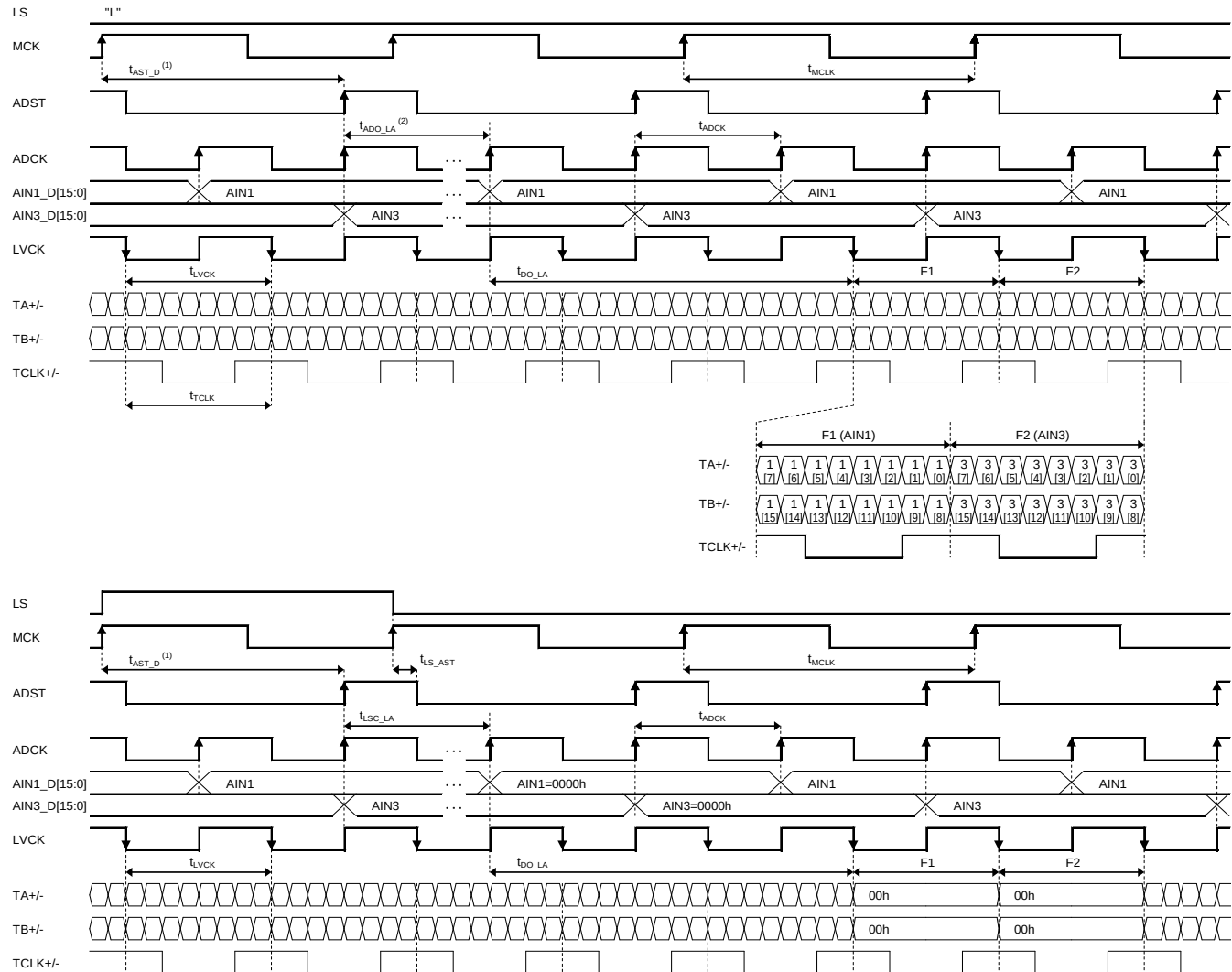
9) 8-Bit Serialize, 2 Channel, x2 Sample, 1 Pair LVDS, Single Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/8$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot 3/48$		$t_{MCLK} \cdot 26/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

- (1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing
(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing
(3) Unit Basis is ADCK
(4) Unit Basis is LVCK

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10) 8-Bit Serialize, 2 Channel, ×1 Sample, 2 Pair LVDS, Single Output

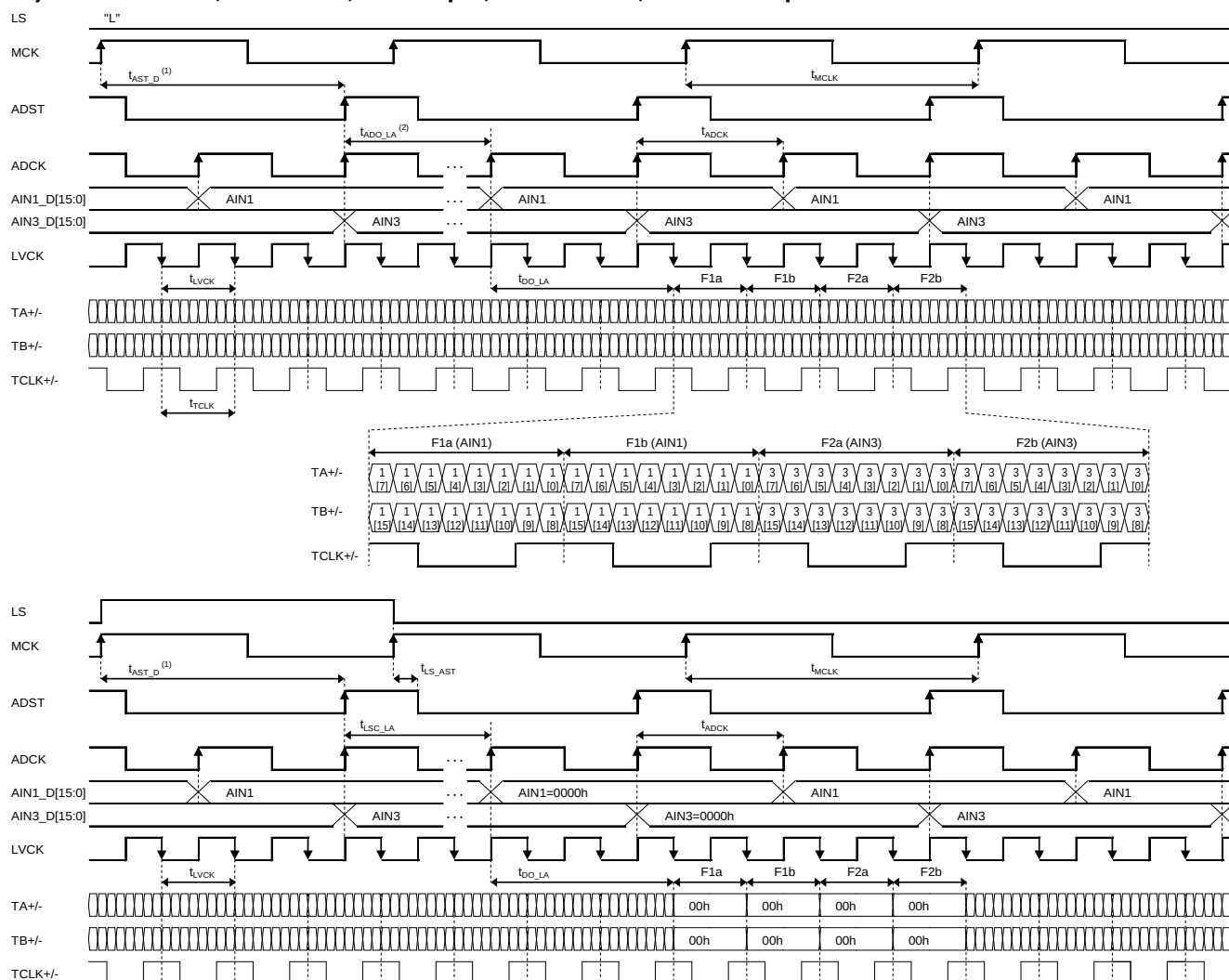
SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/2$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/2$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot 3/48$		$t_{MCLK} \cdot 50/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

11) 8-Bit Serialize, 2 Channel, ×1 Sample, 2 Pair LVDS, Double Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{TCLK}	TCLK cycle period		$t_{MCLK} \cdot 1/4$		ns
t_{LS_AST}	Delay time from LS falling edge to AST falling edge	$t_{MCLK} \cdot 3/48$		$t_{MCLK} \cdot 50/48$	ns
t_{LSC_LA}	LS code data latency		5		clock ⁽³⁾
t_{DO_LA}	LVDS data latency		2.5		clock ⁽⁴⁾

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

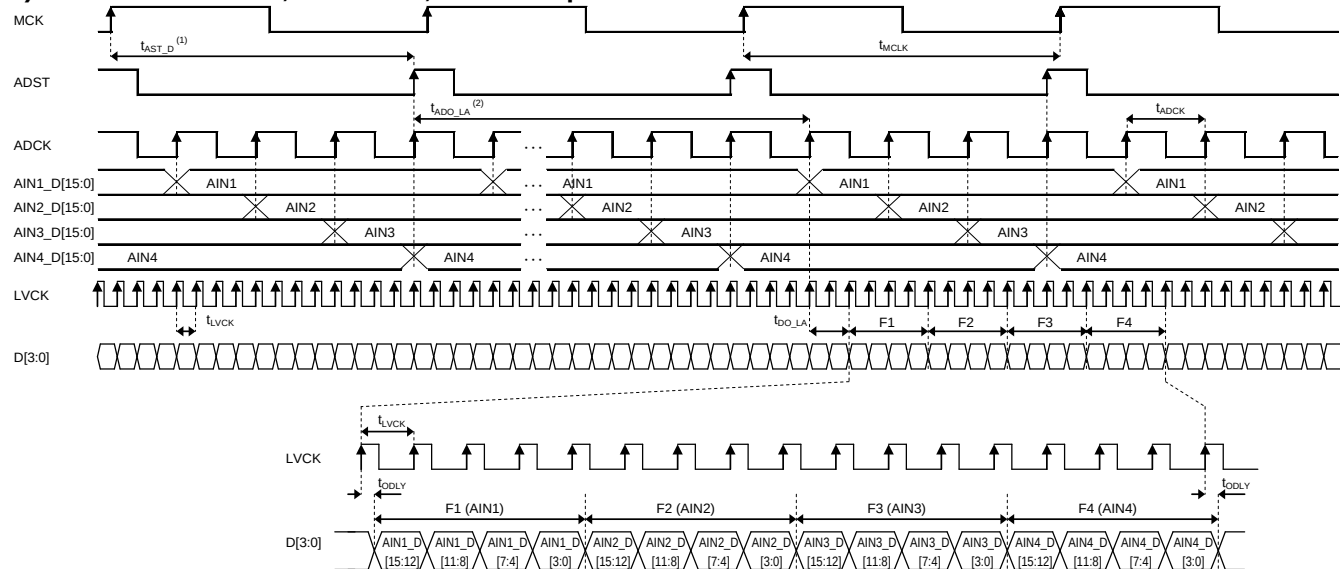
(3) Unit Basis is ADCK

(4) Unit Basis is LVCK

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10.4. 4-Bit CMOS Mode Data Output

1) 4-Bit CMOS Mode, 4 Channel, 16-Bit Output



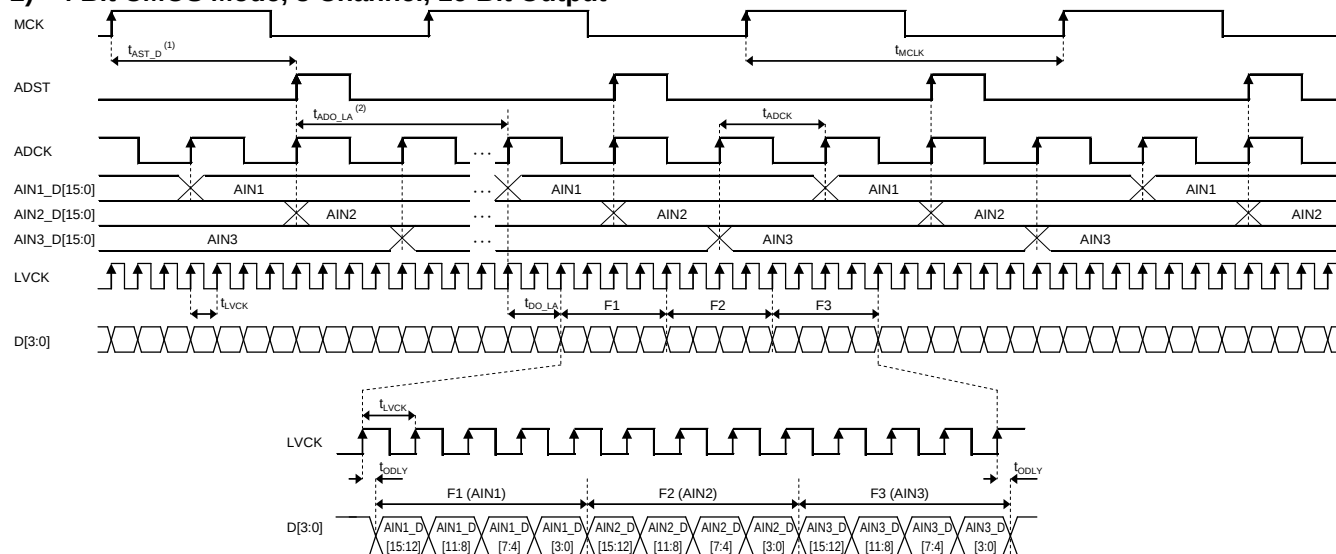
SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/16$		ns
	LVCK clock duty		33		%
t_{DO_LA}	Data output latency		2		clock ⁽³⁾
t_{ODLY}	Data output delay from LVCK		12		ns

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is LVCK

2) 4 Bit CMOS Mode, 3 Channel, 16-Bit Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/12$		ns
	LVCK clock duty		50		%
t_{DO_LA}	Data output latency		2		clock ⁽³⁾
t_{ODLY}	Data output delay from LVCK		12		ns

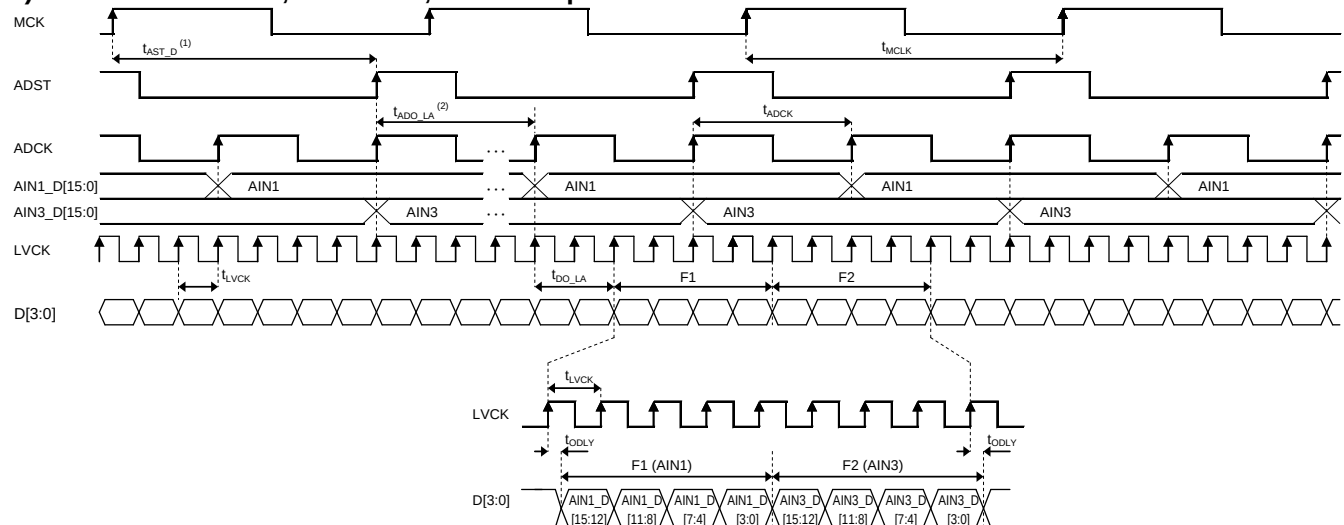
(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is LVCK

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3) 4-Bit CMOS Mode, 2 Channel, 16-Bit Output



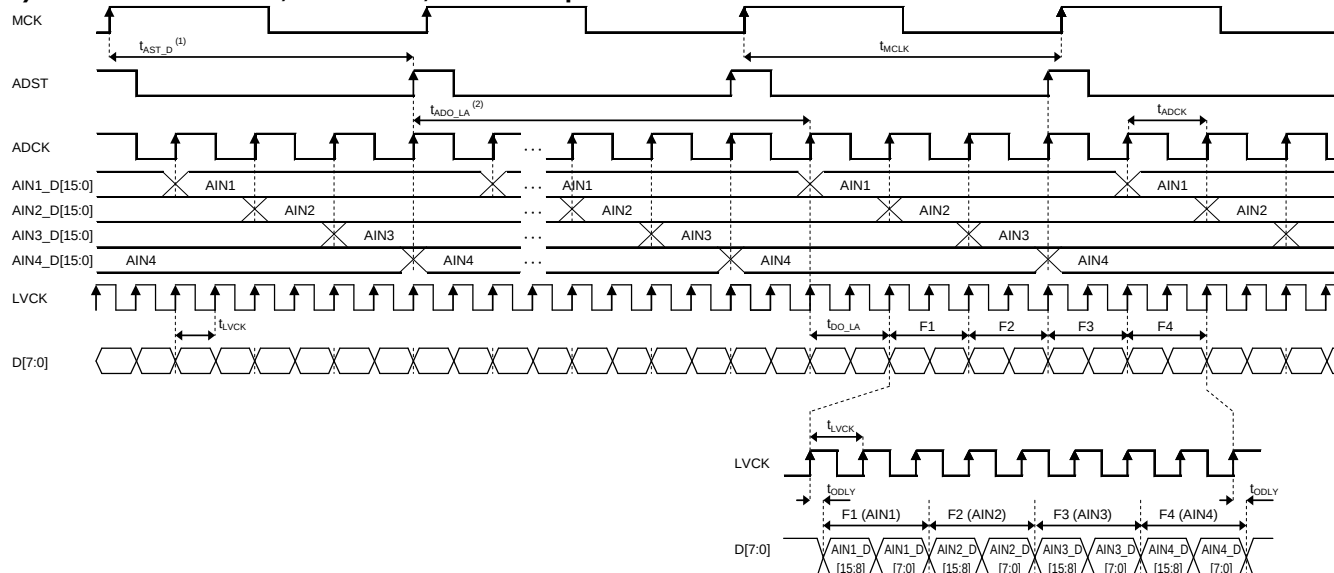
SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/8$		ns
	LVCK clock duty		50		%
t_{DO_LA}	Data output latency		2		clock ⁽³⁾
t_{ODLY}	Data output delay from LVCK		12		ns

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is LVCK

10.5. 8-Bit CMOS Mode Data Output

1) 8-Bit CMOS Mode, 4 Channel, 16-Bit Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/8$		ns
	LVCK clock duty		50		%
t_{DO_LA}	Data output latency		2		clock ⁽³⁾
t_{ODLY}	Data output delay from LVCK		12		ns

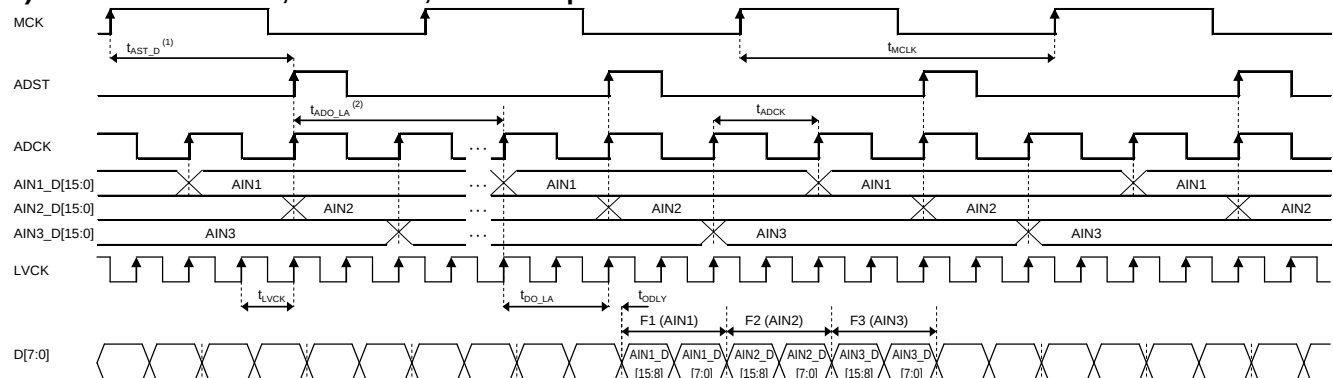
(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing

(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is LVCK

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2) 8-Bit CMOS Mode, 3 Channel, 16-Bit Output

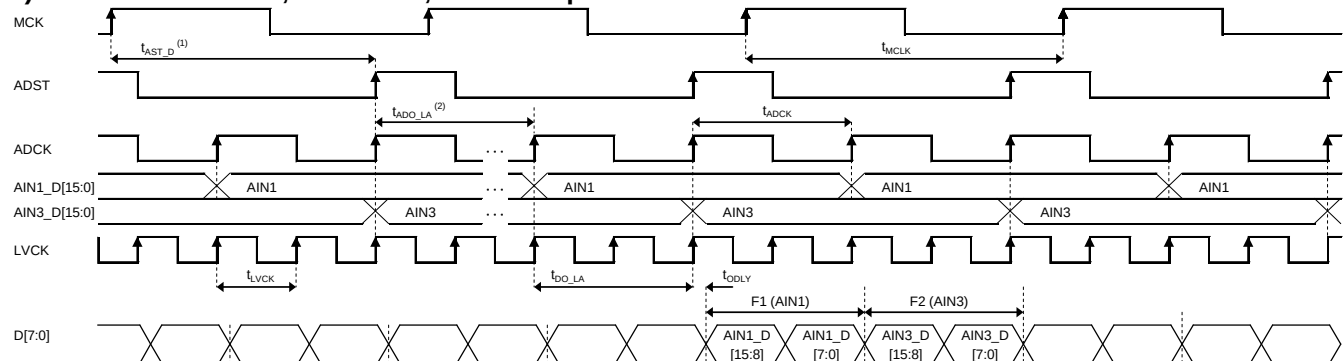


SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/6$		ns
	LVCK clock duty		50		%
t_{DO_LA}	Data output latency		2		clock ⁽³⁾
t_{ODLY}	Data output delay from LVCK		12		ns

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is LVCK

3) 8-Bit CMOS Mode, 2 Channel, 16-Bit Output



SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
t_{LVCK}	LVCK cycle period		$t_{MCLK} \cdot 1/4$		ns
	LVCK clock duty		50		%
t_{DO_LA}	Data output latency		2		clock ⁽³⁾
t_{ODLY}	Data output delay from LVCK		12		ns

(1) t_{AST_D} : ADST delay from MCK, please refer the CDS/SH timing(2) t_{ADO_LA} : ADC data output latency, please refer the CDS/SH timing

(3) Unit Basis is LVCK

11. TIMING GENERATOR

The TG (Timing Generator) for image sensor is incorporated into this product. The TG provides 6 signals as slow transfer clocks and 2 signals as fast transfer clocks. In addition, all fast clock signals can also be used as slow clock signals. All TG signals are synchronized with "LS", which is the line cycle for image sensor, and are completely controlled by internal registers. As the TG output is locked under default setting, EN_OUT (Add2, Bit10) should be set to "1" to enable the outputs.

11.1. Line Synchronous Function

VSP5640 has 2 modes for synchronizing sensor line cycle. One is internal line synchronous mode. In this mode, the line cycle signal "LS" is generated after a certain number of MCLK cycles (counted by an internal counter "PIX_CNT"). The number of MCLK cycles is determined by the register LINT[19:0] and the counter is cleared after LS is generated. The active period of "LS" is equal to one cycle period of MCLK.

Another is external line synchronous mode which need external signal "XLSYNC". In this mode, if logic circuit detects "XLSYNC" active period for more than 3 MCLK cycles, the internal line synchronous signal "LS" is generated. This mode has a function to mask "XLSYNC" for avoiding noise interference. The duration of the mask for "XLSYNC" can be set by register "LINT[19:0]" which is also used in the internal line synchronous mode.

The line synchronous modes and their polarity can be selected by registers "XLSYNC_SEL" and "XLSYNC_POL" respectively. The default settings are external mode and active high polarity. XLSYNC can be used to output some internal signals. Table 20 shows the register setting required to select the desired output signals.

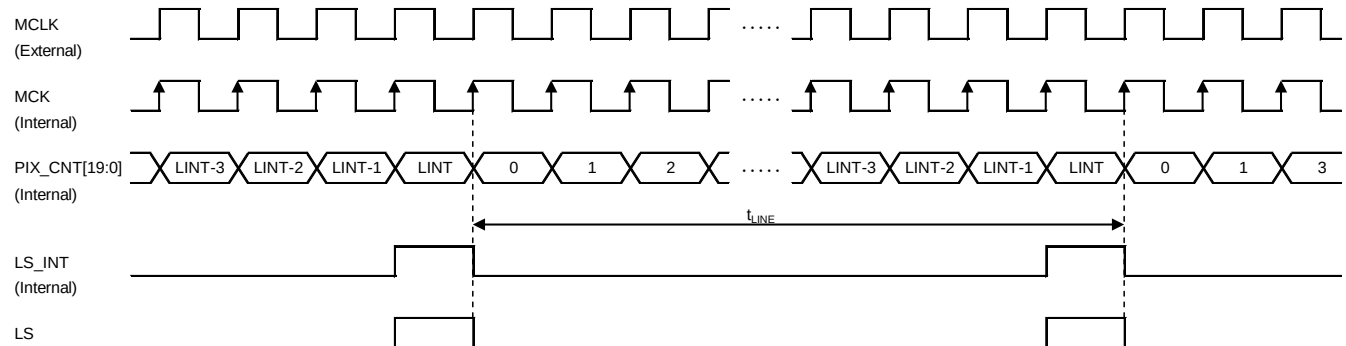
PIX_CNT can be mandatorily reset by "LS_CNT_RST" which is internal register. Before doing this function, you need to execute the software reset which "RST_ALL" sets to "1". If "LS_CNT_RST" sets to "1" after software reset, then pixel counter is held to "0". For making the counter active, "LS_CNT_RST" should return to "0".

Table 19. "XLSYNC" output signal (XLSYNC_SEL = "1")

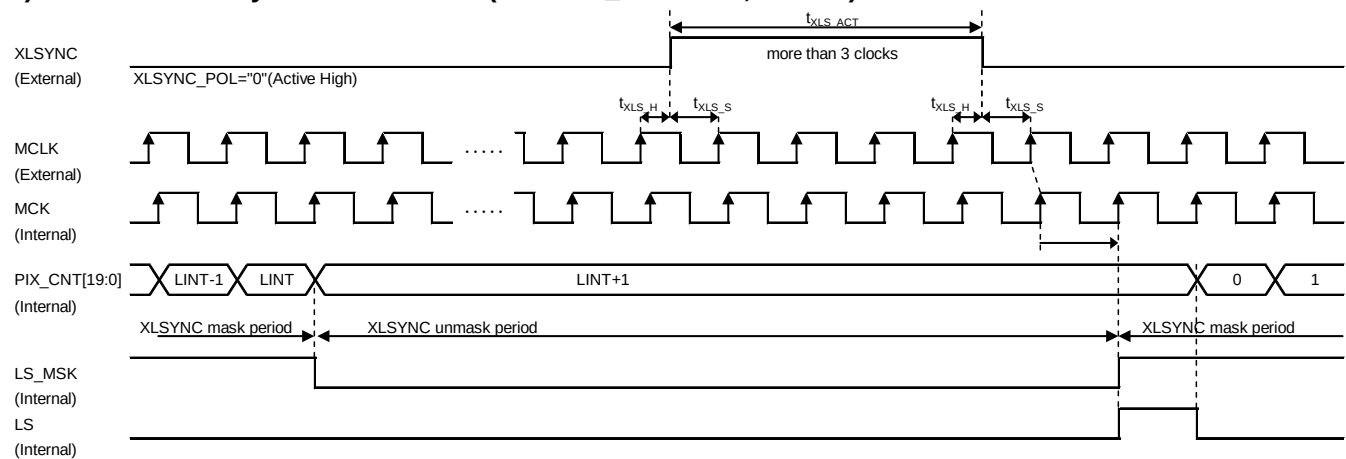
REGISTER SETTING XLSYNC_OUT	OUTPUT SIGNAL
0	LS
1	CLPDM
2	Reserved
3	Reserved

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1) Internal Line Synchronous Mode (XLSYNC_SEL = "1")



2) External Line Synchronous Mode (XLSYNC_SEL = "0", Default)



SYMBOL	PARAMETER	CONDITION	MIN	TYP	MAX	UNIT
t _{LINE}	Line cycle period setting	XLSYNC = "1"	3	LINT + 1	2 ²⁰ - 1	clock
t _{XLS_ACT}	XLSYNC active period	XLSYNC = "0"	3			clock
t _{XLS_S}	XLSYNC setup to MCLK	XLSYNC = "0"	10			ns
t _{XLS_H}	XLSYNC hold to MCLK	XLSYNC = "0"	10			ns

11.2. Slow Transfer Clock Setting (XST)

XST is slow transfer clocks that can be configured by setting initial polarity and toggle points. Table 21 shows the number of toggle setting and generated pulse. Since the two toggles generate one pulse, the number of pulses is half the number of toggles.

Table 20. Toggle number and generated pulse

SIGNAL	TOGGLE	PULSE
XST	8	4

Each toggle position is defined by a register that is exclusive for each signal. The toggle position is synchronized with "LS" and the gap between toggle-position and "LS" falling edge defined in terms of t_{MCLK} , the cycle period of "MCLK". This gap is set by register settings and is defined by the following expression:

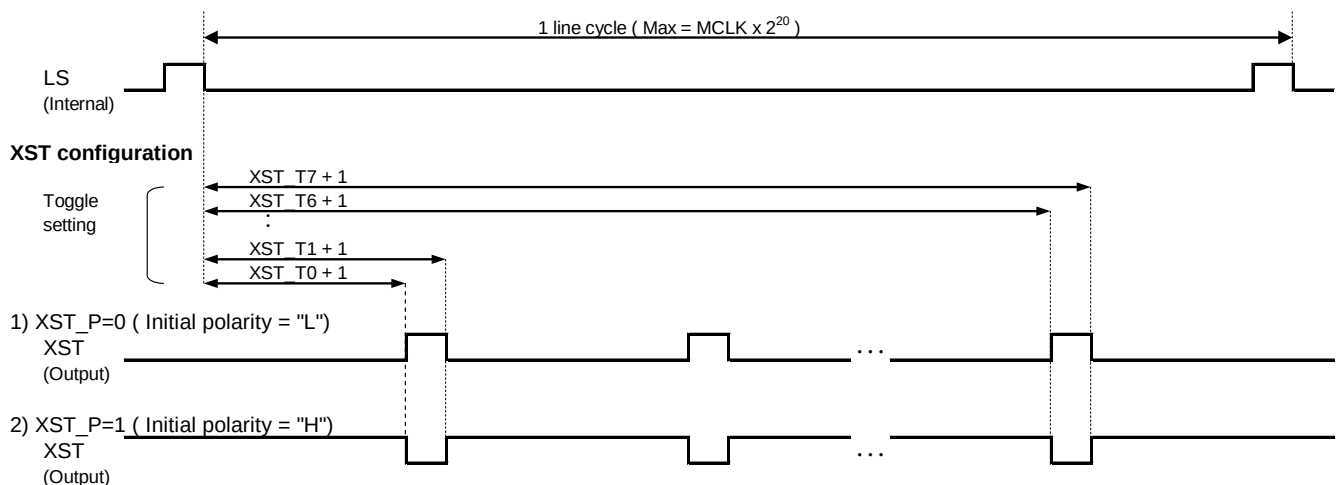
$$t = (XST_T(k) + 1) \times t_{MCLK} \quad (XST_T(k) : \text{Register value of toggle setting})$$

Note:

- (1) $k = 0$ to 7
- (2) $XST_T(k) < LINT$

The toggle for "XST" signal can be disabled with register settings. To make toggle active, TG_OUT_EN should be set to "1". However, as "XST" shares a pin with "GPIO3", pin function should be configured using the register "GPIO3_XST_SEL".

Figure 20 shows the configuration regarding slow transfer clock.



Note:

- (1) If XST_Tn is set to "0" except XST_T0 , toggle position is ignored
- (2) The period between toggle position and LS falling edge = $(XST_T(k) + 1) \times t_{MCLK}$
- (3) The following requirement should be satisfied: $XST_T(k) < XST_T(k+1)$
- (4) The signal is set to the desired polarity settings at the falling edge of "LS".

Figure 21. Slow Transfer Gate Signal Setting for "XST"

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11.3. Fast Transfer Clock Pulse Setting

XP1 and X1L are fast transfer clock signals with configurable rising and falling edges using register settings. Figure 21 shows the block diagram of the fast clock configuration. In this figure, "DLL tap selector" is used to select both the rise and the fall edges of each signal from among 48 tap positions.

In addition, it is possible to change the clock rate of each signal with register settings. The clock rate is based on the frequency of "MCLK". "XP1" can select x1, x2 or x4 modes with common settings. "X1L" can choose between x1 and x2 rate settings.

Note: Two independent set of registers are available to set the clock rate, the rise edge and the fall edge of clocks for operation in x1-mode and x2-mode.

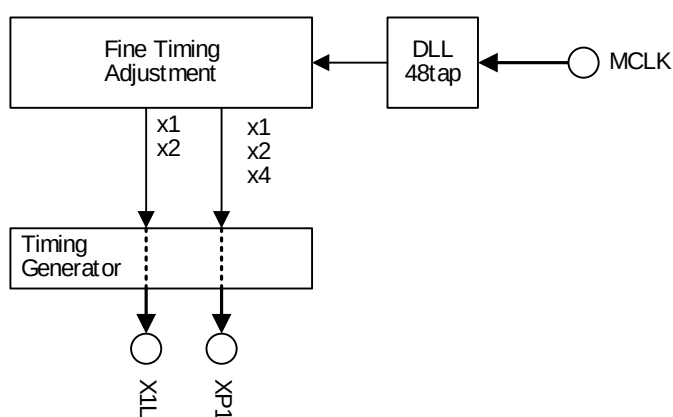


Figure 22. Fast Transfer Clock Pulse Generator

1) Fast Transfer Clock Pulse for “XP1”

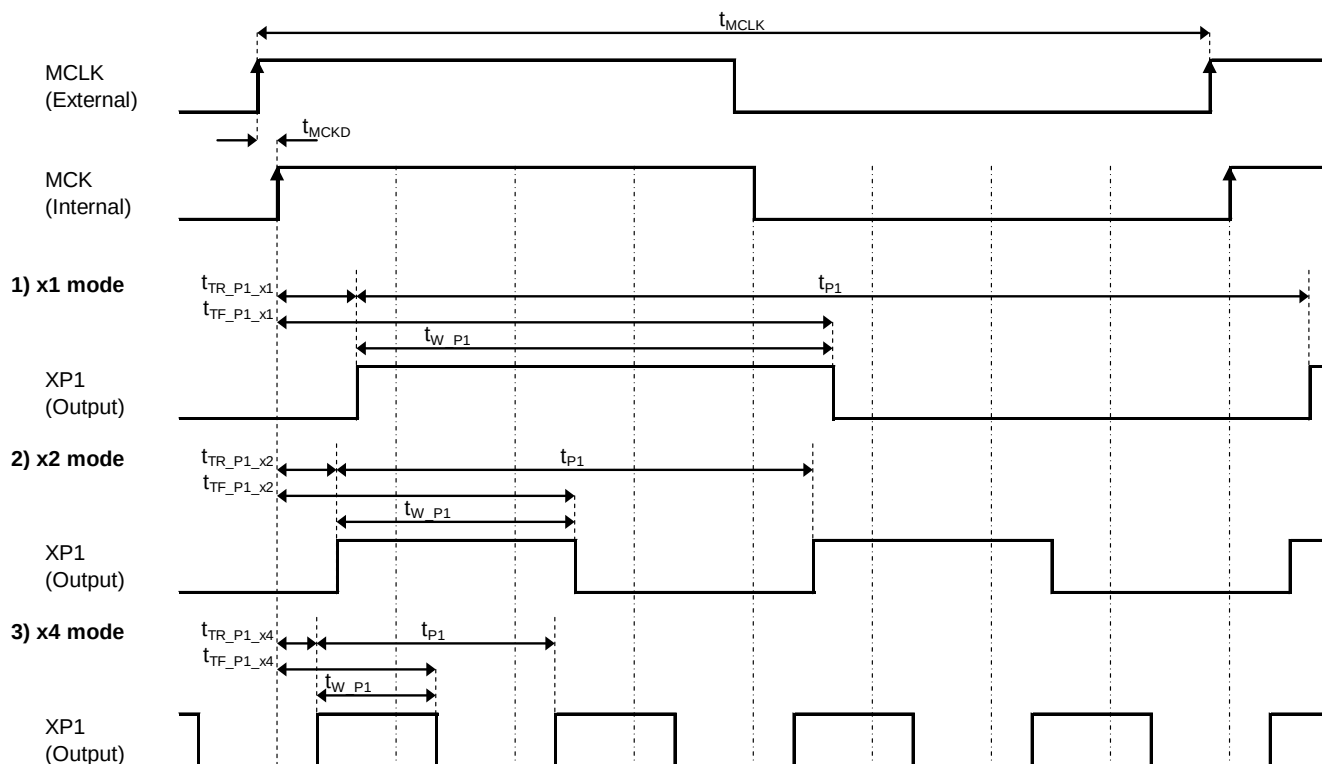


Figure 23. “XP1” Fast Transfer Clock Pulse Setting

SYMBOL	PARAMETER		MIN	TYP	MAX	UNIT
f_{MCLK}	MCLK frequency		1		16.66	MHz
t_{MCLK}	MCLK period			$1/f_{MCLK}$		ns
t_{MCKD}	MCLK to MCK delay			2		ns
t_{P1}	XP1 Period	x1 mode		t_{MCLK}		ns
		x2 mode		$t_{MCLK} \cdot 1/2$		ns
		x4 mode		$t_{MCLK} \cdot 1/4$		ns
$t_{TR_P1_x1}$	XP1 Rising Edge delay from MCK	x1 mode	0		$t_{MCLK} \cdot 47/48$	ns
$t_{TR_P1_x2}$		x2 mode	0		$t_{MCLK} \cdot 23/24$	ns
$t_{TR_P1_x3}$		x4 mode	0		$t_{MCLK} \cdot 11/12$	ns
$t_{TF_P1_x1}$	XP1 Falling Edge delay from MCK	x1 mode	0		$t_{MCLK} \cdot 47/48$	ns
$t_{TF_P1_x2}$		x2 mode	0		$t_{MCLK} \cdot 23/24$	ns
$t_{TF_P1_x3}$		x4 mode	0		$t_{MCLK} \cdot 11/12$	ns
t_{W_P1}	XP1 pulse width	x1 mode	2		$t_{MCLK} - 2$	ns
		x2 mode	2		$t_{MCLK} \cdot 1/2 - 2$	ns
		x4 mode	2		$t_{MCLK} \cdot 1/4 - 2$	ns

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2) Fast Transfer Clock Pulse Setting for “X1L”

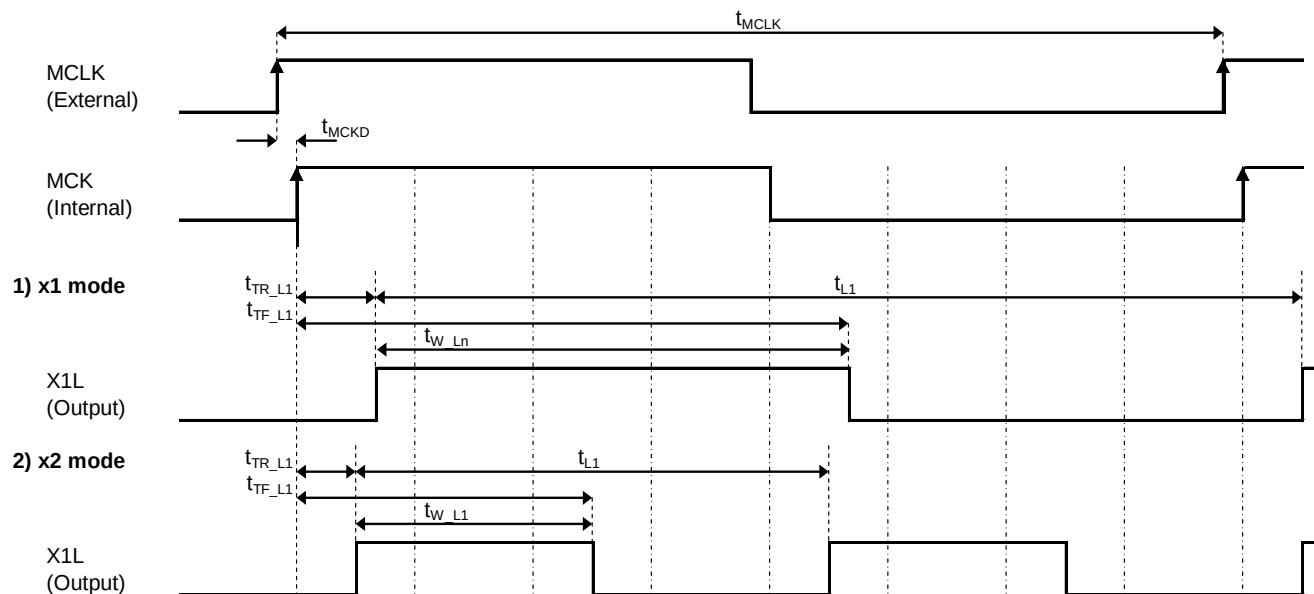


Figure 24. “X1L” Fast Transfer Clock Pulse Setting

SYMBOL	PARAMETER		MIN	TYP	MAX	UNIT
f_{MCLK}	MCLK frequency		1		16.66	MHz
t_{MCLK}	MCLK period			$1/f_{MCLK}$		ns
t_{MCKD}	MCLK to MCK delay			2		ns
t_{L1}	X1L Period	x1 mode		t_{MCLK}		ns
		x2 mode		$t_{MCLK} \cdot 1/2$		ns
t_{TR_L1}	X1L Rising Edge delay from MCK	x1 mode	0		$t_{MCLK} \cdot 47/48$	ns
		x2 mode	0		$t_{MCLK} \cdot 23/24$	ns
t_{TF_L1}	X1L Falling Edge delay from MCK	x1 mode	0		$t_{MCLK} \cdot 47/48$	ns
		x2 mode	0		$t_{MCLK} \cdot 23/24$	ns
t_{W_L1}	X1L pulse width	x1 mode	2		$t_{MCLK} - 2$	ns
		x2 mode	2		$t_{MCLK} \cdot 1/2 - 2$	ns

11.4. Fast Transfer Clock Mask and Toggle Setting

The "Timing Generator", as shown in Figure 21, can mask fast clocks and add some slow pulses in the mask period. The mask period is synchronized with "LS" and is generated by setting the position of start and the end edges. The mask configuration is shared by XP1 and X1L. As shown in Table 22.

The slow pulses can be generated by setting the initial polarity and toggle points within mask period. The predetermined number of toggle points is shown in Table 22. Since the two toggles generate one pulse, the number of pulses is the half the number of toggles.

Table 21. The number of Common Mask and Individual Toggle

SIGNAL GROUP	COMMON MASK	TOGGLE
XP1, X1L	20	16

The toggle and mask positions are defined independently for each signal using different registers. The positions are synchronized with "LS". The period between the falling edge of the "LS" and the position is defined in terms of t_{MCLK} , the cycle period of "MCLK", and is given by the expression given below. The mask and toggle for each signal can be disabled with register settings. In order to make the functions of mask and toggle active, registers $X\#_\text{TGL_EN}$ and $X\#_\text{MSK_EN}$ should be set to "1".

$$\text{Toggle position} = (X\#_\text{T}(k) + 1) \times t_{MCLK} \quad (X\#_\text{T}(k) : \text{Register value of toggle setting})$$

(1) $\#$: XP1, X1L

(2) $k = 0$ to 15

(3) The toggle position should be set in the period of mask. If it is set out of mask, it is ignored.

(4) $X\#_\text{T}(k) < \text{LINT}$

$$\text{Mask start position} = (\text{TC_M}(k)_\text{S} + 1) \times t_{MCLK} \quad (\text{TC_M}(k)_\text{S} : \text{Register value of mask start setting})$$

$$\text{Mask end position} = (\text{TC_M}(k)_\text{E} + 1) \times t_{MCLK} \quad (\text{TC_M}(k)_\text{E} : \text{Register value of mask end setting})$$

(1) $k = 0$ to 19

(2) Mask start should be more than 1.

$$\text{TC_M}(k)_\text{S} \geq 1$$

(3) Mask period should be greater than 2.

$$\text{TC_M}(k)_\text{E} - \text{TC_M}(k)_\text{S} > 2$$

(4) Mask interval should be greater than 2.

$$\text{TC_M}(k+1)_\text{S} - \text{TC_M}(k)_\text{E} > 2$$

(5) $\text{XTC_T}(k) < \text{LINT}$

For signals XP1, the frequency of fast clock can be changed after a mask period. In order to use this function, the DLL taps corresponding to the new frequency should be previously set. In addition, intended clock rate should also be set by using the appropriate register corresponding to a mask period. For example, if clock rate of "XP1" changes after "TC_M0_E" which is the end position of mask 0, "XP1_M0_F" should be set to the new clock rate value. As an additional requirement, with "n" mask defined as the last mask, "n+1" clock rate should be set equal to the clock rate of "n".



2) Fast Transfer Clock Mask and Toggle Setting for "X1L"

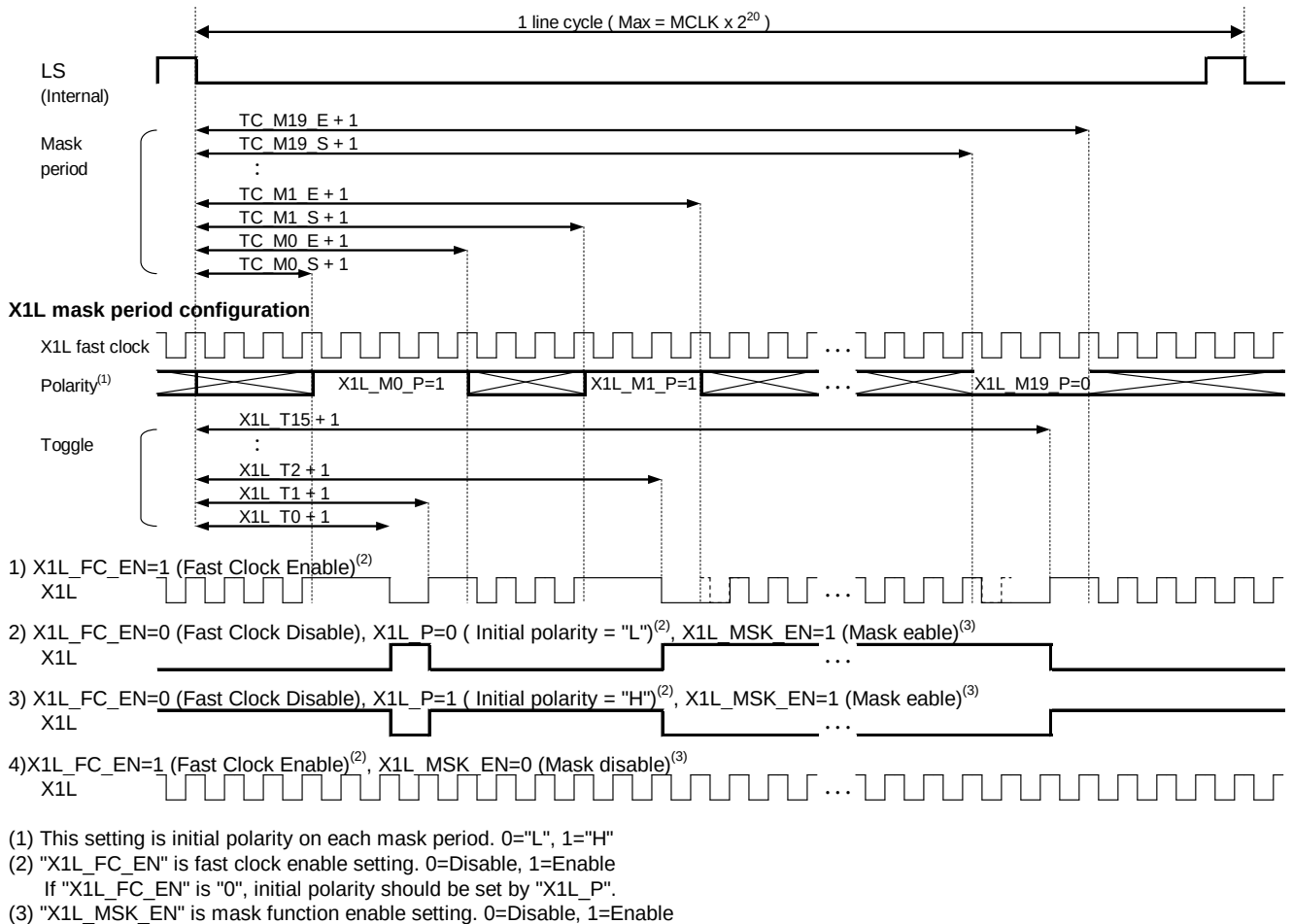


Figure 26. "X1L" Mask Setting

11.5. TG Output Mode

All TG signals can select one of the four output states as shown in the Table 29. The output state for each signal can be configured independently using different registers.

Table 22. The state of TG signals

X##_MODE[1:0]	STATE OF SIGNAL
0	Normal
1	Invert
2	Low
3	High

(1) ## : TG signal name

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12. LED DRIVER

This product has three channels of constant-current sink LED driver. Using register settings, the output current for each channel can be independently within 0mA to 60mA range with 4mA step-size. The maximum allowable total constant-current value for all three channel drivers is 180mA. Since circuitry to limit the total constant-current is not incorporated, the current value should be carefully set when multi-channels are concurrently active.

Figure 26 shows LED driver block diagram and the connection with the external LED device. This driver supports V_{LD} (driver voltage) within a range of 0.7-4.5V. V_{LD} depends on the V_f (forward voltage of LED) because this driver is a constant-current source. LED driver power consumption which is defined as $V_{LD} \times I_{LD}$ (LED current) affects the package temperature. Therefore, in case of small V_f , it is recommendation to insert an external resistor between LEDn (LED driver pin) and LEDVDD (the power supply of LED) for reducing the thermal affects caused by high power consumption.

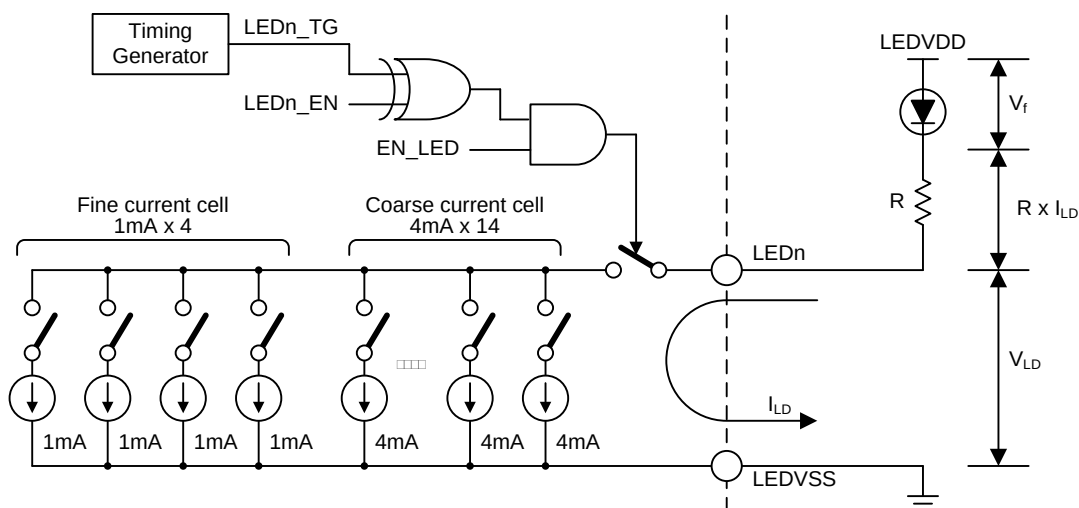


Figure 27. LED Driver Block Diagram

This driver can be turned on or off with a combination of logic circuitry driven by internal-register settings and internal timing signals which is configured by register as shown in Table 24. Please note that the internal register "EN_LED" should be set to "1" to enable the logic circuit.

Table 23. LED Driver Control Register and TG Setting

EN_LED REGISTER	LEDN_TG SIGNAL	LEDN_EN REGISTER	LED DRIVER STATE
0	X	X	Disable
1	0	0	Disable
	0	1	Enable
	1	0	Enable
	1	1	Disable

n=1, 2 and 3

12.1. LED Timing Control

LED on/off timing can be controlled by the internal signal, "LEDn_TG"(n=1,2 and 3), generated from "Timing Generator". "LEDn_TG" is clocks that can be configured by setting initial polarity and toggle points. Table 21 shows the number of toggle setting and generated pulse. Since the two toggles generate one pulse, the number of pulses is half the number of toggles.

Table 24. Toggle number and generated pulse

SIGNAL	TOGGLE	PULSE
LEDn_TG	16	8

Each toggle position is defined by a register that is exclusive for each signal. The toggle position is synchronized with "LS" and the gap between toggle-position and "LS" falling edge defined in terms of t_{MCLK} , the cycle period of "MCLK". This gap is set by register settings and is defined by the following expression:

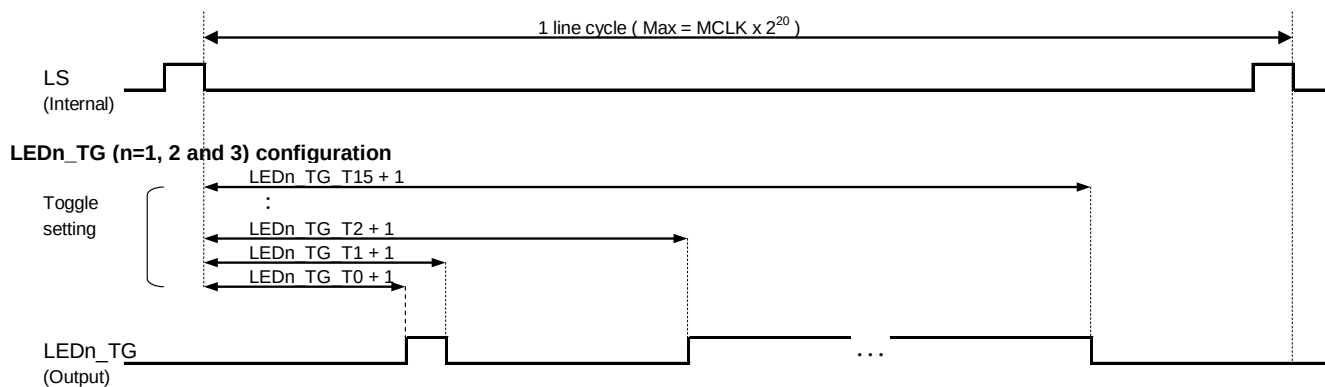
$$t = (LEDn_TG_T(k) + 1) \times t_{MCLK} \quad (LEDn_TG_T(k) : \text{Register value of toggle setting})$$

Note:

- (1) $k = 0$ to 16
- (2) $LEDn_TG_T(k) < LINT$

The toggle for "LEDn_TG" signal can be disabled with register settings. To make toggle active, "LEDn_TGL_EN" should be set to "1".

Figure 27 shows the configuration regarding LED timing control signal.



Note:

- (1) If LEDn_TG_Tn is set to "0" except XST_T0, toggle position is ignored
- (2) The period between toggle position and LS falling edge = $(LEDn_TG_T(k) + 1) \times t_{MCLK}$
- (3) The following requirement should be satisfied: $LEDn_TG_T(k) < LEDn_TG_T(k+1)$

Figure 28. LED Timing Control Signal Setting

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12.2. Step Up/Down of LED Current

When turning LED on and off, the rapid LED current change may contribute some noise to the AFE output code. To avoid this effect, this driver, as shown in Figure 28, has a function that sets the target current by increasing and decreasing current in small steps. The minimum step size is 4mA with an option of selecting step sizes of 4mA, 8mA and 12mA using a 2-bit register. However, it is fixed as 1mA step for the first and final stage. The interval period of one step is defined in terms of t_{MCLK} , the cycle period of MCLK, and is configured with an internal register.

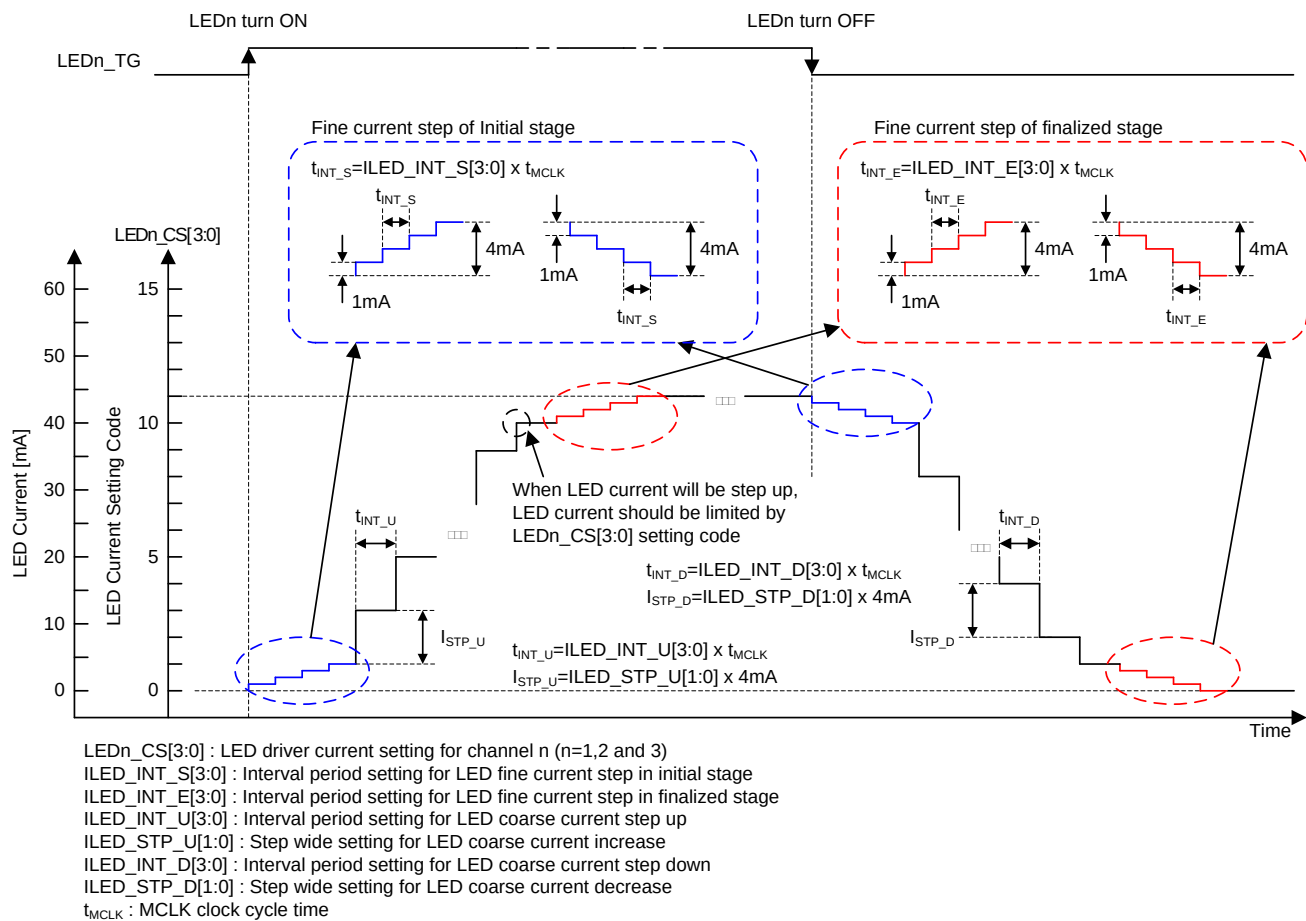


Figure 29. LED Current ON/OFF Timing

12.3. Sequence for enabling LED driver with timing control

The register value of LEDn_TG_Tm is immediately updated by the instruction of serial I/F. If the registers are changed while activating the LED driver, the unintended timing may be executed during only one line cycle. Therefore, If LED driver is controlled with the internal timing signal, LEDn_TG, the following sequence always should be kept before activating LED driver.

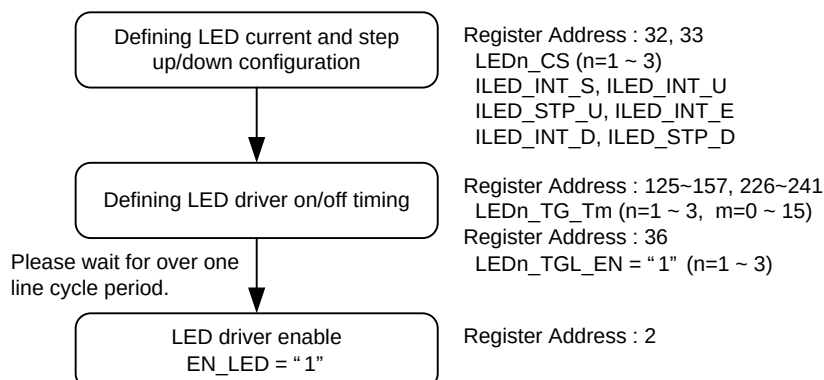


Figure 30. Flow chart for activating LED driver

12.4. Protection circuit for LED driver

In the actual application, the location of LED may be different from the PCB mounting AFE, and the power source may be supplied through the cable which connects Power device on the different PCB. If ESD event is occurred on the cable connecting both PCB while LED power is active, LED driver may suffer a critical damage by sinking large current. For avoiding this risk, TI recommends to add the external ESD protection circuit for LED driver.

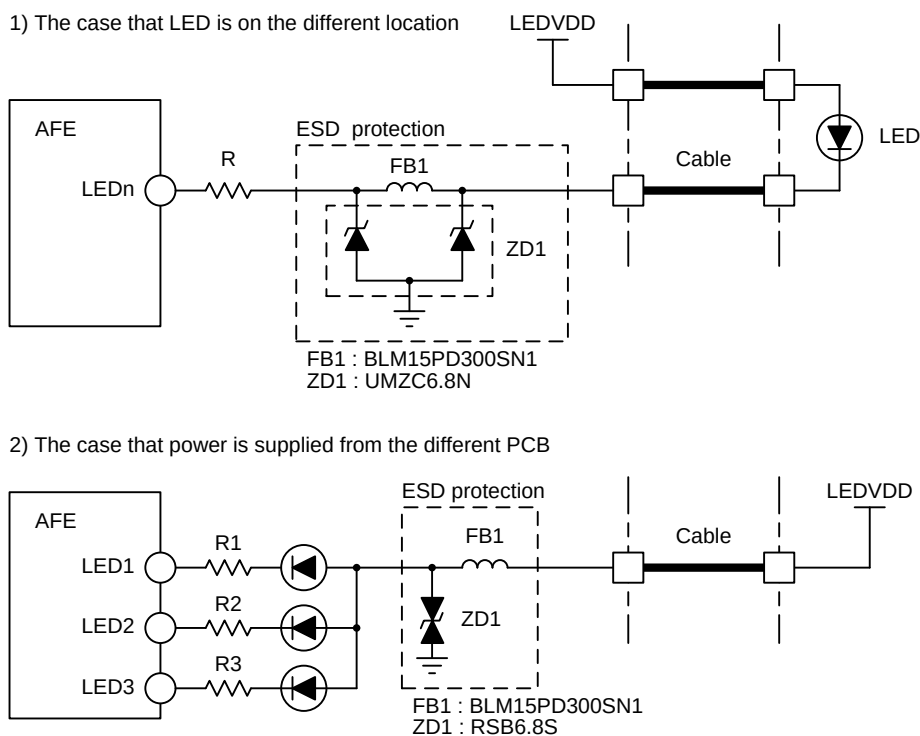


Figure 31. Recommendation circuit for protecting LED driver

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13. STANDBY MODE

Normal operation mode and standby mode can be switched using the serial interface. It is possible to standby all blocks by using the flag "STB_ALL". In this case, all blocks except LDO for Digital, SPI and Registers are in standby mode. If MCLK is stopped, further reduction in power consumption can be achieved. The recovery time from "STB_ALL" is 10ms as maximum.

It is possible to make the output standby with specific registers. All outputs, including TG outputs, can be enabled by setting "EN_OUT" to "1". The default value of this register is "0". It means that all outputs are non-active after reset. Therefore, "EN_OUT" needs to be set to "1" after reset.

Table 26 is the standby condition matrix.

Table 25. Standby Condition Matrix

Condition Register Block		Reset	Standby Function									Mode function									
		All Reset	All Standby	All Output	DLL Standby	PLL Standby	Analog				TG Buffer	LED Driver	Channel mode			Output mode		LVDS mode			
		RST_ ALL	STB_ ALL	EN_ OUT	STB_ DLL	STB_ PLL	STB_ AIN1	STB_ AIN2	STB_ AIN3	STB_ AIN4	TG_OUT _EN	EN_ LED	2ch	3ch	4ch	LVDS	CMOS	Non	1pair	2pair	3pair
		= "1" ⁽²⁾	= "1"	= "0"	= "1"	= "1"	= "1"	= "1"	= "1"	= "1"	= "0"	= "0"	= "2"	= "1"	= "0"	= "0"	= "1"	= "0"	= "1"	= "2"	= "3"
LDO	Analog																				
	Digital																				
Reference																					
SH, CDS	1ch																				
	2ch																				
	3ch																				
	4ch																				
Clamp Buffer	1ch																				
	2ch																				
	3ch																				
	4ch																				
ADC																					
Offset DAC	1ch																				
	2ch																				
	3ch																				
	4ch																				
Clock Generator																					
DLL																					
PLL																					
Serial I/F																					
Register		Clear																			
MCK																					
Logic Block																					
Logic Clock																					
MUX																					
LVDS out	TA	Hi-Z	Hi-Z	Hi-Z													Hi-Z	Hi-Z			
	TB	Hi-Z	Hi-Z	Hi-Z													Hi-Z	Hi-Z	Hi-Z		
	TC	Hi-Z	Hi-Z	Hi-Z													Hi-Z	Hi-Z	Hi-Z	Hi-Z	
	TCLK	Hi-Z	Hi-Z	Hi-Z													Hi-Z	Hi-Z			
CMOS out	D[1:0]	Hi-Z	Hi-Z	Hi-Z												Hi-Z					
	D[3:2]	Hi-Z	Hi-Z	Hi-Z												Hi-Z					
	D[5:4]	Hi-Z	Hi-Z	Hi-Z												Hi-Z					
	CK[1:0]	Hi-Z	Hi-Z	Hi-Z												Hi-Z					
LED Driver	1ch	Hi-Z	Hi-Z	Hi-Z								Hi-Z									
	2ch	Hi-Z	Hi-Z	Hi-Z								Hi-Z									
	3ch	Hi-Z	Hi-Z	Hi-Z								Hi-Z									
TG Buffer		Hi-Z	Hi-Z	Hi-Z							Hi-Z										

(1) Gray is standby status

(2) This flag will be automatically cleared and then standby status will be canceled.

14. PIN CONTROL

14.1. Digital Output Driver Ability

For digital outputs, the driver current can be changed by register settings. The available current choices are 2mA, 4mA and 8mA. Table 27 shows relationship between register and pin-name and Table 28 shows the relationship between driver current and register code.

Table 26. Driver Control Setting Register vs. Pin

REGISTER	PIN NAME
DRV_SEL0[1:0]	GPIO1/SDO, GPIO3/XST
DRV_SEL1[1:0]	XP1
DRV_SEL3[1:0]	X1L
DRV_DOUT[1:0]	D0, D1, D2, D3, D4, D5, CK0, CK1

Table 27. Output Port Driver Setting

REGISTER SET	PORT DRIVABILITY
0 (default)	2 mA
1	4 mA
2	8 mA
3	Reserved

14.2. GPIO (General Purpose Input and Output)

VSP5640 has two GPIO pins. GPIO pins are bi-directional and the direction can be set by register settings. When these pins are used as input, the external signal information can be obtained by reading the internal register GPIOx_RD. When GPIO pin is set as output, it can output various signals, including some internal clocks, based on register settings shown in Table 29.

Table 28. General Purpose Output

REGISTER SETTING GPIOX_OUT_SEL	GPIO1	GPIO3
0	"Low"	
1	"High"	
2	LVCK	
3	CLP_B	CLP_A
4	SHP_A	
5	SHP_B	
6	SHD_A	
7	SHD_B	
8	ADCK	
9	LS	
10	CLPDM	
11	Reserved	
12	ADST	

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14.3. Master Clock input

This device allows MCLK input frequency within the range of 1 to 11.66MHz. Appropriate register setting (MCLK_FRQ), corresponding to the MCLK frequency, should be done for optimizing main DLL circuit.

Table 29. Allowable Input Frequency Range Setting for MCLK

REGISTER SETTING MCLK_FRQ	INPUT FREQUENCY MCLK
0	1 to 4.6 MHz
1	4 to 11MHz
2	9 to 16MHz
3	14 to 21MHz
4	19 to 25MHz

VSP5640 support CMOS and LVDS for master clock supply.



Figure 32. Master Clock Input

15. SERIAL INTERFACE

All functions and settings of this device are controlled through serial interface. Serial interface is composed of three signals (SCLK, SEN and SDI) for register writing, and a fourth signal (SDO) for read-back. SDO shares the terminal with GPIO signal and thus register setting is required to activate SDO function. Other signals are assigned to individual terminals.

Serial data is composed of total 30 bits as shown in Figure 32. 10 bits are assigned for register address and 20 bits for register data. The input serial data at SDI is sequentially stored in a shift register at SCLK rising edge. Data shift operation is performed at the rising edges of SCLK with SEN set at LOW. All 30 bits of input data are loaded to a parallel latch in an internal register at the rising edge of SEN.

This device has two modes – read and write. The mode selection can be made by internal register “SPL_RW” located at Bit 0 of Address 0. SPL_RW equal to “0” implies write mode and SPL_RW equal to “1” implies read mode.

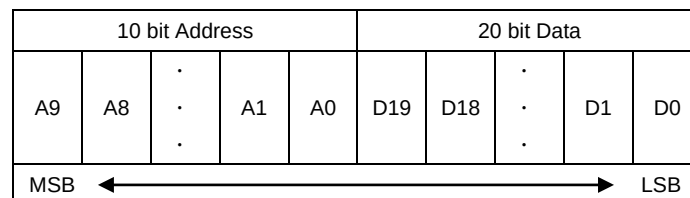


Figure 33. Serial I/F Data Format

15.1. Write mode (SPL_RW = “0”, default)

Normally, one serial interface command is sent by one combination of address and data. The address should be sent as MSB first. Data is stored into respective register indicated by address. If serial-data at the end of the data stream is less than 30 bits, then the last incomplete serial-data is discarded.

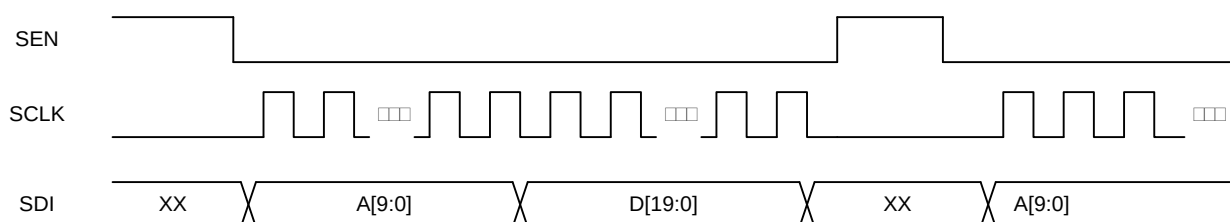


Figure 34. SPI signal flow of Write mode

15.2. Read mode (SPL_RW = “1”)

In read mode, two types of connections are possible between AFE and external systems as ASIC/CPU. One is the four-wired connection in which SDI and SDO pins are separately connected to the system as shown in Figure 34.

The other is 3-wired connection in which only the SDI pin is connected to the bidirectional I/O port of the external system. In this case, “SDI_BUFF_CTRL” should be set to “1” for making SPI bidirectional port. The bit flow of 3 wire connection is shown in Figure 36. As shown in this figure, SDI changes from input to output at the falling edge of SCLK after the end of A[9:0] input. Since SDI port is always in pull-down mode, the external pull-down resistance is unnecessary.

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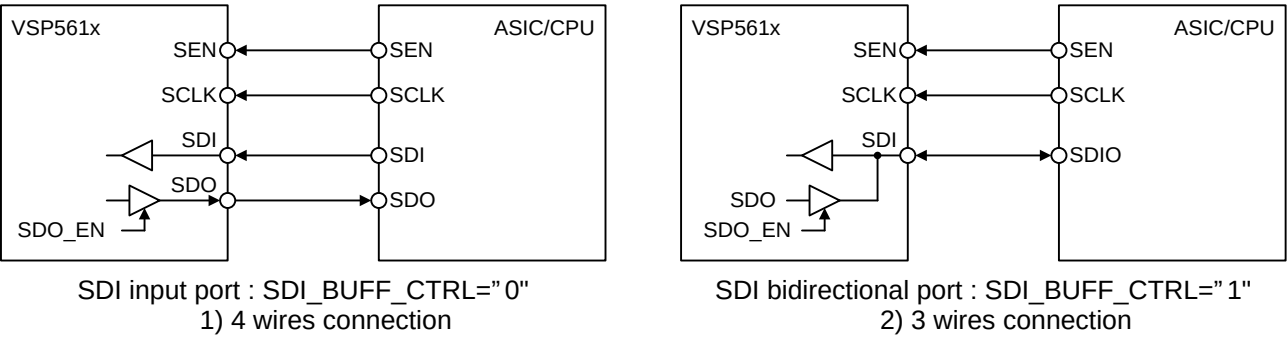


Figure 35. SPI connection between AFE and System

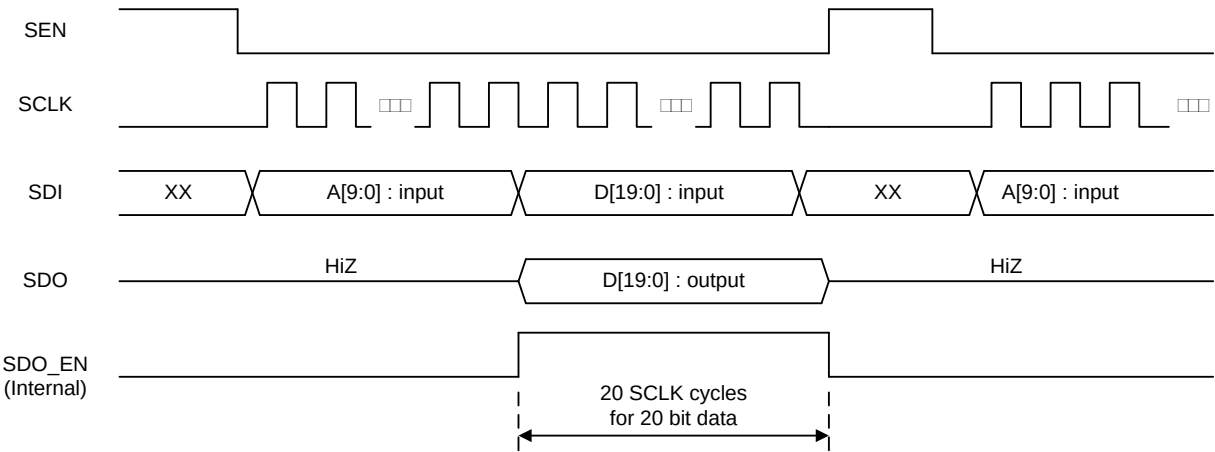


Figure 36. SPI signal flow of Read mode for 4 wires connection

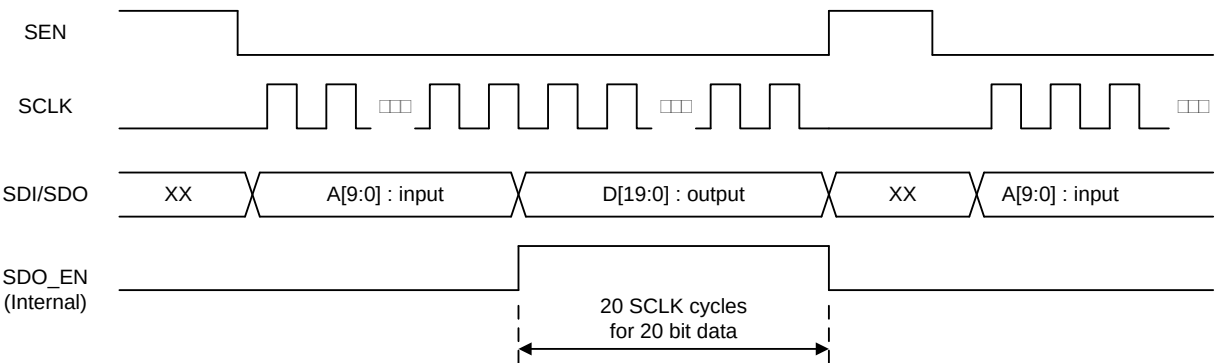


Figure 37. SPI signal flow of Read mode for 3 wires connection

16. TEST PATTERN FUNCTION

In order to check the handshake between AFE and receiver system, test pattern function is provided in this device. Test pattern function can be used in two modes. In first mode, the test pattern is directly input to output port (CMOS output) or serializer (LVDS output) for checking buffer and serializer functions. In the second mode, the test pattern is used instead of the ADC output for checking the output function. Figure 37 shows the block diagram indicating the connection choices of test pattern generator.

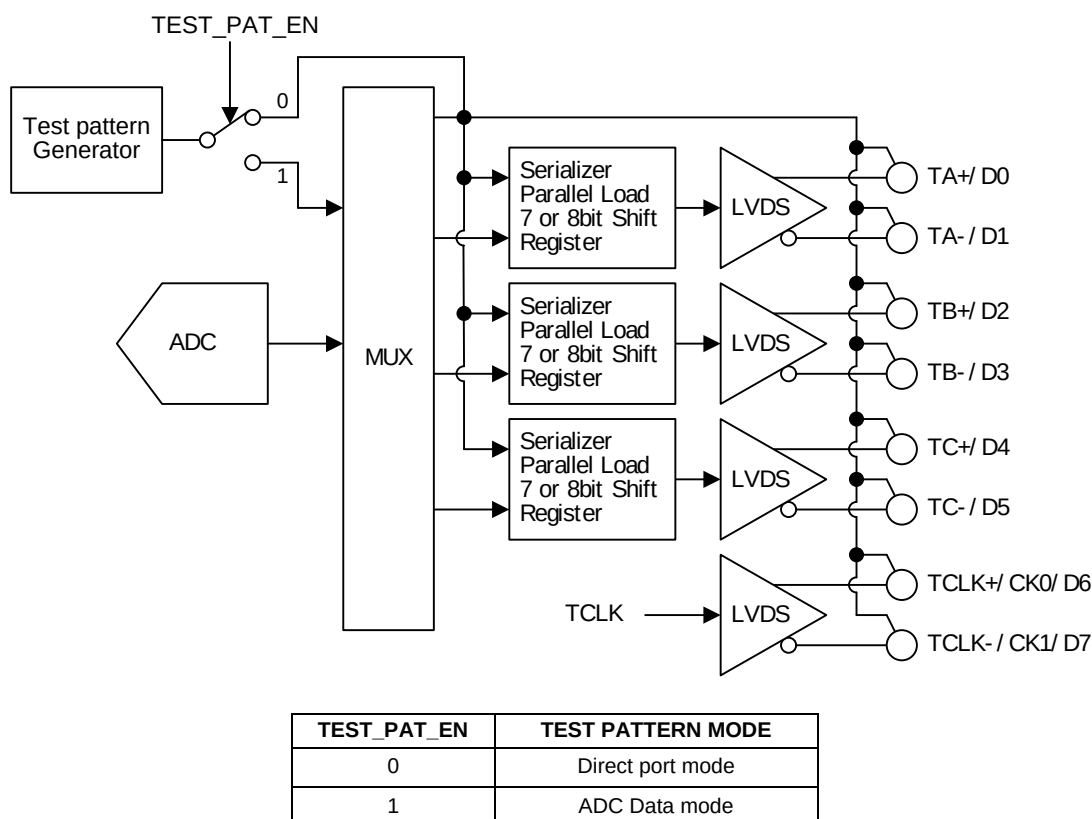


Figure 38. Connection of test pattern generator

16.1. Direct Port Mode (TEST_PAT_EN = "0")

In case of direct port mode, the test pattern directly inputs to serializer (for LVDS output) or to output port (for CMOS output). As shown in Table 31, different test patterns can be selected by setting register "TEST_PAT_SEL[3:0]".

16.2. ADC Data Mode (TEST_PAT_EN = "1")

As shown in Table 32, this mode has various patterns of 16bits output data that are selectable by setting register "TEST_PAT_SEL[3:0]".

In case of LVDS output with this mode, test pattern can support LS (line synchronize) code similar to the normal output. This means that the output data, besides the LS code, is not less than 3LSB. But when register "DIS_TEST_LS_SYNC" is set to "1", LS code is not provided and the data can be less than 3LSB.

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Table 30. Test pattern selection in Direct Port Mode

TEST_PAT_SEL[3:0] ⁽¹⁾		BITS ASSIGNMENT OF TEST PATTERN		
		7BIT LVDS	8BIT LVDS/CMOS	4BIT CMOS
0	0000'b	Normal output		
1	0001'b	All "0"		
2	0010'b	All "1"		
3	0011'b	00001111'b	00001111'b	0011'b
4	0100'b	0101010'b	01010101'b	0101'b
5	0101'b	TEST_PAT_VAL1[7:1] ⁽²⁾	TEST_PAT_VAL1[7:0] ⁽²⁾	TEST_PAT_VAL1[3:0] ⁽²⁾
6	0110'b	Ramp (1LSB will be constantly increased)		
7	0111'b	Alternate 2A'h and 55'h	Alternate AA'h and 55'h	Alternate A'h and 5'h
8	1000'b	Alternate ⁽²⁾ TEST_PAT_VAL1[7:1] To TEST_PAT_VAL2[7:1]	Alternate ⁽²⁾ TEST_PAT_VAL1[7:0] To TEST_PAT_VAL2[7:0]	Alternate ⁽²⁾ TEST_PAT_VAL1[3:0] To TEST_PAT_VAL2[3:0]
9	1001'b	N/A		
10	1010'b	Select TEST_PAT_VAL1[7:1] or TEST_PAT_VAL2[7:1] with SEL_TEST_VAL[2:0] ⁽³⁾	Select TEST_PAT_VAL1[7:0] or TEST_PAT_VAL2[7:0] with SEL_TEST_VAL[2:0] ⁽³⁾	Select TEST_PAT_VAL1[3:0] or TEST_PAT_VAL2[3:0] with SEL_TEST_VAL[2:0] ⁽³⁾

(1) TEST_PAT_SEL[3:0] : {Add407[16], Add409[19:17]}

(2) TEST_PAT_VAL1 : Add409[15:0], TEST_PAT_VAL2 : Add433[15:0]

(3) SEL_TEST_VAL[3:0]: Add434[3:0]

Table 31. Test pattern selection

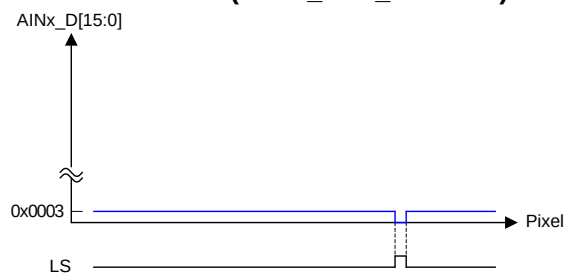
TEST_PAT_SEL[3:0] ⁽¹⁾		TEST PATTERN DATA
0	0000'b	Normal output
1	0001'b	All "0"
2	0010'b	All "1"
3	0011'b	00FFh constant
4	0100'b	5555h constant
5	0101'b	Constant TEST_PAT_VAL1[15:0] ⁽²⁾
6	0110'b	Ramp (1LSB will be constantly increased)
7	0111'b	Alternate AAAAh and 5555h
8	1000'b	Alternate TEST_PAT_VAL1[15:0] to TEST_PAT_VAL2[15:0] ⁽²⁾
9	1001'b	Ramp between TEST_PAT_VAL1[15:0] to TEST_PAT_VAL2[15:0] ⁽²⁾
10	1010'b	Select TEST_PAT_VAL1 or 2 with SEL_TEST_VAL[3:0] ⁽³⁾

(1) TEST_PAT_SEL[3:0] : {Add407[16], Add409[19:17]}

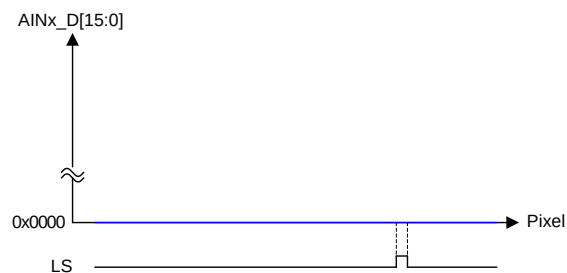
(2) TEST_PAT_VAL1 : Add409[15:0], TEST_PAT_VAL2 : Add433[15:0]

(3) SEL_TEST_VAL[3:0]: Add434[3:0]

1) Constant of All "0" (TEST_PAT_SEL="1")

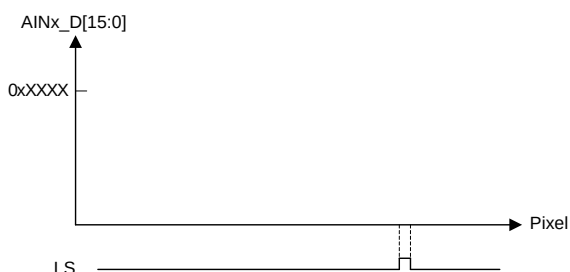


a) LVDS with DIS_TEST_LS_SNC="0"

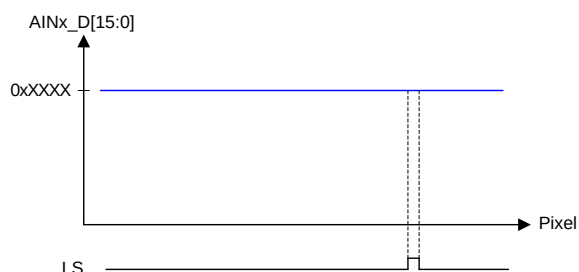


b) LVDS with DIS_TEST_LS_SNC="1" or CMOS

2) Constant of All "1", 00FFh or 5555h (TEST_PAT_SEL="2" to "4")

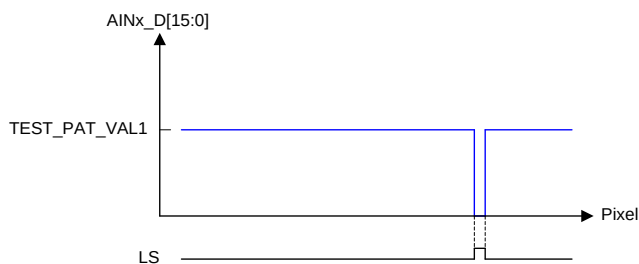


a) LVDS output

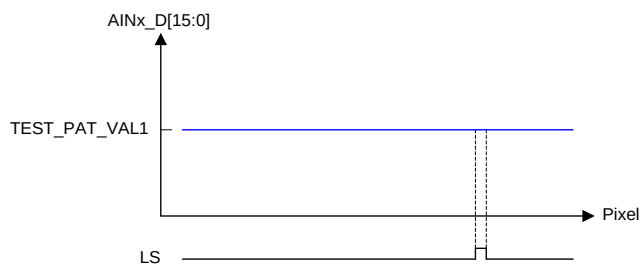


b) CMOS output

3) Constant of TEST_PAT_VAL1 (TEST_PAT_SEL="5")



a) LVDS output

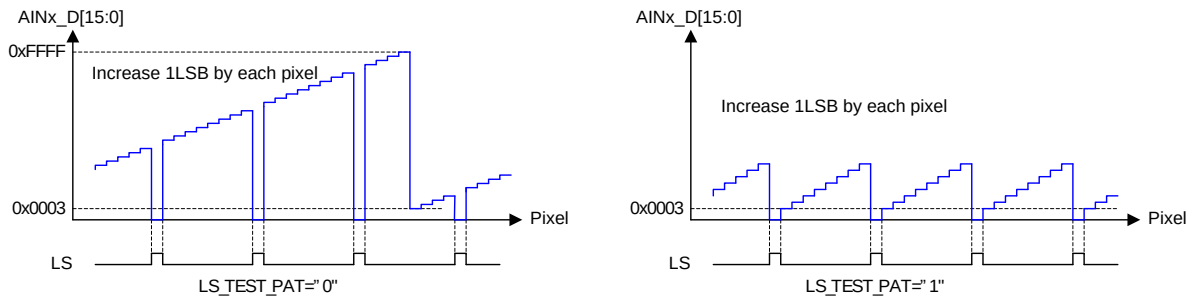


b) CMOS output

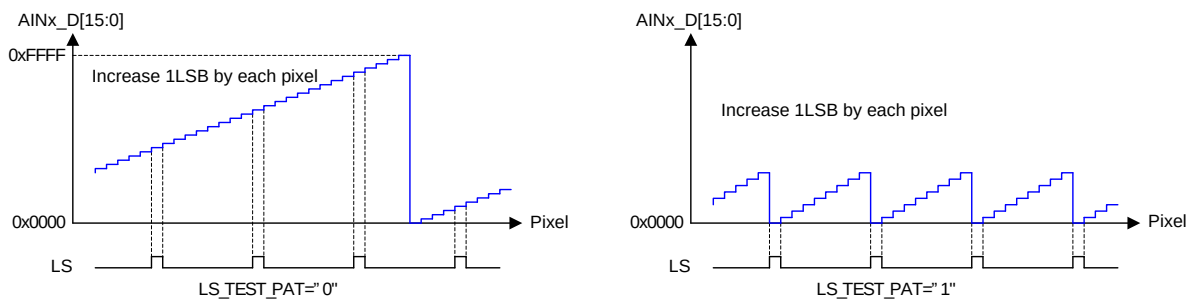
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4) Ramp (TEST_PAT_SEL="6")

Ramp test pattern can synchronize with LS by register "LS_TEST_PAT" setting.

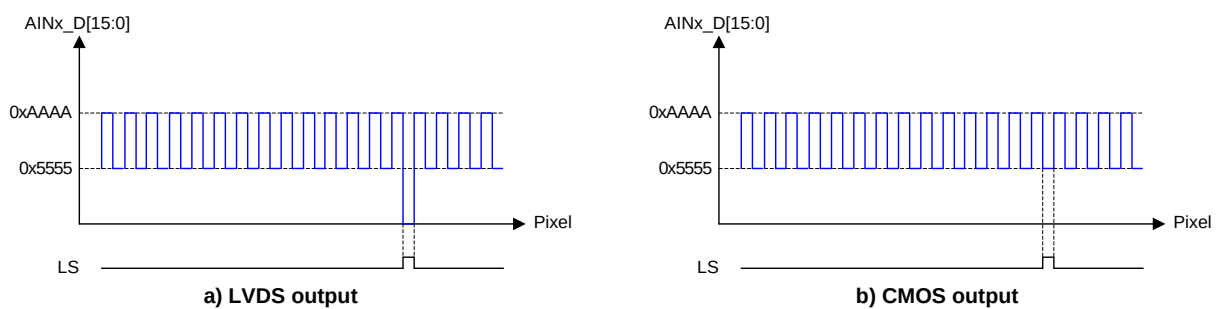


a) LVDS output



b) CMOS output

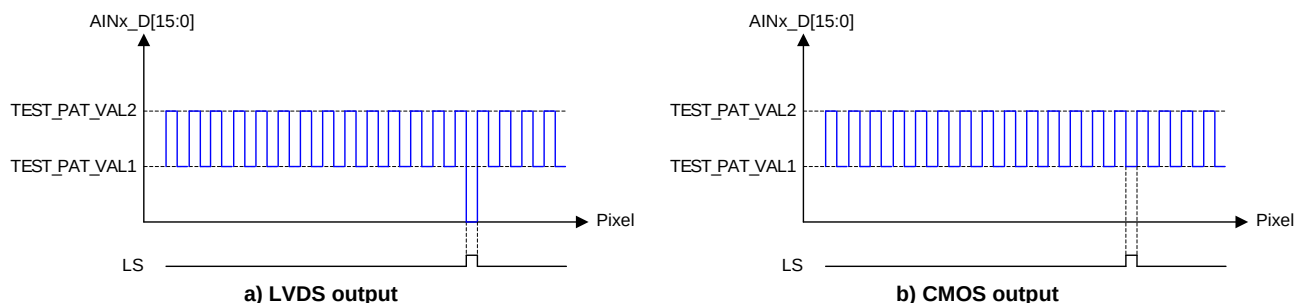
5) Alternate AAAAh and 5555h (TEST_PAT_SEL="7")



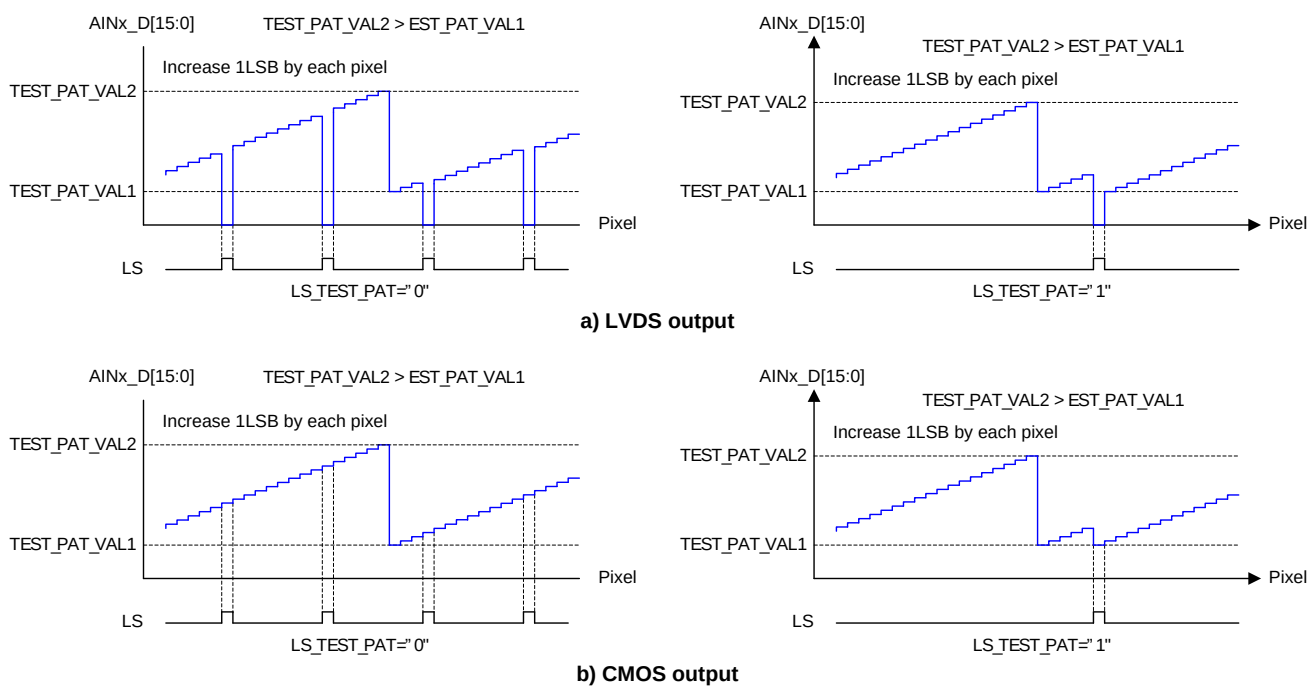
a) LVDS output

b) CMOS output

6) Alternate TEST_PAT_VAL1 and TEST_PAT_VAL2 (TEST_PAT_SEL="8")



7) Ramp between TEST_PAT_VAL1[15:0] to TEST_PAT_VAL2[15:0] (TEST_PAT_SEL="9")



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17. REGISTER SPECIFICATION

17.1. Register Map

Address		Register	Description	Default	
1) Configuration					
0	000h	SPI_CFG	SPI configuration	0	00000h
1	001h	REV_RST_CFG	Chip revision / reset configuration	0	000xxh ¹
2	002h	STB_CFG	Standby configuration	0	00000h
3	003h	OUTPUT_CFG	Output function configuration	0	00000h
4	004h	PIN_FNC_CFG0	Pin function configuration 0	0	00000h
5	005h	PIN_FNC_CFG1	Pin function configuration 1	0	00000h
6	006h	PIN_FNC_CFG2	Pin function configuration 2	0	00000h
7	007h	LINT	Line cycle setting / XLSYNC mask setting	0	00000h
8	008h	DM_STR	CLPDM start period setting	0	00000h
9	009h	DM_END	CLPDM end period setting	0	00000h
10	00Ah	Reserved	Reserved	0	00000h
11	00Bh	Reserved	Reserved	0	00000h
12	00Ch	Reserved	Reserved	0	00000h
13	00Dh	Reserved	Reserved	0	00000h
14	00Eh	Reserved	Reserved	0	00000h
15	00Fh	Reserved	Reserved	0	00000h
2) Analog Setting					
16	010h	AIN_CFG0	Analog input configuration 0	0	00000h
17	011h	AIN_CFG1	Analog input configuration 1	0	00000h
18	012h	SHP_A_ADJ	SHP_A timing adjustment	0	00000h
19	013h	SHD_A_ADJ	SHD_A timing adjustment	0	00000h
20	014h	SHP_B_ADJ	SHP_B timing adjustment	0	00000h
21	015h	SHD_B_ADJ	SHD_B timing adjustment	0	00000h
22	016h	Reserved	Reserved	0	00000h
23	017h	GAIN_1	AIN1 gain setting	0	00000h
24	018h	GAIN_2	AIN2 gain setting	0	00000h
25	019h	GAIN_3	AIN3 gain setting	0	00000h
26	01Ah	GAIN_4	AIN4 gain setting	0	00000h
27	01Bh	OFDAC_12	Offset correction DAC for AIN 1 and 2	0	00000h
28	01Ch	OFDAC_34	Offset correction DAC for AIN 3 and 4	0	00000h
29	01Dh	Reserved	Reserved	0	00000h
30	01Eh	Reserved	Reserved	0	00000h
31	01Fh	Reserved	Reserved	0	00000h
3) LED Driver					
32	020h	LED_DRV_SET	LED driver setting	0	00000h
33	021h	LED_DRV_DLY	LED driver on/off delay setting	0	00000h
34	022h	Reserved	Reserved	0	00000h
35	023h	Reserved	Reserved	0	00000h
4) Timing Generator Configuration					
36	024h	TG_OUT_EN	TG output enable	0	00000h
37	025h	TG_OUT_MD0	TG output mode setting 0	0	00000h
38	026h	TG_OUT_MD1	TG output mode setting 1	0	00000h
39	027h	Reserved	Reserved	0	00000h

¹ x – Means “Undefined”

17.1. Register Map (Continued)

Address		Register	Description	Default	
5) Fast Transfer Clock Setting					
40	028h	XP1_FC_SET0	XP1 fast clock setting register 0	0	00000h
41	029h	XP1_FC_SET1	XP1 fast clock setting register 1	0	00000h
42	02Ah	XP1_M_F0	XP1 fast transfer clock frequency rate setting 0	0	00000h
43	02Bh	XP1_M_F1	XP1 fast transfer clock frequency rate setting 1	0	00000h
44	02Ch	X1L_FC_SET	X1L fast clock setting	0	00000h
45	02Dh	Reserved	Reserved	0	00000h
46	02Eh	Reserved	Reserved	0	00000h
47	02Fh	Reserved	Reserved	0	00000h
48	030h	Reserved	Reserved	0	00000h
49	031h	Reserved	Reserved	0	00000h
50	032h	Reserved	Reserved	0	00000h
51	033h	Reserved	Reserved	0	00000h
6) XST Setting					
52	034h	XST_T0	XST toggle setting period 0	0	00000h
53	035h	XST_T1	XST toggle setting period 1	0	00000h
54	036h	XST_T2	XST toggle setting period 2	0	00000h
55	037h	XST_T3	XST toggle setting period 3	0	00000h
56	038h	XST_T4	XST toggle setting period 4	0	00000h
57	039h	XST_T5	XST toggle setting period 5	0	00000h
58	03Ah	XST_T6	XST toggle setting period 6	0	00000h
59	03Bh	XST_T7	XST toggle setting period 7	0	00000h
60 to 124	03Ch to 07Ch	Reserved	Reserved	0	00000h
7) LED2_TG Setting					
125	07Dh	LED2_TG_T0	LED2_TG toggle setting period 0	0	00000h
126	07Eh	LED2_TG_T1	LED2_TG toggle setting period 1	0	00000h
127	07Fh	LED2_TG_T2	LED2_TG toggle setting period 2	0	00000h
128	080h	LED2_TG_T3	LED2_TG toggle setting period 3	0	00000h
129	081h	LED2_TG_T4	LED2_TG toggle setting period 4	0	00000h
130	082h	LED2_TG_T5	LED2_TG toggle setting period 5	0	00000h
131	083h	LED2_TG_T6	LED2_TG toggle setting period 6	0	00000h
132	084h	LED2_TG_T7	LED2_TG toggle setting period 7	0	00000h
133	085h	LED2_TG_T8	LED2_TG toggle setting period 8	0	00000h
134	086h	LED2_TG_T9	LED2_TG toggle setting period 9	0	00000h
135	087h	LED2_TG_T10	LED2_TG toggle setting period 10	0	00000h
136	088h	LED2_TG_T11	LED2_TG toggle setting period 11	0	00000h
137	089h	LED2_TG_T12	LED2_TG toggle setting period 12	0	00000h
138	08Ah	LED2_TG_T13	LED2_TG toggle setting period 13	0	00000h
139	08Bh	LED2_TG_T14	LED2_TG toggle setting period 14	0	00000h
140	08Ch	LED2_TG_T15	LED2_TG toggle setting period 15	0	00000h
141	08Dh	Reserved	Reserved	0	00000h

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17.1. Register Map (Continued)

Address		Register	Description	Default	
8) LED1_TG Setting					
142	08Eh	LED1_TG_T0	LED1_TG toggle setting period 0	0	00000h
143	08Fh	LED1_TG_T1	LED1_TG toggle setting period 1	0	00000h
144	090h	LED1_TG_T2	LED1_TG toggle setting period 2	0	00000h
145	091h	LED1_TG_T3	LED1_TG toggle setting period 3	0	00000h
146	092h	LED1_TG_T4	LED1_TG toggle setting period 4	0	00000h
147	093h	LED1_TG_T5	LED1_TG toggle setting period 5	0	00000h
148	094h	LED1_TG_T6	LED1_TG toggle setting period 6	0	00000h
149	095h	LED1_TG_T7	LED1_TG toggle setting period 7	0	00000h
150	096h	LED1_TG_T8	LED1_TG toggle setting period 8	0	00000h
151	097h	LED1_TG_T9	LED1_TG toggle setting period 9	0	00000h
152	098h	LED1_TG_T10	LED1_TG toggle setting period 10	0	00000h
153	099h	LED1_TG_T11	LED1_TG toggle setting period 11	0	00000h
154	09Ah	LED1_TG_T12	LED1_TG toggle setting period 12	0	00000h
155	09Bh	LED1_TG_T13	LED1_TG toggle setting period 13	0	00000h
156	09Ch	LED1_TG_T14	LED1_TG toggle setting period 14	0	00000h
157	09Dh	LED1_TG_T15	LED1_TG toggle setting period 15	0	00000h
158 to 207	09Eh 0CFh	Reserved	Reserved	0	00000h
9) XP1 Setting					
208	0D0h	XP1_M_P	XP1 mask initial polarity setting	0	00000h
209	0D1h	XP1_T0	XP1 toggle setting period 0	0	00000h
210	0D2h	XP1_T1	XP1 toggle setting period 1	0	00000h
211	0D3h	XP1_T2	XP1 toggle setting period 2	0	00000h
212	0D4h	XP1_T3	XP1 toggle setting period 3	0	00000h
213	0D5h	XP1_T4	XP1 toggle setting period 4	0	00000h
214	0D6h	XP1_T5	XP1 toggle setting period 5	0	00000h
215	0D7h	XP1_T6	XP1 toggle setting period 6	0	00000h
216	0D8h	XP1_T7	XP1 toggle setting period 7	0	00000h
217	0D9h	XP1_T8	XP1 toggle setting period 8	0	00000h
218	0DAh	XP1_T9	XP1 toggle setting period 9	0	00000h
219	0DBh	XP1_T10	XP1 toggle setting period 10	0	00000h
220	0DCh	XP1_T11	XP1 toggle setting period 11	0	00000h
221	0DDh	XP1_T12	XP1 toggle setting period 12	0	00000h
222	0DEh	XP1_T13	XP1 toggle setting period 13	0	00000h
223	0DFh	XP1_T14	XP1 toggle setting period 14	0	00000h
224	0E0h	XP1_T15	XP1 toggle setting period 15	0	00000h
225	0E1h	Reserved	Reserved	0	00000h

17.1. Register Map (Continued)

Address		Register	Description	Default	
10) LED3_TG Setting					
226	0E2h	LED3_TG_T0	LED3_TG toggle setting period 0	0	00000h
227	0E3h	LED3_TG_T1	LED3_TG toggle setting period 1	0	00000h
228	0E4h	LED3_TG_T2	LED3_TG toggle setting period 2	0	00000h
229	0E5h	LED3_TG_T3	LED3_TG toggle setting period 3	0	00000h
230	0E6h	LED3_TG_T4	LED3_TG toggle setting period 4	0	00000h
231	0E7h	LED3_TG_T5	LED3_TG toggle setting period 5	0	00000h
232	0E8h	LED3_TG_T6	LED3_TG toggle setting period 6	0	00000h
233	0E9h	LED3_TG_T7	LED3_TG toggle setting period 7	0	00000h
234	0EAh	LED3_TG_T8	LED3_TG toggle setting period 8	0	00000h
235	0EBh	LED3_TG_T9	LED3_TG toggle setting period 9	0	00000h
236	0ECh	LED3_TG_T10	LED3_TG toggle setting period 10	0	00000h
237	0EDh	LED3_TG_T11	LED3_TG toggle setting period 11	0	00000h
238	0EEh	LED3_TG_T12	LED3_TG toggle setting period 12	0	00000h
239	0EFh	LED3_TG_T13	LED3_TG toggle setting period 13	0	00000h
240	0F0h	LED3_TG_T14	LED3_TG toggle setting period 14	0	00000h
241	0F1h	LED3_TG_T15	LED3_TG toggle setting period 15	0	00000h
11) X1L Setting					
242	0F2h	X1L_M_P	X1L mask initial polarity setting	0	00000h
243	0F3h	X1L_T0	X1L toggle setting period 0	0	00000h
244	0F4h	X1L_T1	X1L toggle setting period 1	0	00000h
245	0F5h	X1L_T2	X1L toggle setting period 2	0	00000h
246	0F6h	X1L_T3	X1L toggle setting period 3	0	00000h
247	0F7h	X1L_T4	X1L toggle setting period 4	0	00000h
248	0F8h	X1L_T5	X1L toggle setting period 5	0	00000h
249	0F9h	X1L_T6	X1L toggle setting period 6	0	00000h
250	0FAh	X1L_T7	X1L toggle setting period 7	0	00000h
251	0FBh	X1L_T8	X1L toggle setting period 8	0	00000h
252	0FCh	X1L_T9	X1L toggle setting period 9	0	00000h
253	0FDh	X1L_T10	X1L toggle setting period 10	0	00000h
254	0FEh	X1L_T11	X1L toggle setting period 11	0	00000h
255	0FFh	X1L_T12	X1L toggle setting period 12	0	00000h
256	100h	X1L_T13	X1L toggle setting period 13	0	00000h
257	101h	X1L_T14	X1L toggle setting period 14	0	00000h
258	102h	X1L_T15	X1L toggle setting period 15	0	00000h
259 to 323	103h to 143h	Reserved	Reserved	0	00000h

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17.1. Register Map (Continued)

Address		Register	Description	Default	
12) XP and X1L Mask Setting					
324	144h	TC_M0_S	XP1, X1L transfer clock mask 0 start period	0	00000h
325	145h	TC_M0_E	XP1, X1L transfer clock mask 0 end period	0	00000h
326	146h	TC_M1_S	XP1, X1L transfer clock mask 1 start period	0	00000h
327	147h	TC_M1_E	XP1, X1L transfer clock mask 1 end period	0	00000h
328	148h	TC_M2_S	XP1, X1L transfer clock mask 2 start period	0	00000h
329	149h	TC_M2_E	XP1, X1L transfer clock mask 2 end period	0	00000h
330	14Ah	TC_M3_S	XP1, X1L transfer clock mask 3 start period	0	00000h
331	14Bh	TC_M3_E	XP1, X1L transfer clock mask 3 end period	0	00000h
332	14Ch	TC_M4_S	XP1, X1L transfer clock mask 4 start period	0	00000h
333	14Dh	TC_M4_E	XP1, X1L transfer clock mask 4 end period	0	00000h
334	14Eh	TC_M5_S	XP1, X1L transfer clock mask 5 start period	0	00000h
335	14Fh	TC_M5_E	XP1, X1L transfer clock mask 5 end period	0	00000h
336	150h	TC_M6_S	XP1, X1L transfer clock mask 6 start period	0	00000h
337	151h	TC_M6_E	XP1, X1L transfer clock mask 6 end period	0	00000h
338	152h	TC_M7_S	XP1, X1L transfer clock mask 7 start period	0	00000h
339	153h	TC_M7_E	XP1, X1L transfer clock mask 7 end period	0	00000h
340	154h	TC_M8_S	XP1, X1L transfer clock mask 8 start period	0	00000h
341	155h	TC_M8_E	XP1, X1L transfer clock mask 8 end period	0	00000h
342	156h	TC_M9_S	XP1, X1L transfer clock mask 9 start period	0	00000h
343	157h	TC_M9_E	XP1, X1L transfer clock mask 9 end period	0	00000h
344	158h	TC_M10_S	XP1, X1L transfer clock mask 10 start period	0	00000h
345	159h	TC_M10_E	XP1, X1L transfer clock mask 10 end period	0	00000h
346	15Ah	TC_M11_S	XP1, X1L transfer clock mask 11 start period	0	00000h
347	15Bh	TC_M11_E	XP1, X1L transfer clock mask 11 end period	0	00000h
348	15Ch	TC_M12_S	XP1, X1L transfer clock mask 12 start period	0	00000h
349	15Dh	TC_M12_E	XP1, X1L transfer clock mask 12 end period	0	00000h
350	15Eh	TC_M13_S	XP1, X1L transfer clock mask 13 start period	0	00000h
351	15Fh	TC_M13_E	XP1, X1L transfer clock mask 13 end period	0	00000h
352	160h	TC_M14_S	XP1, X1L transfer clock mask 14 start period	0	00000h
353	161h	TC_M14_E	XP1, X1L transfer clock mask 14 end period	0	00000h
354	162h	TC_M15_S	XP1, X1L transfer clock mask 15 start period	0	00000h
355	163h	TC_M15_E	XP1, X1L transfer clock mask 15 end period	0	00000h
356	164h	TC_M16_S	XP1, X1L transfer clock mask 16 start period	0	00000h
357	165h	TC_M16_E	XP1, X1L transfer clock mask 16 end period	0	00000h
358	166h	TC_M17_S	XP1, X1L transfer clock mask 17 start period	0	00000h
359	167h	TC_M17_E	XP1, X1L transfer clock mask 17 end period	0	00000h
360	168h	TC_M18_S	XP1, X1L transfer clock mask 18 start period	0	00000h
361	169h	TC_M18_E	XP1, X1L transfer clock mask 18 end period	0	00000h
362	16Ah	TC_M19_S	XP1, X1L transfer clock mask 19 start period	0	00000h
363	16Bh	TC_M19_E	XP1, X1L transfer clock mask 19 end period	0	00000h
364 to 395	16Ch to 18Bh	Reserved	Reserved	0	00000h

17.1. Register Map (Continued)

Address		Register	Description	Default	
13) Configuration & Test Register Setting					
396	18Ch	Reserved	Reserved	0	00000h
397	18Dh	Reserved	Reserved	0	00000h
398	18Eh	Reserved	Reserved	0	00000h
399	18Fh	CLP_CFG0	CLP configuration 0	0	00000h
400	190h	ADC_CFG0	ADC configuration 0	0	00000h
401	191h	TEST_PAT1	Test pattern configuration 1	0	00000h
402	192h	Reserved	Reserved	0	00000h
403	193h	CLK_CFG0	Clock configuration 0	0	00000h
404	194h	CLK_CFG1	Clock configuration 1	0	00000h
405	195h	CLK_CFG2	Clock configuration 2	0	00000h
406	196h	Reserved	Reserved	0	00000h
407	197h	TEST_PAT2	Test pattern configuration 2	0	00000h
408	198h	Reserved	Reserved	0	00000h
409	199h	TEST_PAT0	Test pattern configuration 0	0	00000h
410	19Ah	Reserved	Reserved	0	00000h
411	19Bh	Reserved	Reserved	0	00000h
412	19Ch	Reserved	Reserved	0	00000h
413	19Dh	CLK_CFG4	Clock configuration 4	0	00000h
414	19Eh	Reserved	Reserved	0	00000h
415	19Fh	CLK_CFG5	Clock configuration 5	0	00000h
416	1A0h	Reserved	Reserved	0	00000h
417	1A1h	Reserved	Reserved	0	00000h
418	1A2h	Reserved	Reserved	0	00000h
419	1A3h	Reserved	Reserved	0	00000h
420	1A4h	Reserved	Reserved	0	00000h
421	1A5h	Reserved	Reserved	0	00000h
422	1A6h	Reserved	Reserved	0	00000h
423	1A7h	SYS_TM0	System test configuration 0	0	00000h
424	1A8h	Reserved	Reserved	0	00000h
425	1A9h	Reserved	Reserved	0	00000h
426	1AAh	Reserved	Reserved	0	00000h
427	1ABh	Reserved	Reserved	0	00000h
428	1ACh	Reserved	Reserved	0	00000h
429	1ADh	Reserved	Reserved	0	00000h
430	1AEh	Reserved	Reserved	0	00000h
431	1AFh	Reserved	Reserved	0	00000h
432	1B0h	Reserved	Reserved	0	00000h
433	1B1h	TEST_PAT3	Test pattern configuration 3	0	00000h
434	1B2h	TEST_PAT4	Test pattern configuration 4	0	00000h

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17.2. Register Definition

1) Configuration

Address		Name	Description	Default	
0	000h	SPI_CFG	SPI configuration	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	SPI_RW	SPI Read / write selection "0": Write mode, "1": Read mode	Immediate	W	0
1	SDI_BUFF_CTRL	I/O buffer control for "SDI" pin "0" : Input mode "1" : Bidirectional mode ("SDI" outputs register data with SPI_R/W="1")	Immediate	W	0
19:2		Reserved			0

Address		Name	Description	Default	
1	001h	REV_RST_CFG	Chip revision / reset configuration	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
1:0	RESERVE_REV[1:0]	Reserved revision	-	R	X ¹
4:2	MINOR_REV[2:0]	Minor revision	-	R	X ¹
7:5	MAJOR_REV[2:0]	Major revision	-	R	X ¹
8	RST_ALL	All register reset This flag will be automatically cleared. "0": Normal, "1": Reset	Immediate	R/W	0
10:9	Reserved	Reserved			0
11	MCK_DET_DIS	MCLK stop detection (MCLK frequency < 0.5 MHz) "0": Enable, "1": Disable	Immediate	R/W	0
16:12	Reserved	Reserved			0
19:17	MCLK_FRQ[2:0]	MCLK input frequency range setting "0": 1–4.6 MHz "1": 4–11 MHz "2": 9–16 MHz "3": 14–21 MHz "4" to "7" : Reserved	Immediate	R/W	0

Note:

1. X – means "undefined"

Address		Name	Description	Default	
2	002h	STB_CFG	Standby configuration	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	STB_ALL	All standby without SPI "0": Normal, "1": Standby	Immediate	R/W	0
1	STB_DLL	DLL standby "0": Normal, "1": Standby	Immediate	R/W	0
2	STB_PLL	PLL standby "0": Normal, "1": Standby	Immediate	R/W	0
3	STB_AIN1	1st analog channel standby (AIN1_D = "0") "0": Normal, "1": Standby	Immediate	R/W	0
4	STB_AIN2	2nd analog channel standby (AIN2_D = "0") "0": Normal, "1": Standby	Immediate	R/W	0
5	STB_AIN3	3rd analog channel standby (AIN3_D = "0") "0": Normal, "1": Standby	Immediate	R/W	0
6	STB_AIN4	4th analog channel standby (AIN4_D = "0") "0": Normal, "1": Standby	Immediate	R/W	0
9:7		Reserved			0
10	EN_OUT	All output enable "0": Standby condition, "1": Normal	Immediate	R/W	0
11	EN_LED	LED driver enable "0": Standby, "1": Normal	Immediate	R/W	0
19:12	Reserved	Reserved			0

Address		Name	Description	Default	
3	003h	OUTPUT_CFG	Output function configuration	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	CMOS_OUT	Output mode selection "0": LVDS output (TA, TB, TC, TCLK) "1": CMOS output (D[5:0], CK[1:0])	Immediate	R/W	0
2:1	LVDS_CH_SEL[1:0]	LVDS output channel selection "0": All standby (TCLK, TA, TB, TC: Standby Hi-Z) "1": 1 pair LVDS output (TA: Enable, TB and TC: Standby Hi-Z) "2": 2 pair LVDS output (TA, TB: Enable, TC Standby Hi-Z) "3": 3 pair LVDS output (all Enable)	Immediate	R/W	0
3	LVDS_DAT_2X	LVDS output data rate selection "0": ×1 mode, "1": ×2 mode	Immediate	R/W	0
4	LVDS_7BIT	LVDS serialize bit selection "0": 8-bit mode, "1": 7-bit mode	Immediate	R/W	0
6:5	LVDS_VS[1:0]	LVDS output differential voltage selection (Terminal resistor: 100 Ω) "0": 350 mV, "1": 400 mV, "2": reserved, "3": 300 mV	Immediate	R/W	0
8:7	STO_SEL[1:0]	Status bit output mode setting (LVDS 3pair and 7bit mode only) "0": Mode 0, "1": Mode 1, "2": Mode 2, "3": Mode 3	Immediate	R/W	0
9	Reserved	Reserved			0
10	CMOS_PORT_SEL	CMOS output port selection "0": 4-bit port mode (4-bit port × 4) "1": 8-bit port mode (8-bit port × 2)	Immediate	R/W	0
12:11	Reserved	Reserved			0
13	CMOS_CK0_POL	CMOS CK0 output polarity "0": Normal, "1": Inverse	Immediate	R/W	0
15:14	Reserved	Reserved			0
16	CMOS_CK1_POL	CMOS CK1 output polarity "0": Normal, "1": Inverse	Immediate	R/W	0
18:17	DRV_DOUT[1:0]	D[5:0], D6/CK0, D7/CK1 output driver current setting "0": 2 mA, "1": 4 mA, "2": 8 mA, "3": reserved	Immediate	R/W	0
19	Reserved	Reserved			0

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Address		Name	Description	Default	
4	004h	PIN_FNC_CFG0	Pin function configuration 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	Reserved	Reserved			0
2:1	GPIO1_SDO_SEL[1:0]	SDO/GPIO1 pin function setting "0": GPI1 (General Purpose Input) "1": GPO1 (General purpose Output) "2": Reserved (Internal test 1) "3": SDO (Serial I/F data output)	Immediate	R/W	0
4:3	Reserved	Reserved			0
6:5	GPIO3_XST_SEL[1:0]	XST/GPIO3 pin function setting "0": GPI3 (General purpose Input 3) "1": GPO3 (General purpose Output 3) "2": Reserved (Internal test 2) "3": XST	Immediate	R/W	0
7	XLSYNC_SEL	Line synchronous mode selection "0": Input, External synchronous mode "1": Output, Internal synchronous mode	Immediate	R/W	0
8	REF_DAC_IN	Internal reference DAC level (V_{rnt}) output disable "0": Enable, "REF_AIN" is V_{rnt} monitor output "1": Disable, "REF_AIN" is input mode	Immediate	R/W	0
9	XLSYNC_POL	XLSYNC polarity selection "0": Active high, "1": Active low	Immediate	R/W	0
10	LED_EN_POL	LEDn_EN (n = 1 to 3) polarity selection "0": Active high, "1": Active low	Immediate	R/W	0
16:11	Reserved	Reserved			0
17	GPI1_RD	GP1 input read	-	R	0
18	Reserved	Reserved			0
19	GPI3_RD	GP3 input read	-	R	0

Address		Name	Description	Default	
5	005h	PIN_FNC_CFG1	Pin function configuration 1	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
3:0	Reserved	Reserved			0
7:4	GPIO1_OUT_SEL[3:0]	GPIO1 output signal selection (GPIO1_SDO_SEL[1:0]="1") "0" : Low, "1" : High, "2" : LVCK, "3" : CLP_B, "4" : SHP_A, "5" : SHP_B, "6" : SHD_A, "7" : SHD_B, "8" : ADCK, "9" : LS, "10" : CLPDM, "11" : Reserved, "12" : ADST	Immediate	R/W	0
11:8	Reserved	Reserved			0
15:12	GPIO3_OUT_SEL[3:0]	GPIO3 output signal selection (GPIO3_XST_SEL[1:0]="1") "0" : Low, "1" : High, "2" : LVCK, "3" : CLP_A, "4" : SHP_A, "5" : SHP_B, "6" : SHD_A, "7" : SHD_B, "8" : ADCK, "9" : LS, "10" : CLPDM, "11" : Reserved, "12" : ADST	Immediate	R/W	0
17:16	XLSYNC_OUT[1:0]	XLSYNC output signal selection "0": LS (internal synchronous signal) "1": CLPDM (clamp period signal) "2": Reserved "3": Reserved	Immediate	R/W	0
19:18	Reserved	Reserved			0

Address		Name	Description	Default	
6	006h	PIN_FNC_CFG2	Pin function configuration 2	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
1:0	DRV_SEL0[1:0]	GPIO1/SDO, GPIO3/XSToutput driver current setting "0": 2 mA, "1": 4 mA, "2": 8 mA, "3": reserved	Immediate	R/W	0
3:2	DRV_SEL1[1:0]	XP1, XP2 and XRS output driver current setting "0": 2 mA, "1": 4 mA, "2": 8 mA, "3": reserved	Immediate	R/W	0
5:4	Reserved	Reserved			0
7:6	DRV_SEL3[1:0]	X1L, X2L output driver current setting "0": 2 mA, "1": 4 mA, "2": 8 mA, "3": reserved	Immediate	R/W	0
19:8	Reserved	Reserved			0

Address		Name	Description	Default	
7	007h	LINT	Line cycle setting / XLSYNC mask setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
19:0	LINT[19:0]	- Internal line synchronous (XLSYNC_SEL = "1") Line cycle setting Line cycle = (LINT[19:0] + 1) × t_{MCLK} - External line synchronous mode (XLSYNC_SEL = "0") XLSYNC mask period setting Mask period = (LINT[19:0]) × t_{MCLK}	LS	R/W	0

Address		Name	Description	Default	
8	008h	DM_STR	CLPDM start period setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
19:0	DM_STR[19:0]	Number of CLPDM start period = (DM_STR[19:0] + 1) × t _{MCLK}	LS	R/W	0

Address		Name	Description	Default	
9	009h	DM_END	CLPDM end period setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
19:0	DM_END[19:0]	Number of CLPDM end period = (DM_END[19:0] + 1) × t _{MCLK}	LS	R/W	0

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2) Analog Setting

Address		Name	Description	Default	
16	010h	AIN_CFG0	Analog input configuration 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
1:0	AIN_CH_SEL	Analog input channel mode selection "0": 4-channel mode (All channel) "1": 3-channel mode (AIN1, AIN2 and AIN3) "2": 2-channel mode (AIN1 and AIN3) "3": Reserved	Immediate	R/W	0
2	AIN1_POL	1st analog channel polarity "0": Negative input (for CCD) "1": Positive input (for CMOS)	Immediate	R/W	0
3	AIN1_SH_CDS	1st analog channel sampling mode selection "0": CDS mode, "1": SH mode	Immediate	R/W	0
5:4	AIN1_REF_SEL[1:0]	1st analog channel reference and clamp level selection "0": Internal reference 2.2 V "1": Internal reference DAC ($V_{\text{rnt}} = 0.5, 1.1, 1.5, 2.0 \text{ V}$) "2": External input reference ($V_{\text{ext}} = \text{"REF_AIN"}$) "3": Reserved	Immediate	R/W	0
7:6	AIN1_CLP_SEL[1:0]	1st analog channel (AIN1) clamp mode selection "0": Pixel clamp "1": Line clamp "2": Non clamp and Internal reference (SH_REF_EN on) "3": Non clamp and External reference (SH_REF_EN off)	Immediate	R/W	0
8	AIN2_POL	2nd analog channel polarity "0": Negative input (for CCD) "1": Positive input (for CMOS)	Immediate	R/W	0
9	AIN2_SH_CDS	2nd analog channel sampling mode selection "0": CDS mode, "1": SH mode	Immediate	R/W	0
11:10	AIN2_REF_SEL[1:0]	2nd analog channel reference and clamp level selection "0": Internal reference 2.2V "1": Internal reference DAC ($V_{\text{rnt}} = 0.5, 1.1, 1.5, 2.0 \text{ V}$) "2": External input reference ($V_{\text{ext}} = \text{"REF_AIN"}$) "3": Reserved	Immediate	R/W	0
13:12	AIN2_CLP_SEL[1:0]	2nd analog channel (AIN2) clamp mode selection "0": Pixel clamp "1": Line clamp "2": Non clamp and internal reference (SH_REF_EN on) "3": Non clamp and external reference (SH_REF_EN off)	Immediate	R/W	0
14	AIN3_POL	3rd analog channel polarity "0": Negative input (for CCD) "1": Positive input (for CMOS)	Immediate	R/W	0
15	AIN3_SH_CDS	3rd analog channel sampling mode selection "0": CDS mode, "1": SH mode	Immediate	R/W	0
17:16	AIN3_REF_SEL[1:0]	3rd analog channel reference and clamp level selection "0": Internal reference 2.2 V "1": Internal reference DAC ($V_{\text{rnt}} = 0.5, 1.1, 1.5, 2.0 \text{ V}$) "2": External input reference ($V_{\text{ext}} = \text{"REF_AIN"}$) "3": Reserved	Immediate	R/W	0
19:18	AIN3_CLP_SEL[1:0]	3rd analog channel (AIN3) clamp mode selection "0": Pixel clamp "1": Line clamp "2": Non clamp and internal reference (SH_REF_EN on) "3": Non clamp and external reference (SH_REF_EN off)	Immediate	R/W	0

Address		Name	Description	Default	
17	011h	AIN_CFG1	Analog input configuration 1	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	AIN4_POL	4th analog channel polarity "0": Negative input (for CCD) "1": Positive input (for CMOS)	Immediate	R/W	0
1	AIN4_SH_CDS	4th analog channel sampling mode selection "0": CDS mode, "1": SH mode	Immediate	R/W	0
3:2	AIN4_REF_SEL[1:0]	4th analog channel reference and clamp level selection "0": Internal reference 2.2 V "1": Internal reference DAC ($V_{\text{int}} = 0.5, 1.1, 1.5, 2.0 \text{ V}$) "2": External input reference ($V_{\text{ext}} = \text{"REF_AIN"}$) "3": Reserved	Immediate	R/W	0
5:4	AIN4_CLP_SEL[1:0]	4th analog channel (AIN4) clamp mode selection "0": Pixel clamp "1": Line clamp "2": Non clamp and Internal reference (SH_REF_EN on) "3": Non clamp and External reference (SH_REF_EN off)	Immediate	R/W	0
7:6	VRINT_SEL[1:0]	Internal reference DAC level (V_{int}) selection "0": 1.5 V, "1": 2.0 V, "2": 0.5 V, "3": 1.1 V	Immediate	R/W	0
8	SAM_RATE	Sample rate mode setting "0": $\times 1$ mode, "1": $\times 2$ mode	Immediate	R/W	0
19:9	Reserved	Reserved			0

Address		Name	Description	Default	
18	012h	SHP_A_ADJ	SHP_A timing adjustment	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	SHP_A_TR[5:0]	SHP_A rising Edge delay from MCK $t_{\text{RP_A}} = \text{SHP_A_TR}[5:0] \times t_{\text{MCLK}}/48 \text{ [ns]}$	Immediate	R/W	0
7:6	Reserved	Reserved			0
13:8	SHP_A_TF[5:0]	SHP_A falling Edge delay from MCK $t_{\text{FP_A}} = \text{SHP_A_TF}[5:0] \times t_{\text{MCLK}}/48 \text{ [ns]}$	Immediate	R/W	0
19:14		Reserved			0

Address		Name	Description	Default	
19	013h	SHD_A_ADJ	SHD_A timing adjustment	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	SHD_A_TR[5:0]	SHD_A rising edge delay from MCK $t_{\text{RP_A}} = \text{SHD_A_TR}[5:0] \times t_{\text{MCLK}}/48 \text{ [ns]}$	Immediate	R/W	0
7:6	Reserved	Reserved			0
13:8	SHD_A_TF[5:0]	SHD_A falling Edge delay from MCK $t_{\text{FP_A}} = \text{SHD_A_TF}[5:0] \times t_{\text{MCLK}}/48 \text{ [ns]}$	Immediate	R/W	0
19:14		Reserved			0

Address		Name	Description	Default	
20	014h	SHP_B_ADJ	SHP_B timing adjustment	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	SHP_B_TR[5:0]	SHP_B rising edge delay from MCK $t_{\text{RP_B}} = \text{SHP_B_TR}[5:0] \times t_{\text{MCLK}}/48 \text{ [ns]}$	Immediate	R/W	0
7:6	Reserved	Reserved			0
13:8	SHP_B_TF[5:0]	SHP_B falling Edge delay from MCK $t_{\text{FP_B}} = \text{SHP_B_TF}[5:0] \times t_{\text{MCLK}}/48 \text{ [ns]}$	Immediate	R/W	0
19:14		Reserved			0

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Address		Name	Description	Default	
21	015h	SHD_B_ADJ	SHD_B timing adjustment	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	SHD_B_TR[5:0]	SHD_B rising edge delay from MCK $t_{RP_B} = SHD_B_TR[5:0] \times t_{MCLK}/48$ [ns]	Immediate	R/W	0
7:6	Reserved	Reserved			0
13:8	SHD_B_TF[5:0]	SHD_B falling edge delay from MCK $t_{FP_B} = SHD_B_TF[5:0] \times t_{MCLK}/48$ [ns]	Immediate	R/W	0
19:14		Reserved			0

Address		Name	Description	Default	
23	017h	GAIN_1	AIN1 gain setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	APG_1[5:0]	Analog gain setting for AIN1 Gain = $APG_1[5:0] \times 3/64 + 0.5$ [V/V]	Immediate	R/W	0
7:6	Reserved	Reserved			0
15:8	DPG_1[7:0]	Digital gain setting for AIN1 Gain = $DPG_1[7:0] \times 1/256 + 1$ [V/V]	Immediate	R/W	0
19:16	Reserved	Reserved			0

Address		Name	Description	Default	
24	018h	GAIN_2	AIN2 gain setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	APG_2[5:0]	Analog gain setting for AIN2 Gain = $APG_2[5:0] \times 3/64 + 0.5$ [V/V]	Immediate	R/W	0
7:6	Reserved	Reserved			0
15:8	DPG_2[7:0]	Digital gain setting for AIN2 Gain = $DPG_2[7:0] \times 1/256 + 1$ [V/V]	Immediate	R/W	0
19:16	Reserved	Reserved			0

Address		Name	Description	Default	
25	019h	GAIN_3	AIN3 gain setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	APG_3[5:0]	Analog gain setting for AIN3 Gain = $APG_3[5:0] \times 3/64 + 0.5$ [V/V]	Immediate	R/W	0
7:6	Reserved	Reserved			0
15:8	DPG_3[7:0]	Digital gain setting for AIN3 Gain = $DPG_3[7:0] \times 1/256 + 1$ [V/V]	Immediate	R/W	0
19:16	Reserved	Reserved			0

Address		Name	Description	Default	
26	01Ah	GAIN_4	AIN4 gain setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	APG_4[5:0]	Analog gain setting for AIN4 $\text{Gain} = \text{APG_4}[5:0] \times 3/64 + 0.5 \text{ [V/V]}$	Immediate	R/W	0
7:6	Reserved	Reserved			0
15:8	DPG_4[7:0]	Digital gain setting for AIN4 $\text{Gain} = \text{DPG_4}[7:0] \times 1/256 + 1 \text{ [V/V]}$	Immediate	R/W	0
19:16	Reserved	Reserved			0

Address		Name	Description	Default	
27	01Bh	OFDAC_12	Offset correction DAC for AIN1 and 2	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
7:0	OFDAC_1[7:0]	Offset correction DAC output setting for AIN1 $\text{DAC} = \text{OFDAC_1}[7:0] \times 250/128 \text{ [mV]}$ Code is two's complement	Immediate	R/W	0
9:8	Reserved	Reserved			0
17:10	OFDAC_2[7:0]	Offset correction DAC output setting for AIN2 $\text{DAC} = \text{OFDAC_2}[7:0] \times 250/128 \text{ [mV]}$ Code is two's complement	Immediate	R/W	0
19:18	Reserved	Reserved			0

Address		Name	Description	Default	
28	01Ch	OFDAC_34	Offset correction DAC for AIN3 and 4	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
7:0	OFDAC_3[7:0]	Offset correction DAC output setting for AIN3 $\text{DAC} = \text{OFDAC_3}[7:0] \times 250/128 \text{ [mV]}$ Code is two's complement	Immediate	R/W	0
9:8	Reserved	Reserved			0
17:10	OFDAC_4[7:0]	Offset correction DAC output setting for AIN4 $\text{DAC} = \text{OFDAC_4}[7:0] \times 250/128 \text{ [mV]}$ Code is two's complement	Immediate	R/W	0
19:18	Reserved	Reserved			0

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3) LED Driver

Address		Name	Description	Default	
32	020h	LED_DRV_SET	LED driver setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	LED1_EN	LED driver 1 enable "0": Disable, "1": Enable	Immediate	R/W	0
1	LED2_EN	LED driver 2 enable "0": Disable, "1": Enable	Immediate	R/W	0
2	LED3_EN	LED driver 3 enable "0": Disable, "1": Enable	Immediate	R/W	0
3	Reserved	Reserved			0
7:4	LED1_CS[3:0]	Current adjustment setting for 1-channel LED driver $ILD = LED_CS1[3:0] \times 4 \text{ [mA]}$	Immediate	R/W	0
11:8	LED2_CS[3:0]	Current adjustment setting for 2-channel LED driver $ILD = LED_CS2[3:0] \times 4 \text{ [mA]}$	Immediate	R/W	0
15:12	LED3_CS[3:0]	Current adjustment setting for 3-channel LED driver $ILD = LED_CS3[3:0] \times 4 \text{ [mA]}$	Immediate	R/W	0
19:16	Reserved	Reserved			0

Address		Name	Description	Default	
33	021h	LED_DRV_DLY	LED driver on/off delay setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
3:0	ILED_INT_S[3:0]	Interval period setting for LED fine current step in initial stage $t_{INT_S} = LED_INT_S[3:0] \times t_{MCLK}$	Immediate	R/W	0
7:4	ILED_INT_U[3:0]	Interval period setting for LED coarse current step up $t_{INT_U} = LED_INT_U[3:0] \times t_{MCLK}$	Immediate	R/W	0
9:8	ILED_STP_U[1:0]	Step wide setting for LED coarse current increase $ISTP_U = ILED_STP_U[1:0] \times 4 \text{ mA}$	Immediate	R/W	0
13:10	ILED_INT_E[3:0]	Interval period setting for LED fine current step in finalized stage $t_{INT_E} = LED_INT_E[3:0] \times t_{MCLK}$	Immediate	R/W	0
17:14	ILED_INT_D[3:0]	Interval period setting for LED coarse current step down $t_{INT_D} = LED_INT_D[3:0] \times t_{MCLK}$	Immediate	R/W	0
19:18	ILED_STP_D[1:0]	Step wide setting for LED coarse current decrease $ISTP_D = ILED_STP_D[1:0] \times 4 \text{ mA}$	Immediate	R/W	0

4) Timing Generator Configuration

Address		Name	Description	Default	
36	024h	TG_OUT_EN	TG output enable	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	TG_OUT_EN	All TG output enable, XST toggle setting enable "0" : Disable (HiZ), "1" : Enable	Immediate	R/W	0
4:1	Reserved	Reserved			0
5	LED2_TGL_EN	LED2_TG toggle setting enable "0" : Disable, "1" : Enable	Immediate	R/W	0
6	LED1_TGL_EN	LED1_TG toggle setting enable "0" : Disable, "1" : Enable	Immediate	R/W	0
8:7	Reserved	Reserved			0
9	XP1_TGL_EN	XP1 toggle setting enable "0" : Disable, "1" : Enable	Immediate	R/W	0
10	LED3_TGL_EN	LED3_TG toggle setting enable "0" : Disable, "1" : Enable	Immediate	R/W	0
11	X1L_TGL_EN	X1L toggle setting enable "0" : Disable, "1" : Enable	Immediate	R/W	0
17:12	Reserved	Reserved			0
18	XP1_P	XP1 initial polarity setting (XP1_FC_EN = "0") "0": Low, "1": High	Immediate	R/W	0
19	Reserved	Reserved			0

Address		Name	Description	Default	
37	025h	TG_OUT_MD0	TG output mode setting 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
1:0	XST_MODE[1:0]	XST mode setting "0": Normal, "1": Invert, "2": Low, "3": High	Immediate	R/W	0
19:2	Reserved	Reserved			0

Address		Name	Description	Default	
38	026h	TG_OUT_MD1	TG output mode setting 1	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
1:0	XP1_MODE[1:0]	XP1 mode setting "0": Normal, "1": Invert, "2": Low, "3": High	Immediate	R/W	0
3:2	Reserved	Reserved			0
5:4	X1L_MODE[1:0]	X1L mode setting "0": Normal, "1": Invert, "2": Low, "3": High	Immediate	R/W	0
7:6	Reserved	Reserved			0
8	XST_P	XST initial polarity setting "0": Low, "1": High	Immediate	R/W	0
17:9	Reserved	Reserved			0
18	X1L_P	X1L initial polarity setting (X1L_FC_EN = "0") "0": Low, "1": High	Immediate	R/W	0
19	Reserved	Reserved			0

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5) Fast Transfer Clock Setting

Address		Name	Description	Default	
40	028h	XP1_FC_SET0	XP1 fast clock setting register 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	XP1_TR_X1[5:0]	XP1 fast transfer clock rising edge delay from MCK (x1 mode) $t_{TR_P1} = XP1_TR[5:0] \times t_{MCLK}/48$ [ns] (XP1_TR_X1 = 0 to 47)	Immediate	R/W	0
7:6	Reserved	Reserved			0
13:8	XP1_TF_X1[5:0]	XP1 fast transfer clock falling edge delay from MCK (x1 mode) $t_{TF_P1} = XP1_TF[5:0] \times t_{MCLK}/48$ [ns] (XP1_TF_X1 = 0 to 47)	Immediate	R/W	0
17:14	Reserved	Reserved			0
18	XP1_MSK_EN	XP1 fast transfer clock Mask function enable setting "0": Disable (continuous clock output), "1": Enable	Immediate	R/W	0
19	XP1_FC_EN	XP1 fast transfer clock enable setting "0": Disable, "1": Enable	Immediate	R/W	0

Address		Name	Description	Default	
41	029h	XP1_FC_SET1	XP1 fast clock setting register 1	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
4:0	XP1_TR_X2[4:0]	XP1 fast transfer clock rising edge delay from MCK (x2 mode) $t_{TR_P1} = XP1_TR[4:0] \times t_{MCLK}/48$ [ns] (XP1_TR_X2 = 0 to 23)	Immediate	R/W	0
9:5	XP1_TF_X2[4:0]	XP1 fast transfer clock falling edge delay from MCK (x2 mode) $t_{TF_P1} = XP1_TF[4:0] \times t_{MCLK}/48$ [ns] (XP1_TF_X2 = 0 to 23)	Immediate	R/W	0
13:10	XP1_TR_X4[3:0]	XP1 fast transfer clock rising edge delay from MCK (x4 mode) $t_{TR_P1} = XP1_TR[3:0] \times t_{MCLK}/48$ [ns] (XP1_TR_X4 = 0 to 11)	Immediate	R/W	0
17:14	XP1_TF_X4[3:0]	XP1 fast transfer clock falling edge delay from MCK (x4 mode) $t_{TF_P1} = XP1_TF[3:0] \times t_{MCLK}/48$ [ns] (XP1_TF_X4 = 0 to 11)	Immediate	R/W	0
19:18	Reserved	Reserved			0

Address		Name	Description	Default	
42	02Ah	XP1_M_F0	XP1 fast transfer clock frequency rate setting 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
1:0	XP1_M0_F[1:0]	XP1 fast transfer clock frequency rate after mask 0 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
3:2	XP1_M1_F[1:0]	XP1 fast transfer clock frequency rate after mask 1 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
5:4	XP1_M2_F[1:0]	XP1 fast transfer clock frequency rate after mask 2 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
7:6	XP1_M3_F[1:0]	XP1 fast transfer clock frequency rate after mask 3 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
9:8	XP1_M4_F[1:0]	XP1 fast transfer clock frequency rate after mask 4 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
11:10	XP1_M5_F[1:0]	XP1 fast transfer clock frequency rate after mask 5 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
13:12	XP1_M6_F[1:0]	XP1 fast transfer clock frequency rate after mask 6 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
15:14	XP1_M7_F[1:0]	XP1 fast transfer clock frequency rate after mask 7 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
17:16	XP1_M8_F[1:0]	XP1 fast transfer clock frequency rate after mask 8 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
19:18	XP1_M9_F[1:0]	XP1 fast transfer clock frequency rate after mask 9 "0": x1, "1": x2, "3": x4	Immediate	R/W	0

Address		Name	Description	Default	
43	02Bh	XP1_M_F1	XP1 fast transfer clock frequency rate setting 1	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
1:0	XP1_M10_F[1:0]	XP1 fast transfer clock frequency rate after mask 10 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
3:2	XP1_M11_F[1:0]	XP1 fast transfer clock frequency rate after mask 11 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
5:4	XP1_M12_F[1:0]	XP1 fast transfer clock frequency rate after mask 12 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
7:6	XP1_M13_F[1:0]	XP1 fast transfer clock frequency rate after mask 13 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
9:8	XP1_M14_F[1:0]	XP1 fast transfer clock frequency rate after mask 14 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
11:10	XP1_M15_F[1:0]	XP1 fast transfer clock frequency rate after mask 15 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
13:12	XP1_M16_F[1:0]	XP1 fast transfer clock frequency rate after mask 16 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
15:14	XP1_M17_F[1:0]	XP1 fast transfer clock frequency rate after mask 17 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
17:16	XP1_M18_F[1:0]	XP1 fast transfer clock frequency rate after mask 18 "0": x1, "1": x2, "3": x4	Immediate	R/W	0
19:18	XP1_M19_F[1:0]	XP1 fast transfer clock frequency rate after mask 19 "0": x1, "1": x2, "3": x4	Immediate	R/W	0

Address		Name	Description	Default	
44	02Ch	X1L_FC_SET	X1L fast clock setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	X1L_TR[5:0]	X1L fast transfer clock rising edge delay from MCK $t_{TR_1L} = X1L_TR[5:0] \times t_{MCLK}/48$ [ns]	Immediate	R/W	0
7:6	Reserved	Reserved			0
13:8	X1L_TF[5:0]	X1L fast transfer clock falling edge delay from MCK $t_{TF_1L} = X1L_TF[5:0] \times t_{MCLK}/48$ [ns]	Immediate	R/W	0
16:14	Reserved	Reserved			0
17	X1L_MSK_EN	X1L fast transfer clock Mask function enable setting "0": Disable (Continuous clock output), "1": Enable	Immediate	R/W	0
18	X1L_FC_F	X1L fast transfer clock frequency rate setting "0": x1, "1": x2	Immediate	R/W	0
19	X1L_FC_EN	X1L Fast transfer clock enable setting "0": Disable, "1": Enable	Immediate	R/W	0

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6) XST Setting

Address		Name	Description	Default	
52	034h	XST_T0	XST toggle setting period 0	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XST_T0[19:0]		XST toggle setting period 0 $= (XST_T0[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
53	035h	XST_T1	XST toggle setting period 1	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XST_T1[19:0]		XST toggle setting period 1 $= (XST_T1[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
54	036h	XST_T2	XST toggle setting period 2	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XST_T2[19:0]		XST toggle setting period 2 $= (XST_T2[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
55	037h	XST_T3	XST toggle setting period 3	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XST_T3[19:0]		XST toggle setting period 3 $= (XST_T3[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
56	038h	XST_T4	XST toggle setting period 4	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XST_T4[19:0]		XST toggle setting period 4 $= (XST_T4[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
57	039h	XST_T5	XST toggle setting period 5	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XST_T5[19:0]		XST toggle setting period 5 $= (XST_T5[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
58	03Ah	XST_T6	XST toggle setting period 6	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XST_T6[19:0]		XST toggle setting period 6 $= (XST_T6[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
59	03Bh	XST_T7	XST toggle setting period 7	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XST_T7[19:0]		XST toggle setting period 7 $= (XST_T7[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

7) LED2_TG Setting

Address		Name	Description	Default	
125	07Dh	LED2_TG_T0	LED2_TG toggle setting period 0	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED2_TG_T0[19:0]		LED2_TG toggle setting period 0 $= (\text{LED2_TG_T0}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
126	07Eh	LED2_TG_T1	LED2_TG toggle setting period 1	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED2_TG_T1[19:0]		LED2_TG toggle setting period 1 $= (\text{LED2_TG_T1}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
127	07Fh	LED2_TG_T2	LED2_TG toggle setting period 2	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED2_TG_T2[19:0]		LED2_TG toggle setting period 2 $= (\text{LED2_TG_T2}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
128	080h	LED2_TG_T3	LED2_TG toggle setting period 3	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED2_TG_T3[19:0]		LED2_TG toggle setting period 3 $= (\text{LED2_TG_T3}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
129	081h	LED2_TG_T4	LED2_TG toggle setting period 4	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED2_TG_T4[19:0]		LED2_TG toggle setting period 4 $= (\text{LED2_TG_T4}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
130	082h	LED2_TG_T5	LED2_TG toggle setting period 5	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED2_TG_T5[19:0]		LED2_TG toggle setting period 5 $= (\text{LED2_TG_T5}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
131	083h	LED2_TG_T6	LED2_TG toggle setting period 6	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED2_TG_T6[19:0]		LED2_TG toggle setting period 6 $= (\text{LED2_TG_T6}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
132	084h	LED2_TG_T7	LED2_TG toggle setting period 7	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED2_TG_T7[19:0]		LED2_TG toggle setting period 7 $= (\text{LED2_TG_T7}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

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Address		Name	Description	Default	
133	085h	LED2_TG_T8	LED2_TG toggle setting period 8	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED2_TG_T8[19:0]	LED2_TG toggle setting period 8 = (LED2_TG_T8[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
134	086h	LED2_TG_T9	LED2_TG toggle setting period 9	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED2_TG_T9[19:0]	LED2_TG toggle setting period 9 = (LED2_TG_T9[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
135	087h	LED2_TG_T10	LED2_TG toggle setting period 10	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED2_TG_T10[19:0]	LED2_TG toggle setting period 10 = (LED2_TG_T10[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
136	088h	LED2_TG_T11	LED2_TG toggle setting period 11	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED2_TG_T11[19:0]	LED2_TG toggle setting period 11 = (LED2_TG_T11[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
137	089h	LED2_TG_T12	LED2_TG toggle setting period 12	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED2_TG_T12[19:0]	LED2_TG toggle setting period 12 = (LED2_TG_T12[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
138	08Ah	LED2_TG_T13	LED2_TG toggle setting period 13	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED2_TG_T13[19:0]	LED2_TG toggle setting period 13 = (LED2_TG_T13[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
139	08Bh	LED2_TG_T14	LED2_TG toggle setting period 14	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED2_TG_T14[19:0]	LED2_TG toggle setting period 14 = (LED2_TG_T14[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
140	08Ch	LED2_TG_T15	LED2_TG toggle setting period 15	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED2_TG_T15[19:0]	LED2_TG toggle setting period 15 = (LED2_TG_T15[19:0] + 1) x t _{MCLK}		Immediate	W 0

8) LED1_TG Setting

Address		Name	Description	Default	
142	08Eh	LED1_TG_T0	LED1_TG toggle setting period 0	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED1_TG_T0[19:0]		LED1_TG toggle setting period 0 $= (\text{LED1_TG_T0}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
143	08Fh	LED1_TG_T1	LED1_TG toggle setting period 1	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED1_TG_T1[19:0]		LED1_TG toggle setting period 1 $= (\text{LED1_TG_T1}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
144	090h	LED1_TG_T2	LED1_TG toggle setting period 2	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED1_TG_T2[19:0]		LED1_TG toggle setting period 2 $= (\text{LED1_TG_T2}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
145	091h	LED1_TG_T3	LED1_TG toggle setting period 3	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED1_TG_T3[19:0]		LED1_TG toggle setting period 3 $= (\text{LED1_TG_T3}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
146	092h	LED1_TG_T4	LED1_TG toggle setting period 4	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED1_TG_T4[19:0]		LED1_TG toggle setting period 4 $= (\text{LED1_TG_T4}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
147	093h	LED1_TG_T5	LED1_TG toggle setting period 5	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED1_TG_T5[19:0]		LED1_TG toggle setting period 5 $= (\text{LED1_TG_T5}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
148	094h	LED1_TG_T6	LED1_TG toggle setting period 6	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED1_TG_T6[19:0]		LED1_TG toggle setting period 6 $= (\text{LED1_TG_T6}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

Address		Name	Description	Default	
149	095h	LED1_TG_T7	LED1_TG toggle setting period 7	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	LED1_TG_T7[19:0]		LED1_TG toggle setting period 7 $= (\text{LED1_TG_T7}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W 0

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Address		Name	Description	Default	
150	096h	LED1_TG_T8	LED1_TG toggle setting period 8	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED1_TG_T8[19:0]	LED1_TG toggle setting period 8 = (LED1_TG_T8[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
151	097h	LED1_TG_T9	LED1_TG toggle setting period 9	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED1_TG_T9[19:0]	LED1_TG toggle setting period 9 = (LED1_TG_T9[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
152	098h	LED1_TG_T10	LED1_TG toggle setting period 10	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED1_TG_T10[19:0]	LED1_TG toggle setting period 10 = (LED1_TG_T10[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
153	099h	LED1_TG_T11	LED1_TG toggle setting period 11	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED1_TG_T11[19:0]	LED1_TG toggle setting period 11 = (LED1_TG_T11[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
154	09Ah	LED1_TG_T12	LED1_TG toggle setting period 12	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED1_TG_T12[19:0]	LED1_TG toggle setting period 12 = (LED1_TG_T12[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
155	09Bh	LED1_TG_T13	LED1_TG toggle setting period 13	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED1_TG_T13[19:0]	LED1_TG toggle setting period 13 = (LED1_TG_T13[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
156	09Ch	LED1_TG_T14	LED1_TG toggle setting period 14	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED1_TG_T14[19:0]	LED1_TG toggle setting period 14 = (LED1_TG_T14[19:0] + 1) x t _{MCLK}		Immediate	W 0

Address		Name	Description	Default	
157	09Dh	LED1_TG_T15	LED1_TG toggle setting period 15	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED1_TG_T15[19:0]	LED1_TG toggle setting period 15 = (LED1_TG_T15[19:0] + 1) x t _{MCLK}		Immediate	W 0

9) XP1 Setting

Address		Name	Description	Default	
208	0D0h	XP1_M_P	XP1 mask initial polarity setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	XP1_M0_P	XP1 mask 0 initial polarity setting "0": Low, "1": High	Immediate	W	0
1	XP1_M1_P	XP1 mask 1 initial polarity setting "0": Low, "1": High	Immediate	W	0
2	XP1_M2_P	XP1 mask 2 initial polarity setting "0": Low, "1": High	Immediate	W	0
3	XP1_M3_P	XP1 mask 3 initial polarity setting "0": Low, "1": High	Immediate	W	0
4	XP1_M4_P	XP1 mask 4 initial polarity setting "0": Low, "1": High	Immediate	W	0
5	XP1_M5_P	XP1 mask 5 initial polarity setting "0": Low, "1": High	Immediate	W	0
6	XP1_M6_P	XP1 mask 6 initial polarity setting "0": Low, "1": High	Immediate	W	0
7	XP1_M7_P	XP1 mask 7 initial polarity setting "0": Low, "1": High	Immediate	W	0
8	XP1_M8_P	XP1 mask 8 initial polarity setting "0": Low, "1": High	Immediate	W	0
9	XP1_M9_P	XP1 mask 9 initial polarity setting "0": Low, "1": High	Immediate	W	0
10	XP1_M10_P	XP1 mask 10 initial polarity setting "0": Low, "1": High	Immediate	W	0
11	XP1_M11_P	XP1 mask 11 initial polarity setting "0": Low, "1": High	Immediate	W	0
12	XP1_M12_P	XP1 mask 12 initial polarity setting "0": Low, "1": High	Immediate	W	0
13	XP1_M13_P	XP1 mask 13 initial polarity setting "0": Low, "1": High	Immediate	W	0
14	XP1_M14_P	XP1 mask 14 initial polarity setting "0": Low, "1": High	Immediate	W	0
15	XP1_M15_P	XP1 mask 15 initial polarity setting "0": Low, "1": High	Immediate	W	0
16	XP1_M16_P	XP1 mask 16 initial polarity setting "0": Low, "1": High	Immediate	W	0
17	XP1_M17_P	XP1 mask 17 initial polarity setting "0": Low, "1": High	Immediate	W	0
18	XP1_M18_P	XP1 mask 18 initial polarity setting "0": Low, "1": High	Immediate	W	0
19	XP1_M19_P	XP1 mask 19 initial polarity setting "0": Low, "1": High	Immediate	W	0

Address		Name	Description	Default	
209	0D1h	XP1_T0	XP1 toggle setting period 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
19:0	XP1_T0[19:0]	XP1 toggle setting period 0 = (XP1_T0[19:0] + 1) × t _{MCLK}	Immediate	W	0

Address		Name	Description	Default	
210	0D2h	XP1_T1	XP1 toggle setting period 1	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
19:0	XP1_T1[19:0]	XP1 toggle setting period 1 = (XP1_T1[19:0] + 1) × t _{MCLK}	Immediate	W	0

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Address		Name	Description	Default	
211	0D3h	XP1_T2	XP1 toggle setting period 2	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XP1_T2[19:0]		XP1 toggle setting period 2 $= (XP1_T2[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
212	0D4h	XP1_T3	XP1 toggle setting period 3	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XP1_T3[19:0]		XP1 toggle setting period 3 $= (XP1_T3[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
213	0D5h	XP1_T4	XP1 toggle setting period 4	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XP1_T4[19:0]		XP1 toggle setting period 4 $= (XP1_T4[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
214	0D6h	XP1_T5	XP1 toggle setting period 5	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XP1_T5[19:0]		XP1 toggle setting period 5 $= (XP1_T5[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
215	0D7h	XP1_T6	XP1 toggle setting period 6	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XP1_T6[19:0]		XP1 toggle setting period 6 $= (XP1_T6[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
216	0D8h	XP1_T7	XP1 toggle setting period 7	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XP1_T7[19:0]		XP1 toggle setting period 7 $= (XP1_T7[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
217	0D9h	XP1_T8	XP1 toggle setting period 8	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XP1_T8[19:0]		XP1 toggle setting period 8 $= (XP1_T8[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
218	0DAh	XP1_T9	XP1 toggle setting period 9	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	XP1_T9[19:0]		XP1 toggle setting period 9 $= (XP1_T9[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
219	0DBh	XP1_T10	XP1 toggle setting period 10	0	00000h
Bit	Symbol	Function		Update Timing	R/W
19:0	XP1_T10[19:0]	XP1 toggle setting period 10 $= (XP1_T10[19:0] + 1) \times t_{MCLK}$		Immediate	W

Address		Name	Description	Default	
220	0DCh	XP1_T11	XP1 toggle setting period 11	0	00000h
Bit	Symbol	Function		Update Timing	R/W
19:0	XP1_T11[19:0]	XP1 toggle setting period 11 $= (XP1_T11[19:0] + 1) \times t_{MCLK}$		Immediate	W

Address		Name	Description	Default	
221	0DDh	XP1_T12	XP1 toggle setting period 12	0	00000h
Bit	Symbol	Function		Update Timing	R/W
19:0	XP1_T12[19:0]	XP1 toggle setting period 12 $= (XP1_T12[19:0] + 1) \times t_{MCLK}$		Immediate	W

Address		Name	Description	Default	
222	0DEh	XP1_T13	XP1 toggle setting period 13	0	00000h
Bit	Symbol	Function		Update Timing	R/W
19:0	XP1_T13[19:0]	XP1 toggle setting period 13 $= (XP1_T13[19:0] + 1) \times t_{MCLK}$		Immediate	W

Address		Name	Description	Default	
223	0DFh	XP1_T14	XP1 toggle setting period 14	0	00000h
Bit	Symbol	Function		Update Timing	R/W
19:0	XP1_T14[19:0]	XP1 toggle setting period 14 $= (XP1_T14[19:0] + 1) \times t_{MCLK}$		Immediate	W

Address		Name	Description	Default	
224	0E0h	XP1_T15	XP1 toggle setting period 15	0	00000h
Bit	Symbol	Function		Update Timing	R/W
19:0	XP1_T15[19:0]	XP1 toggle setting period 15 $= (XP1_T15[19:0] + 1) \times t_{MCLK}$		Immediate	W

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10) LED3_TG Setting

Address		Name	Description	Default		
226	0E2h	LED3_TG_T0	LED3_TG toggle setting period 0	0	00000h	
Bit	Symbol	Function		Update Timing	R/W	Default
19:0	LED3_TG_T0[19:0]	LED3_TG toggle setting period 0 = (LED3_TG_T0[19:0] + 1) x t _{MCLK}		Immediate	W	0

Address		Name	Description	Default	
227	0E3h	LED3_TG_T1	LED3_TG toggle setting period 1		0 00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	LED3_TG_T1[19:0]	LED3_TG toggle setting period 1 = (LED3_TG_T1[19:0] + 1) x t_MCLK		Immediate	W 0

Address		Name	Description	Default		
228	0E4h	LED3_TG_T2	LED3_TG toggle setting period 2	0	00000h	
Bit	Symbol	Function		Update Timing	R/W	Default
19:0	LED3_TG_T2[19:0]	LED3_TG toggle setting period 2 = (LED3_TG_T2[19:0] + 1) x t _{MCLK}		Immediate	W	0

Address		Name	Description	Default		
229	0E5h	LED3_TG_T3	LED3_TG toggle setting period 3	0	00000h	
Bit	Symbol	Function		Update Timing	R/W	Default
19:0	LED3_TG_T3[19:0]	LED3_TG toggle setting period 3 = (LED3_TG_T3[19:0] + 1) x t _{MCLK}		Immediate	W	0

Address		Name	Description	Default		
230	0E6h	LED3_TG_T4	LED3_TG toggle setting period 4	0	00000h	
Bit	Symbol	Function		Update Timing	R/W	Default
19:0	LED3_TG_T4[19:0]	LED3_TG toggle setting period 4 = (LED3_TG_T4[19:0] + 1) x t_MCLK		Immediate	W	0

Address		Name	Description	Default		
231	0E7h	LED3_TG_T5	LED3_TG toggle setting period 5	0	00000h	
Bit	Symbol	Function		Update Timing	R/W	Default
19:0	LED3_TG_T5[19:0]	LED3_TG toggle setting period 5 = (LED3_TG_T5[19:0] + 1) x t _{MCLK}		Immediate	W	0

Address		Name	Description	Default		
232	0E8h	LED3_TG_T6	LED3_TG toggle setting period 6	0	00000h	
Bit	Symbol	Function		Update Timing	R/W	Default
19:0	LED3_TG_T6[19:0]	LED3_TG toggle setting period 6 = (LED3_TG_T6[19:0] + 1) x t _{MCLK}		Immediate	W	0

Address		Name	Description	Default		
233	0E9h	LED3_TG_T7	LED3_TG toggle setting period 7		0	00000h
Bit	Symbol	Function		Update Timing	R/W	Default
19:0	LED3_TG_T7[19:0]	LED3_TG toggle setting period 7 = (LED3_TG_T7[19:0] + 1) × tMCLK		Immediate	W	0

Address		Name	Description	Default	
234	0EAh	LED3_TG_T8	LED3_TG toggle setting period 8	0	00000h
Bit	Symbol		Function	Update Timing	R/W
19:0	LED3_TG_T8[19:0]		LED3_TG toggle setting period 8 $= (\text{LED3_TG_T8}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W

Address		Name	Description	Default	
235	0EBh	LED3_TG_T9	LED3_TG toggle setting period 9	0	00000h
Bit	Symbol		Function	Update Timing	R/W
19:0	LED3_TG_T9[19:0]		LED3_TG toggle setting period 9 $= (\text{LED3_TG_T9}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W

Address		Name	Description	Default	
236	0ECh	LED3_TG_T10	LED3_TG toggle setting period 10	0	00000h
Bit	Symbol		Function	Update Timing	R/W
19:0	LED3_TG_T10[19:0]		LED3_TG toggle setting period 10 $= (\text{LED3_TG_T10}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W

Address		Name	Description	Default	
237	0EDh	LED3_TG_T11	LED3_TG toggle setting period 11	0	00000h
Bit	Symbol		Function	Update Timing	R/W
19:0	LED3_TG_T11[19:0]		LED3_TG toggle setting period 11 $= (\text{LED3_TG_T11}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W

Address		Name	Description	Default	
238	0EEh	LED3_TG_T12	LED3_TG toggle setting period 12	0	00000h
Bit	Symbol		Function	Update Timing	R/W
19:0	LED3_TG_T12[19:0]		LED3_TG toggle setting period 12 $= (\text{LED3_TG_T12}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W

Address		Name	Description	Default	
239	0EFh	LED3_TG_T13	LED3_TG toggle setting period 13	0	00000h
Bit	Symbol		Function	Update Timing	R/W
19:0	LED3_TG_T13[19:0]		LED3_TG toggle setting period 13 $= (\text{LED3_TG_T13}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W

Address		Name	Description	Default	
240	0F0h	LED3_TG_T14	LED3_TG toggle setting period 14	0	00000h
Bit	Symbol		Function	Update Timing	R/W
19:0	LED3_TG_T14[19:0]		LED3_TG toggle setting period 14 $= (\text{LED3_TG_T14}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W

Address		Name	Description	Default	
241	0F1h	LED3_TG_T15	LED3_TG toggle setting period 15	0	00000h
Bit	Symbol		Function	Update Timing	R/W
19:0	LED3_TG_T15[19:0]		LED3_TG toggle setting period 15 $= (\text{LED3_TG_T15}[19:0] + 1) \times t_{\text{MCLK}}$	Immediate	W

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11) X1L Setting

Address		Name	Description	Default	
242	0F2h	X1L_M_P	X1L mask initial polarity setting	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	X1L_M0_P	X1L mask 0 initial polarity setting "0": Low, "1": High	Immediate	W	0
1	X1L_M1_P	X1L mask 1 initial polarity setting "0": Low, "1": High	Immediate	W	0
2	X1L_M2_P	X1L mask 2 initial polarity setting "0": Low, "1": High	Immediate	W	0
3	X1L_M3_P	X1L mask 3 initial polarity setting "0": Low, "1": High	Immediate	W	0
4	X1L_M4_P	X1L mask 4 initial polarity setting "0": Low, "1": High	Immediate	W	0
5	X1L_M5_P	X1L mask 5 initial polarity setting "0": Low, "1": High	Immediate	W	0
6	X1L_M6_P	X1L mask 6 initial polarity setting "0": Low, "1": High	Immediate	W	0
7	X1L_M7_P	X1L mask 7 initial polarity setting "0": Low, "1": High	Immediate	W	0
8	X1L_M8_P	X1L mask 8 initial polarity setting "0": Low, "1": High	Immediate	W	0
9	X1L_M9_P	X1L mask 9 initial polarity setting "0": Low, "1": High	Immediate	W	0
10	X1L_M10_P	X1L mask 10 initial polarity setting "0": Low, "1": High	Immediate	W	0
11	X1L_M11_P	X1L mask 11 initial polarity setting "0": Low, "1": High	Immediate	W	0
12	X1L_M12_P	X1L mask 12 initial polarity setting "0": Low, "1": High	Immediate	W	0
13	X1L_M13_P	X1L mask 13 initial polarity setting "0": Low, "1": High	Immediate	W	0
14	X1L_M14_P	X1L mask 14 initial polarity setting "0": Low, "1": High	Immediate	W	0
15	X1L_M15_P	X1L mask 15 initial polarity setting "0": Low, "1": High	Immediate	W	0
16	X1L_M16_P	X1L mask 16 initial polarity setting "0": Low, "1": High	Immediate	W	0
17	X1L_M17_P	X1L mask 17 initial polarity setting "0": Low, "1": High	Immediate	W	0
18	X1L_M18_P	X1L mask 18 initial polarity setting "0": Low, "1": High	Immediate	W	0
19	X1L_M19_P	X1L mask 19 initial polarity setting "0": Low, "1": High	Immediate	W	0

Address		Name	Description	Default	
243	0F3h	X1L_T0	X1L toggle setting period 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
19:0	X1L_T0[19:0]	X1L toggle setting period 0 = (X1L_T0[19:0] + 1) x t _{MCLK}	Immediate	W	0

Address		Name	Description	Default	
244	0F4h	X1L_T1	X1L toggle setting period 1	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
19:0	X1L_T1[19:0]	X1L toggle setting period 1 = (X1L_T1[19:0] + 1) x t _{MCLK}	Immediate	W	0

Address		Name	Description	Default	
245	0F5h	X1L_T2	X1L toggle setting period 2	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T2[19:0]	X1L toggle setting period 2 $= (X1L_T2[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
246	0F6h	X1L_T3	X1L toggle setting period 3	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T3[19:0]	X1L toggle setting period 3 $= (X1L_T3[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
247	0F7h	X1L_T4	X1L toggle setting period 4	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T4[19:0]	X1L toggle setting period 4 $= (X1L_T4[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
248	0F8h	X1L_T5	X1L toggle setting period 5	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T5[19:0]	X1L toggle setting period 5 $= (X1L_T5[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
249	0F9h	X1L_T6	X1L toggle setting period 6	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T6[19:0]	X1L toggle setting period 6 $= (X1L_T6[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
250	0FAh	X1L_T7	X1L toggle setting period 7	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T7[19:0]	X1L toggle setting period 7 $= (X1L_T7[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
251	0FBh	X1L_T8	X1L toggle setting period 8	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T8[19:0]	X1L toggle setting period 8 $= (X1L_T8[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
252	0FCh	X1L_T9	X1L toggle setting period 9	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T9[19:0]	X1L toggle setting period 9 $= (X1L_T9[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

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Address		Name	Description	Default	
253	0FDh	X1L_T10	X1L toggle setting period 10	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T10[19:0]	X1L toggle setting period 10 $= (X1L_T10[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
254	0FEh	X1L_T11	X1L toggle setting period 11	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T11[19:0]	X1L toggle setting period 11 $= (X1L_T11[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
255	0FFh	X1L_T12	X1L toggle setting period 12	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T12[19:0]	X1L toggle setting period 12 $= (X1L_T12[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
256	100h	X1L_T13	X1L toggle setting period 13	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T13[19:0]	X1L toggle setting period 13 $= (X1L_T13[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
257	101h	X1L_T14	X1L toggle setting period 14	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T14[19:0]	X1L toggle setting period 14 $= (X1L_T14[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
258	102h	X1L_T15	X1L toggle setting period 15	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	X1L_T15[19:0]	X1L toggle setting period 15 $= (X1L_T15[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

12) XP and X1L Mask Setting

Address		Name	Description	Default	
324	144h	TC_M0_S	XP1, X1L transfer clock mask 0 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M0_S[19:0]		XP1, X1L transfer clock mask 0 start period $= (TC_M0_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
325	145h	TC_M0_E	XP1, X1L transfer clock mask 0 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M0_E[19:0]		XP1, X1L transfer clock mask 0 end period $= (TC_M0_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
326	146h	TC_M1_S	XP1, X1L transfer clock mask 1 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M1_S[19:0]		XP1, X1L transfer clock mask 1 start period $= (TC_M1_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
327	147h	TC_M1_E	XP1, X1L transfer clock mask 1 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M1_E[19:0]		XP1, X1L transfer clock mask 1 end period $= (TC_M1_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
328	148h	TC_M2_S	XP1, X1L transfer clock mask 2 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M2_S[19:0]		XP1, X1L transfer clock mask 2 start period $= (TC_M2_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
329	149h	TC_M2_E	XP1, X1L transfer clock mask 2 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M2_E[19:0]		XP1, X1L transfer clock mask 2 end period $= (TC_M2_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
330	14Ah	TC_M3_S	XP1, X1L transfer clock mask 3 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M3_S[19:0]		XP1, X1L transfer clock mask 3 start period $= (TC_M3_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
331	14Bh	TC_M3_E	XP1, X1L transfer clock mask 3 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M3_E[19:0]		XP1, X1L transfer clock mask 3 end period $= (TC_M3_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

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Address		Name	Description	Default	
332	14Ch	TC_M4_S	XP1, X1L transfer clock mask 4 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M4_S[19:0]		XP1, X1L transfer clock mask 4 start period $= (TC_M4_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
333	14Dh	TC_M4_E	XP1, X1L transfer clock mask 4 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M4_E[19:0]		XP1, X1L transfer clock mask 4 end period $= (TC_M4_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
334	14Eh	TC_M5_S	XP1, X1L transfer clock mask 5 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M5_S[19:0]		XP1, X1L transfer clock mask 5 start period $= (TC_M5_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
335	14Fh	TC_M5_E	XP1, X1L transfer clock mask 5 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M5_E[19:0]		XP1, X1L transfer clock mask 5 end period $= (TC_M5_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
336	150h	TC_M6_S	XP1, X1L transfer clock mask 6 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M6_S[19:0]		XP1, X1L transfer clock mask 6 start period $= (TC_M6_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
337	151h	TC_M6_E	XP1, X1L transfer clock mask 6 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M6_E[19:0]		XP1, X1L transfer clock mask 6 end period $= (TC_M6_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
338	152h	TC_M7_S	XP1, X1L transfer clock mask 7 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M7_S[19:0]		XP1, X1L transfer clock mask 7 start period $= (TC_M7_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
339	153h	TC_M7_E	XP1, X1L transfer clock mask 7 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M7_E[19:0]		XP1, X1L transfer clock mask 7 end period $= (TC_M7_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
340	154h	TC_M8_S	XP1, X1L transfer clock mask 8 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M8_S[19:0]		XP1, X1L transfer clock mask 8 start period $= (TC_M8_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
341	155h	TC_M8_E	XP1, X1L transfer clock mask 8 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M8_E[19:0]		XP1, X1L transfer clock mask 8 end period $= (TC_M8_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
342	156h	TC_M9_S	XP1, X1L transfer clock mask 9 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M9_S[19:0]		XP1, X1L transfer clock mask 9 start period $= (TC_M9_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
343	157h	TC_M9_E	XP1, X1L transfer clock mask 9 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M9_E[19:0]		XP1, X1L transfer clock mask 9 end period $= (TC_M9_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
344	158h	TC_M10_S	XP1, X1L transfer clock mask 10 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M10_S[19:0]		XP1, X1L transfer clock mask 10 start period $= (TC_M10_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
345	159h	TC_M10_E	XP1, X1L transfer clock mask 10 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M10_E[19:0]		XP1, X1L transfer clock mask 10 end period $= (TC_M10_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
346	15Ah	TC_M11_S	XP1, X1L transfer clock mask 11 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M11_S[19:0]		XP1, X1L transfer clock mask 11 start period $= (TC_M11_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
347	15Bh	TC_M11_E	XP1, X1L transfer clock mask 11 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M11_E[19:0]		XP1, X1L transfer clock mask 11 end period $= (TC_M11_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

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Address		Name	Description	Default	
348	15Ch	TC_M12_S	XP1, X1L transfer clock mask 12 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M12_S[19:0]		XP1, X1L transfer clock mask 12 start period $= (TC_M12_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
349	15Dh	TC_M12_E	XP1, X1L transfer clock mask 12 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M12_E[19:0]		XP1, X1L transfer clock mask 12 end period $= (TC_M12_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
350	15Eh	TC_M13_S	XP1, X1L transfer clock mask 13 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M13_S[19:0]		XP1, X1L transfer clock mask 13 start period $= (TC_M13_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
351	15Fh	TC_M13_E	XP1, X1L transfer clock mask 13 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M13_E[19:0]		XP1, X1L transfer clock mask 13 end period $= (TC_M13_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
352	160h	TC_M14_S	XP1, X1L transfer clock mask 14 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M14_S[19:0]		XP1, X1L transfer clock mask 14 start period $= (TC_M14_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
353	161h	TC_M14_E	XP1, X1L transfer clock mask 14 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M14_E[19:0]		XP1, X1L transfer clock mask 14 end period $= (TC_M14_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
354	162h	TC_M15_S	XP1, X1L transfer clock mask 15 start period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M15_S[19:0]		XP1, X1L transfer clock mask 15 start period $= (TC_M15_S[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
355	163h	TC_M15_E	XP1, X1L transfer clock mask 15 end period	0	00000h
Bit	Symbol		Function	Update Timing	R/W Default
19:0	TC_M15_E[19:0]		XP1, X1L transfer clock mask 15 end period $= (TC_M15_E[19:0] + 1) \times t_{MCLK}$	Immediate	W 0

Address		Name	Description	Default	
356	164h	TC_M16_S	XP1, X1L transfer clock mask 16 start period	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	TC_M16_S[19:0]	XP1, X1L transfer clock mask 16 start period $= (TC_M16_S[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
357	165h	TC_M16_E	XP1, X1L transfer clock mask 16 end period	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	TC_M16_E[19:0]	XP1, X1L transfer clock mask 16 end period $= (TC_M16_E[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
358	166h	TC_M17_S	XP1, X1L transfer clock mask 17 start period	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	TC_M17_S[19:0]	XP1, X1L transfer clock mask 17 start period $= (TC_M17_S[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
359	167h	TC_M17_E	XP1, X1L transfer clock mask 17 end period	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	TC_M17_E[19:0]	XP1, X1L transfer clock mask 17 end period $= (TC_M17_E[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
360	168h	TC_M18_S	XP1, X1L transfer clock mask 18 start period	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	TC_M18_S[19:0]	XP1, X1L transfer clock mask 18 start period $= (TC_M18_S[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
361	169h	TC_M18_E	XP1, X1L transfer clock mask 18 end period	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	TC_M18_E[19:0]	XP1, X1L transfer clock mask 18 end period $= (TC_M18_E[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
362	16Ah	TC_M19_S	XP1, X1L transfer clock mask 19 start period	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	TC_M19_S[19:0]	XP1, X1L transfer clock mask 19 start period $= (TC_M19_S[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

Address		Name	Description	Default	
363	16Bh	TC_M19_E	XP1, X1L transfer clock mask 19 end period	0	00000h
Bit	Symbol	Function		Update Timing	R/W Default
19:0	TC_M19_E[19:0]	XP1, X1L transfer clock mask 19 end period $= (TC_M19_E[19:0] + 1) \times t_{MCLK}$		Immediate	W 0

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13) Configuration & Test Register Setting

Address		Name	Description	Default	
399	18Fh	CLP_CFG0	Clamp setting configuration 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	Reserved	Reserved			0
1	EN_CLPDM	Enable the generation fo CLPDM "0" : Disable, "1" : Enable	Immediate	R/W	0
19:2	Reserved	Reserved			0

Address		Name	Description	Default	
400	190h	ADC_CFG0	ADC_CFG0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
15:0	Reserved	Reserved			0
16	ADC_DAT_FRM	ADC data output format selection "0" : Two's complement, "1" : Offset binary	Immediate	R/W	0
17	LS_CNT_RST	Line counter reset "0" : Normal, "1" : Hold counter to "0" Before LS_CNT_RST="1", "ALL_RST"(Add1[8]) should be executed.	Immediate	R/W	0
19:18	Reserved	Reserved			0

Address		Name	Description	Default	
401	191h	TEST_PAT1	Test pattern configuration 1	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
15:0	Reserved				0
16	DIS_TEST_PAT_CH1	TEST_PAT_EN="1", Disable test pattern for ch1 TEST_PAT_EN="0", Disable test pattern for TA ch "0" : Enable, "1" : Disable	Immediate	R/W	0
17	DIS_TEST_PAT_CH2	TEST_PAT_EN="1", Disable test pattern for ch2 TEST_PAT_EN="0", Disable test pattern for TB ch "0" : Enable, "1" : Disable	Immediate	R/W	0
18	DIS_TEST_PAT_CH3	TEST_PAT_EN="1", Disable test pattern for ch3 TEST_PAT_EN="0", Disable test pattern for TC ch "0" : Enable, "1" : Disable	Immediate	R/W	0
19	DIS_TEST_PAT_CH4	TEST_PAT_EN="1", Disable test pattern for ch4 "0" : Enable, "1" : Disable	Immediate	R/W	0

Address		Name	Description	Default	
403	193h	CLK_CFG0	Clock configuration 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
15:0	Reserved	Reserved			0
16	CLP_TF_AT_DIS	The end of clamp period setting mode selection "0" : Automatic "1" : Manual setting by using CLP_A/B_TF	Immediate	R/W	0
19:17	Reserved	Reserved			0

Address		Name	Description	Default	
404	194h	CLK_CFG1	Clock configuration 1	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	Reserved	Reserved			0
11:6	CLP_A_TF[5:0]	Manual setting the end of clamp period for AIN1 and 2 CLP_A_TF[5:0] x tMCLK/48 [ns]	Immediate	R/W	0
17:12	CLP_B_TF[5:0]	Manual setting the end of clamp period for AIN1 and 2 CLP_B_TF[5:0] x tMCLK/48 [ns]	Immediate	R/W	0
19:12	Reserved	Reserved			0

Address		Name	Description	Default	
405	195h	CLK_CFG2	Clock configuration 2	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
7:0	TCLK_FRM_SET[7:0]	TCLK clock frame pattern setting 8bit serialize mode : "11000011" 7bit serialize mode : "1100011x" (x : Don't care)	Immediate	R/W	0
19:8	Reserved	Reserved			0

Address		Name	Description	Default	
407	197h	TEST_PAT2	Test pattern configuration 2	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
15:0	Reserved	Reserved			0
16	TEST_PAT_SEL[3]	Selects the handoff pattern output. (This register is used with Add409[19:17].)	Immediate	R/W	0
19:17	Reserved	Reserved			0

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Address		Name	Description	Default	
409	199h	TEST_PAT0	Test pattern configuration 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
15:0	TEST_PAT_VAL[15:0]	Value setting for constant test pattern	Immediate	R/W	0
16	TEST_PAT_EN	Test pattern enable "0" : Test pattern will be directly put into output "1" : Test pattern will be put instead of ADC data	Immediate	R/W	0
19:17	TEST_PAT_SEL[2:0]	Selects the handoff pattern output. TEST_PAT_SEL[3:0] : Pattern output Note) TEST_PAT_SEL[3] : Add407[16] In case of TEST_PAT_EN="1" "0" : Normal output "1" : All "0" "2" : All "1" "3" : 00FFh "4" : 5555h "5" : Constant of TEST_PAT_VAL1 "6" : Ramp (1LSB will be constantly increased) "7" : Alternate AAAAh and 5555h "8" : Alternate TEST_PAT_VAL1 and VAL2 "9" : Ramp between TEST_PAT_VAL1 and VAL2 "10" : Select TEST_PAT_VAL1/2 with SEL_TEST_VAL[3:0] In case of TEST_PAT_EN="0" "0" : Normal output "1" : All "0" "2" : All "1" "3" : 7bitLVDS: 0000111b, 8bitLVDS/CMOS: 00001111b 4bitCMOS: 0011b "4" : 7bitLVDS: 0101010b, 8bitLVDS/CMOS: 01010101b 4bitCMOS: 0101b "5" : 7bitLVDS: TEST_PAT_VAL1[7:1] 8bitLVDS/CMOS: TEST_PAT_VAL1[7:0] 4bitCMOS: TEST_PAT_VAL1[3:0] "6" : Ramp (1LSB will be constantly increased) "7" : 7bitLVDS: Alternate 2Ah and 55h 8bitLVDS/CMOS: Alternate AAh and 55h 4bitCMOS: Alternate Ah and 5h "8" : 7bitLVDS: Alternate TEST_PAT_VAL1[7:1] and TEST_PAT_VAL2[7:1] 8bitLVDS/CMOS: Alternate TEST_PAT_VAL1[7:0] and TEST_PAT_VAL2[7:0] 4bitCMOS: Alternate TEST_PAT_VAL1[3:0] and TEST_PAT_VAL2[3:0] "9" : N/A "10" : Select TEST_PAT_VAL1/2 with SEL_TEST_VAL[3:0] Note) TEST_PAT_VAL2 : Add433[15:0] Note) SEL_TEST_VAL[3:0] : Add434[3:0]	Immediate	R/W	0

Address		Name	Description	Default	
413	19Dh	CLK_CFG4	Clock configuration 4	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
5:0	Reserved	Reserved			0
6	RCLKN_FLOAT	Internal connection state setting for "RCLKN" pin 0 : Connect, 1 : Disconnect	Immediate	R/W	0
19:7	Reserved	Reserved			0

Address		Name	Description	Default	
415	19Fh	CLK_CFG5	Clock configuration 5	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
2:0	Reserved	Reserved			0
3	CLK_HYST_INC	Hysteresis setting for MCLK clock input port 0 : Default hysteresis for LVDS clock input 1 : Increased hysteresis for CMOS clock input	Immediate	R/W	0
19:4	Reserved	Reserved			0

Address		Name	Description	Default	
423	1A7h	SYS_TM0	System test configuration 0	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
3:0	Reserved	Reserved			0
4	PLLS_RESET	Reset PLL "0": Normal, "1": Reset	Immediate	R/W	0
19:5	Reserved	Reserved			0

Address		Name	Description	Default	
433	1B1h	TEST_PAT3	Test pattern configuration 3	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
15:0	TEST_PAT_VAL2 [15:0]	Value setting for constant test pattern 2	Immediate	R/W	0
16	DIS_TEST_LS_SNC	Disable "LS" code in LVDS mode with test pattern "0" : Normal, "1" : Disable	Immediate	R/W	0
17	LS_TEST_PAT	Synchronize Test pattern with "LS" "0" : ASYNC, "1" : SYNC	Immediate	R/W	0
19:18	Reserved	Reserved	Immediate	R/W	0

Address		Name	Description	Default	
434	1B2h	TEST_PAT4	Test pattern configuration 4	0	00000h
Bit	Symbol	Function	Update Timing	R/W	Default
0	SEL_TEST_VAL[0]	TEST_PAT_EN = "0" , Select TEST_PAT_VAL1 or 2 for TA TEST_PAT_EN = "1" , Select TEST_PAT_VAL1 or 2 for Ch1 "0" : TEST_PAT_VAL1, "1" : TEST_PAT_VAL2	Immediate	R/W	0
1	SEL_TEST_VAL[1]	TEST_PAT_EN = "0" , Select TEST_PAT_VAL1 or 2 for TB TEST_PAT_EN = "1" , Select TEST_PAT_VAL1 or 2 for Ch2 "0" : TEST_PAT_VAL1, "1" : TEST_PAT_VAL2	Immediate	R/W	0
2	SEL_TEST_VAL[2]	TEST_PAT_EN = "0" , Select TEST_PAT_VAL1 or 2 for TC TEST_PAT_EN = "1" , Select TEST_PAT_VAL1 or 2 for Ch3 "0" : TEST_PAT_VAL1, "1" : TEST_PAT_VAL2	Immediate	R/W	0
3	SEL_TEST_VAL[3]	TEST_PAT_EN = "0" , N/A TEST_PAT_EN = "1" , Select TEST_PAT_VAL1 or 2 for Ch4 "0" : TEST_PAT_VAL1, "1" : TEST_PAT_VAL2	Immediate	R/W	0
19:4	Reserved	Reserved	Immediate	R/W	0